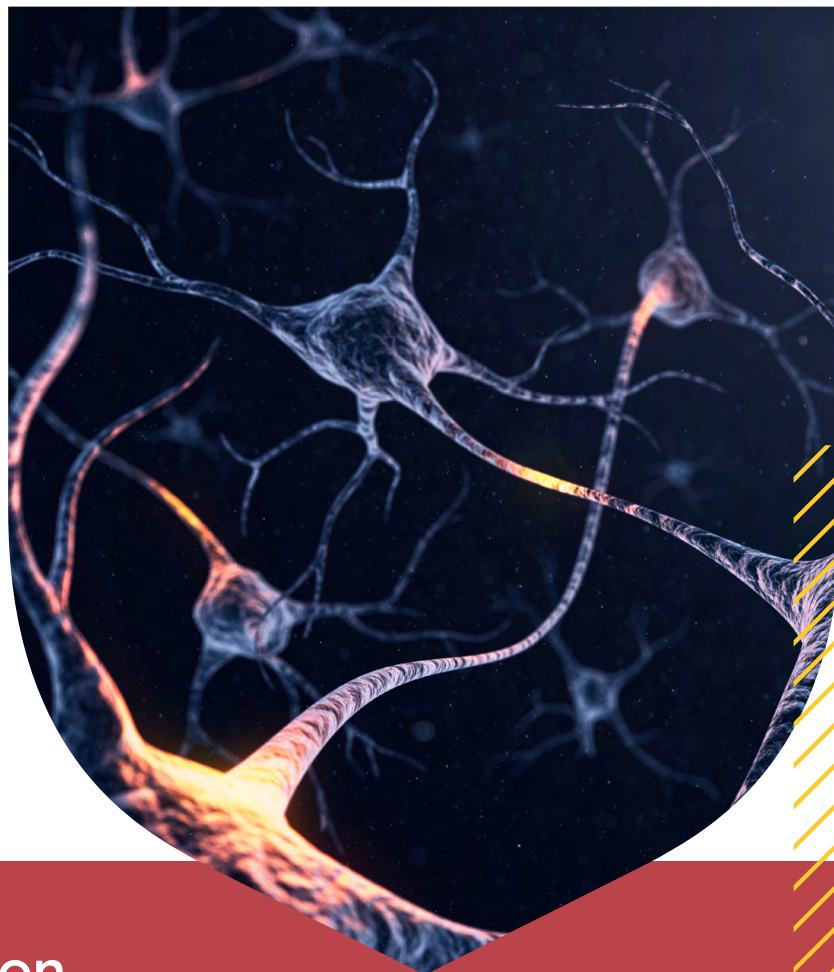




Specification

# Cambridge Advanced National in Human Biology

Cambridge OCR Level 3 Alternative Academic Qualification  
Cambridge Advanced National in Human Biology

**Certificate | H049**  
**Extended Certificate | HI49**  
For first teaching in 2025

Version 6.0 | September 2025  
[ocr.org.uk/cambridge-advanced-nationals](https://ocr.org.uk/cambridge-advanced-nationals)

CAMBRIDGE  
ADVANCED  
NATIONALS



## About our new name

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### As of September 2025, our name is Cambridge OCR.

Students who sat an exam **in summer 2025** will receive a Cambridge OCR branded exam certificate (our new brand), which will be the same for all future exam series.

You'll continue to see the OCR or Oxford Cambridge and RSA Examinations name while we work to update our material to our new name, Cambridge OCR. This will take some time, and you can still access all up-to-date qualification resources and materials via [Teach Cambridge](#).

**Important:** We'll keep the OCR/Oxford Cambridge and RSA name on existing teaching resources while the content of these remains applicable to the specification being taught. **New and refreshed** resources will be produced using the Cambridge OCR name/logo.

## Are you using the latest version of this specification?

The latest version of our specifications will always be on [our website](#) and may differ from printed versions. We will inform centres about changes to specifications.

## Tell us what you think

Your feedback plays an important role in how we develop, market, support, and resource qualifications now and into the future. We want teachers and students to enjoy and get the best out of our qualifications and resources, but to do that we need honest opinions to tell us whether we're on the right track or not. That's where you come in.

You can email your thoughts to [ProductDevelopment@OCR.org.uk](mailto:ProductDevelopment@OCR.org.uk) or visit our [feedback page](#) to learn more about how you can help us improve our qualifications.



Designing and testing in [collaboration with teachers](#) and students



Helping young people develop an [ethical view of the world](#)



Equality, diversity, inclusion and belonging (EDIB) are [part of everything we do](#)

## Specification updates

Key changes have been listed below:

| Section          | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Version and date of issue               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|                  | Draft specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Version 1                               |
| Section 5        | Unit F170 Breadth and depth updated for 2.1.2, 2.2.2, 3.3.1, 4.1.2, 4.1.4<br>Unit F171 Taught content updated for 1.3.1<br>Unit F172 Exemplification updated for 1.3.2<br>Unit F173 Taught content and Exemplification updated for 5.1.2, and M4 Assessment criterion updated<br>Unit F174 M1 Assessment guidance updated<br>Unit F177 Exemplification updated for 2.4.1                                                                                                                                                      | Changes between Version 2 and Version 5 |
| Section 6        | Unit F171 overview table updated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                         |
| Section 8.2      | New section added: Collecting evidence of student performance to ensure resilience in the qualifications system                                                                                                                                                                                                                                                                                                                                                                                                               |                                         |
| Appendix C       | HSW6 updated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                         |
| Appendix D       | M1.5 updated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                         |
| Section 2        | 'Why Choose Cambridge OCR' updated text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Version 6 (September 2025)              |
| Section 3        | Updated information about: <ul style="list-style-type: none"> <li>• Registering students and making entries</li> <li>• Exam resits</li> <li>• NEA submission</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |                                         |
| Section 7        | Enhanced guidance about: <ul style="list-style-type: none"> <li>• The use of set assignments, including templates provided</li> <li>• Appropriate use of the internet when completing assignments</li> <li>• Assessing student work and providing feedback, including informing students of their marks</li> <li>• Digital submission of work</li> <li>• Submitting outcomes and sample requests to us (including timeframes)</li> <li>• Resubmitting moderated work</li> <li>• The use of URS or recording sheets</li> </ul> |                                         |
| Sections 7 and 8 | Updated information to ensure compliance with new JCQ requirements and guidance.                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                         |
| Section 8        | Updated information about: <ul style="list-style-type: none"> <li>• Registering students</li> <li>• Making unit entries</li> <li>• Making certification claims</li> <li>• Unit and qualification resits</li> </ul>                                                                                                                                                                                                                                                                                                            |                                         |

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# 1 Qualifications at a glance

## 1.1 Qualification structure

Key to units for these qualifications:

|                               |                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------|
| EA = External Assessment      | We set and mark the exams for these units.                                                          |
| NEA = Non Examined Assessment | We set the assignment for these units.<br>You assess the assignment and we moderate the assessment. |
| M = Mandatory                 | Students must complete these units.                                                                 |
| O = Optional                  | Students must complete some of these units.                                                         |
| GLH = Guided Learning Hours   | The teacher contact time needed to teach the content, plus the assessment time for the unit.        |

### Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate)

For this qualification, students must complete three units:

- One mandatory externally assessed unit
- Two mandatory NEA units

### Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate)

For this qualification, students must complete six units:

- Two mandatory externally assessed units
- Two mandatory NEA units
- Two optional NEA units

| Unit no | Unit title                    | Unit ref no (URN) | Guided learning hours (GLH) | Assessment method | Certificate | Extended Certificate |
|---------|-------------------------------|-------------------|-----------------------------|-------------------|-------------|----------------------|
| F170    | Fundamentals of human biology | M/651/0641        | 80                          | EA                | M           | M                    |
| F171    | Health and disease            | R/651/0642        | 80                          | EA                | -           | M                    |
| F172    | Genetics                      | T/651/0643        | 50                          | NEA               | M           | M                    |
| F173    | Biomedical techniques         | Y/651/0644        | 50                          | NEA               | M           | M                    |
| F174    | Nutrition and metabolism      | A/651/0645        | 50                          | NEA               | -           | O                    |
| F175    | Human reproduction            | D/651/0646        | 50                          | NEA               | -           | O                    |
| F176    | The brain                     | F/651/0647        | 50                          | NEA               | -           | O                    |
| F177    | Drug development              | H/651/0648        | 50                          | NEA               | -           | O                    |

## 1.2 Comparison between the Cambridge Advanced Nationals Qualifications and the Level 3 Cambridge Technicals qualification models

|   | Area of comparison                          | Approach used in these Level 3 Cambridge Advanced Nationals qualifications                                                                                                                                                                                                                   | Approach used in the Level 3 Cambridge Technicals qualification model                                                                                                                                   | Reasons for the change                                                                                                                                                                                                                   |
|---|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | The size of the qualifications              | Qualifications are available in two sizes <ul style="list-style-type: none"> <li>• 180 GLH</li> <li>• 360 GLH</li> </ul> The 180 GLH qualification includes nested units from the 360 GLH qualification.                                                                                     | Qualifications are typically available in the following sizes: <ul style="list-style-type: none"> <li>• 180 GLH</li> <li>• 360 GLH</li> <li>• 540 GLH</li> <li>• 720 GLH</li> <li>• 1080 GLH</li> </ul> | For this subject, the Department for Education allows: <ul style="list-style-type: none"> <li>• a maximum size of 360 GLH for these qualifications.</li> <li>• a maximum of two qualification sizes.</li> </ul>                          |
| 2 | Number and duration of external assessments | 180 GLH qualification: <ul style="list-style-type: none"> <li>• One externally assessed unit</li> <li>• Exam is 1 hour 15 minutes</li> </ul> 360 GLH qualification: <ul style="list-style-type: none"> <li>• Two externally assessed units</li> <li>• Exams are 1 hour 15 minutes</li> </ul> | There are no exams in the 2012 qualifications.<br><br>In the 2016 suite, there is a minimum requirement of 30% external assessment.                                                                     | It is an Ofqual requirement to have 40% external assessment in these qualifications.<br><br>The exam design is intended to aid accessibility and encourage student engagement while easing the exam burden for students and timetabling. |
| 3 | Format of the exam                          | Each exam is available in January and June and is paper-based.                                                                                                                                                                                                                               | Each exam is available in January and June and is mainly paper-based.                                                                                                                                   | It is an Ofqual requirement to have two assessment opportunities per assessment.                                                                                                                                                         |
| 4 | Setting the NEA assignment                  | We will set all NEA assignments.                                                                                                                                                                                                                                                             | We provide a model assignment, or centres can set their own.                                                                                                                                            | This is a requirement of our Regulator, Ofqual.                                                                                                                                                                                          |
| 5 | Lifespan of the assignment                  | Each assignment will remain live for <b>two</b> years, with a new assignment being released every year.                                                                                                                                                                                      | Assignments can be used for a number of years.                                                                                                                                                          | This is a requirement of our Regulator, Ofqual.                                                                                                                                                                                          |



|   |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                              |
|---|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | The approach to achieving unit grades on the NEA units and its impact on qualification outcomes | <p>These take a 'compensatory' approach. This means that:</p> <ul style="list-style-type: none"> <li>the unit grade students achieve is based on the <b>total</b> number of criteria achieved for that unit.</li> <li>the total number can come from any combination of the pass, merit or distinction criteria.</li> <li>students do not have to achieve all criteria for a grade to achieve that grade (e.g. all pass criteria to achieve a unit pass).</li> <li>if students do not achieve enough total criteria for a unit pass, the criteria they do achieve will still earn uniform marks (UMS) which will count towards their qualification outcome.</li> <li>The qualification outcome is based on the combined total UMS achieved for all units. This means that students may still pass the qualification if they achieve enough total marks, even if they do not pass all units. Every mark counts!</li> </ul> | <p>These take a 'hurdles' approach. This means students must achieve:</p> <ul style="list-style-type: none"> <li>all pass criteria to achieve a unit pass</li> <li>all pass and merit criteria to achieve a unit merit.</li> <li>all pass, merit and distinction criteria to achieve a unit distinction.</li> <li>At least a pass for each NEA unit to achieve the qualification (along with at least a near pass in the examined unit/s).</li> </ul> | <p>The Cambridge Advanced Nationals qualifications are designed for academic progression. A compensatory approach rewards students for what they can do by combining marks achieved to calculate a qualification outcome.</p>                                |
| 7 | Number of NEA Assessment Criteria                                                               | <p>Each NEA unit of the same size has a fixed and consistent number of pass, merit and distinction assessment criteria, within and across qualifications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>The number of pass, merit and distinction assessment criteria differs across units and qualifications.</p>                                                                                                                                                                                                                                                                                                                                         | <p>This is to:</p> <ul style="list-style-type: none"> <li>ensure a consistent approach to the awarding of units within each qualification and across qualifications in the suite.</li> <li>aid familiarity of approach for teachers and students.</li> </ul> |

|    |                                                  |                                                                                                                                                                                                                                                                    |                                                                                                                                                     |                                                                                                                                                                                                                                                              |
|----|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8  | NEA Assessment Criteria design                   | There will be 24 assessment criteria for each NEA unit. Each assessment criterion is designed to: <ul style="list-style-type: none"> <li>• assess one discrete task or activity</li> <li>• provide a yes/no approach to decision-making and achievement</li> </ul> | There may be fewer assessment criteria for each unit, but these are typically broader, and may assess several tasks or activities in one criterion. | This is to: <ul style="list-style-type: none"> <li>• ensure clarity of requirements for students in the form of discrete tasks or activities that they should evidence</li> <li>• simplify decision-making for teachers assessing students' work.</li> </ul> |
| 9  | Introduced Performance Objectives for each unit  | Each exam question and each Assessment Criterion in the NEA units is mapped to one of our four performance objectives.                                                                                                                                             | These qualifications do not contain performance objectives.                                                                                         | To aid consistency of approach and demand to exams and assignments over time.                                                                                                                                                                                |
| 10 | Moderation opportunities for the NEA assignments | Moderation is available twice each year in windows.                                                                                                                                                                                                                | Moderation is available on-demand.                                                                                                                  | Typically, Level 3 Cambridge Advanced Nationals will be delivered in two years. This allows you the opportunity for two moderation activities in each academic year for this subject.                                                                        |
| 11 | Moderation approach                              | Moderation takes the form of face-to-face or virtual visits between the centre and our moderator.                                                                                                                                                                  | Moderation takes the form of face-to-face or virtual visits between the centre and our moderator.                                                   | We have kept this the same to reflect the most requested approach to moderation from centres since the pandemic<br><br>This is to ease the moderation burden on centres, while still providing direct interaction with our moderator.                        |
| 12 | SAMs for NEA                                     | Sample assignments are available for you to use as practice materials with students.                                                                                                                                                                               | We do not provide sample assignments for practice purposes.                                                                                         | This is to ensure that students have access to sample assessment material for both the EA and NEA units.                                                                                                                                                     |

## 2 Why choose Cambridge OCR?

Choose Cambridge OCR and you have the reassurance that you are working with one of the UK's leading examination boards.

We work with teachers, employers and universities to create qualifications that support the needs of all students and help prepare them for their future. We offer A Levels, GCSEs, vocational qualifications and other academic options to schools, colleges, workplaces, and other organisations.

We are part of Cambridge University Press & Assessment, Europe's largest assessment agency and a department of the University of Cambridge. We play a leading role in developing and delivering assessments worldwide, operating in over 150 countries. We listen. The decisions we make when we develop our specifications are based on teacher and student feedback. To tell us more about your experiences of teaching Cambridge OCR, join our teacher panel and help shape the future of our qualifications.

### 2.1 Our specifications

We provide specifications that help you bring the subject to life and inspire your students to achieve more.

We've created teacher-friendly specifications based on extensive research and engagement with the teaching community. Our specifications are designed to be straightforward to deliver and accessible for students. The design allows you to tailor the delivery of the course to suit your needs.

### 2.2 Our support

We provide a range of support services to help you at every stage, from preparation to delivery:

- A wide range of high-quality creative resources including resources created by leading organisations in the industry.
- Textbooks and teaching and learning resources from leading publishers. The Cambridge Advanced Nationals page on our [website](#) has more information about all the published support for the qualifications that we have endorsed.
- Professional development for teachers to meet a range of needs. To join our training (either face-to-face or online) or to search for training materials, go to the [Professional Development page](#) on our website.
- [Active Results](#) which is our free results analysis service. It helps you review the performance of individual students or whole groups.
- [ExamBuilder](#) which is our free question-building platform. It helps you to build your own tests using past OCR exam questions.
- Our Subject Advisors, who give information and support to centres. They can help with specification and non examined assessment (NEA) advice, updates on resources developments and a range of training opportunities. They use networks to work with subject communities and share ideas and expertise to support teachers.

### 2.2.1 More help and support

Whether you are new to us or already teaching with us, you can find useful information, help and support on our [website](#). Or get in touch:

[support@ocr.org.uk](mailto:support@ocr.org.uk)

[@ocrexams](#)

01223 553998

## 2.3 People and Planet

We are part of Cambridge University Press & Assessment, which has clear commitments to champion sustainability, diversity, trust and respect for our people and planet.

We are committed to supporting a curriculum that helps young people develop an ethical view of the world. This enables them to take social responsibility, understand environmental issues and prepare them for the green jobs of the future.

**Our equality, diversity, inclusion and belonging principles are that we:**

- are respectful and considerate
- celebrate differences and promote positive attitudes to belonging
- include perspectives that reflect the diverse cultural and lifestyle backgrounds of our society
- challenge prejudicial views and unconscious biases
- promote a safe and supportive approach to learning
- are accessible and fair, creating positive experiences for all
- provide opportunities for everyone to perform at their best
- are contemporary, relevant and equip everyone to live and thrive in a global, diverse world
- create a shared sense of identity in a modern mixed society with one humanity.

**To learn more, including our work on accessibility in our assessment materials, visit our [People and Planet page](#).**

## 2.4 Aims and learning outcomes

Our Cambridge Advanced Nationals in Human Biology will encourage students to:

- develop key knowledge, understanding and skills, relevant to the subject
- think creatively, innovatively, analytically, logically and critically
- develop valuable communication skills that are important in all aspects of further study and life
- develop transferable learning and skills, such as reflection, planning, presentation and research skills, that are important for progression to HE and can be applied to real-life contexts and work situations
- develop independence and confidence in applying the knowledge and skills that are vital for progression to HE and relevant to the medical science sector and more widely

## 2.5 What are the key features of this specification?

The key features of our Cambridge Advanced Nationals in Human Biology for you and your students are:

- a simple and intuitive assessment model, that has:
  - externally assessed units, which focus on subject knowledge and understanding
  - applied and practical non examined assessment units (NEA)
  - optional NEA units to provide flexibility
- a specification developed with teachers specifically for teachers. The specification lays out the subject content, assessment criteria, teacher guidance and delivery requirements clearly
- a flexible support package made based on teachers' needs. The support package will help teachers to easily understand the qualification and how it is assessed
- a team of Subject Advisors who directly support teachers
- a specification designed to:
  - complement A Levels and/or other Level 3 qualifications in a Post-16 study programme
  - develop wider transferable skills, knowledge and understanding desired by HEIs. More detail about the transferable skills these qualifications may develop is in [Section 6.3](#).

All Cambridge Advanced National qualifications offered by Cambridge OCR are regulated by Ofqual, the Regulator for qualifications offered in England.

The qualification numbers for our Cambridge Advanced Nationals in Human Biology are:

- Certificate: QN 610/3945/7
- Extended Certificate: QN 610/3946/9

## 2.6 Acknowledgements

|                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------|
| We would like to acknowledge the following Higher Education Providers/organisations for their input and support in designing these qualifications: |
| Aston University                                                                                                                                   |
| Cardiff University                                                                                                                                 |
| Coventry University                                                                                                                                |
| Institute of Biomedical Science                                                                                                                    |
| Nottingham Trent University                                                                                                                        |
| Staffordshire University                                                                                                                           |
| Teesside University                                                                                                                                |
| University of Bradford                                                                                                                             |
| University of East Anglia                                                                                                                          |
| University of Gloucester                                                                                                                           |
| University of Lincoln                                                                                                                              |
| University of Manchester                                                                                                                           |
| University of Southampton                                                                                                                          |
| University of the West of England                                                                                                                  |

### 3 Qualification overview

#### 3.1 Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate) overview

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qualification number                             | 610/3945/7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| First entry date                                 | 01 September 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Guided learning hours (GLH)                      | 180                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Total qualification time (TQT)                   | 225                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Entry code                                       | H049                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Approved age range                               | 16-18, 18+, 19+                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Offered in                                       | England only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Performance table information                    | This qualification is designed to meet the Department for Education's requirements for qualifications in the Alternative Academic Qualifications category of the 16-19 performance tables.                                                                                                                                                                                                                                                                                                                                                                                                     |
| Eligibility for funding                          | This qualification meets funding approval criteria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| UCAS points                                      | This qualification is recognised in the UCAS tariff tables.<br>You'll find more information on the <a href="#">UCAS website</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| This qualification is suitable for students who: | <ul style="list-style-type: none"> <li>are age 16-19 and on a full-time study programme</li> <li>want to develop applied knowledge and skills in human biology</li> <li>want to progress onto other related study, such as higher education courses in biological sciences, life sciences and human biology</li> </ul>                                                                                                                                                                                                                                                                         |
| Entry requirements                               | <p>We recommend that students have achieved a science qualification at Level 2, for example:</p> <ul style="list-style-type: none"> <li>GCSEs in science subjects at grade 4 (4-4) or above.</li> <li>a Level 2 vocational qualification such as Cambridge OCR Level 2 Cambridge Technical in Science.</li> </ul> <p>We also recommend that:</p> <ul style="list-style-type: none"> <li>students have grade 4/grade C or above in Maths and English GCSE.</li> <li>you carry out an initial assessment to make sure students can reach the required standards of the qualification.</li> </ul> |
| Qualification requirements                       | <p>Students must complete three units:</p> <ul style="list-style-type: none"> <li>one externally assessed unit</li> <li>two NEA units</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registering students and making entries | <p>You must:</p> <ul style="list-style-type: none"> <li>• register each student for the subject at the start of the course. You will need to do this before you can make unit entries. The registration code is A005.</li> <li>• make exam and NEA unit entries for the relevant assessment series.</li> <li>• make qualification certification entries in the final series at the end of the course.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Assessment method/model                 | <p>Unit F170 is assessed by an exam and marked by us.</p> <p>You will assess the NEA units and we will moderate them.</p> <ul style="list-style-type: none"> <li>• NEA assignments are live for two years. The front cover of each assignment shows the intended student cohort.</li> <li>• You must use and submit work for the live assignment during the period when that assignment is live.</li> </ul> <p>We strongly recommend you use the set assignment that is released in the same calendar year as the new cohort starts. This ensures that students have two years to use the assignment. It gives them access to a resubmission opportunity if needed during their study of the qualification.</p> <p>For example, a cohort starting a two-year course in September 2026 should use the set of assignments labelled for 2026-2028.</p> <p>You should avoid allowing new cohorts to use assignments which have already been live for a year. For example, avoid students starting in September 2027 using assignments for the 2026-2028 cohorts.</p> <p>You must have suitable controls in place to ensure that NEA assignment work is completed by each student independently. You must not allow previously completed work for assignments which are still live to be shared as examples with other students.</p> |
| Exam series each year                   | <ul style="list-style-type: none"> <li>• January</li> <li>• June</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Exam resits                             | <p>Students can resit the examined unit (F170) twice. This means they can have three attempts <b>in total</b> at exams for unit F170.</p> <p>This total number of attempts includes attempts towards both the Certificate and Extended Certificate qualifications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| NEA submission                          | <p>There are two windows each year to submit NEA outcomes and request a moderation visit for this subject.</p> <p>You must register students for the subject and make unit entries before you can submit outcomes for a visit.</p> <p>All dates are on our <a href="#">Administration</a> pages.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resubmission of students' NEA work | <p>If students have not performed at their best in the NEA assignments, they can improve their work and submit it to you again for assessment. They must have your agreement and you must be sure it is in the student's best interests.</p> <p>We use the term 'resubmission' when referring to student work that has previously been submitted for moderation. Following moderation, a student can attempt to improve their work for you to assess and provide the final mark to us. There is one resubmission opportunity per NEA assignment.</p> <p>All work submitted (or resubmitted) must be based on the assignment that is live for assessment.</p> <p>For information about feedback see <a href="#">Section 7.3</a>. The final piece of work must be completed solely by the student and teachers must not detail specifically what amendments should be made.</p> |
| Grading                            | Information about unit and qualification grading is in <a href="#">Section 6</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### 3.2 Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate) overview

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qualification number                             | 610/3946/9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| First entry date                                 | 01 September 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Guided learning hours (GLH)                      | 360                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Total qualification time (TQT)                   | 450                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Entry code                                       | H149                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Approved age range                               | 16-18, 18+, 19+                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Offered in                                       | England only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Performance table information                    | This qualification is designed to meet the Department for Education's requirements for qualifications in the Alternative Academic Qualifications category of the 16-19 performance tables.                                                                                                                                                                                                                                                                                                                                                                                                             |
| Eligibility for funding                          | This qualification meets funding approval criteria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| UCAS points                                      | This qualification is recognised in the UCAS tariff tables.<br>You'll find more information on the <a href="#">UCAS website</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| This qualification is suitable for students who: | <ul style="list-style-type: none"> <li>• are age 16-19 and on a full-time study programme</li> <li>• want to develop applied knowledge and skills in human biology</li> <li>• want to progress onto other related study, such as higher education courses in biological sciences, life sciences and human biology</li> </ul>                                                                                                                                                                                                                                                                           |
| Entry requirements                               | <p>We recommend that students have achieved a science qualification at Level 2, for example:</p> <ul style="list-style-type: none"> <li>• GCSEs in science subjects at grade 4 (4-4) or above.</li> <li>• a Level 2 vocational qualification such as Cambridge OCR Level 2 Cambridge Technical in Science.</li> </ul> <p>We also recommend that:</p> <ul style="list-style-type: none"> <li>• students have grade 4/grade C or above in Maths and English GCSE.</li> <li>• you carry out an initial assessment to make sure students can reach the required standards of the qualification.</li> </ul> |
| Qualification requirements                       | <p>Students must complete six units:</p> <ul style="list-style-type: none"> <li>• two externally assessed units</li> <li>• four NEA units</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registering students and making entries | <p>You must:</p> <ul style="list-style-type: none"> <li>• register each student for the subject at the start of the course. You will need to do this before you can make unit entries. The registration code is A005.</li> <li>• make exam and NEA unit entries for the relevant assessment series.</li> <li>• make qualification certification entries in the final series at the end of the course.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Assessment method/model                 | <p>Units F170 and F171 are assessed by an exam and marked by us.</p> <p>You will assess the NEA units and we will moderate them.</p> <ul style="list-style-type: none"> <li>• NEA assignments are live for two years. The front cover of each assignment shows the intended student cohort.</li> <li>• You must use and submit work for the live assignment during the period when that assignment is live.</li> </ul> <p>We strongly recommend you use the set assignment that is released in the same calendar year as the new cohort starts. This ensures that students have two years to use the assignment. It gives them access to a resubmission opportunity if needed during their study of the qualification.</p> <p>For example, a cohort starting a two-year course in September 2026 should use the set of assignments labelled for 2026-2028.</p> <p>You should avoid allowing new cohorts to use assignments which have already been live for a year. For example, avoid students starting in September 2027 using assignments for the 2026-2028 cohorts.</p> <p>You must have suitable controls in place to ensure that NEA assignment work is completed by each student independently. You must not allow previously completed work for assignments which are still live to be shared as examples with other students.</p> |
| Exam series each year                   | <ul style="list-style-type: none"> <li>• January</li> <li>• June</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Exam resits                             | <p>Students can resit each examined unit twice. This means they can have three attempts <b>in total</b> at exams <b>for each</b> of units F170 and F171.</p> <p>For unit F170, this total number of attempts includes attempts towards both the Certificate and Extended Certificate qualifications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| NEA Submission                          | <p>There are two windows each year to submit NEA outcomes and request a moderation visit for this subject.</p> <p>You must register students for the subject and make unit entries before you can submit outcomes for a visit.</p> <p>All dates are on our <a href="#">Administration</a> pages.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resubmission of students' NEA work | <p>If students have not performed at their best in the NEA assignments, they can improve their work and submit it to you again for assessment. They must have your agreement and you must be sure it is in the student's best interests.</p> <p>We use the term 'resubmission' when referring to student work that has previously been submitted for moderation. Following moderation, a student can attempt to improve their work for you to assess and provide the final mark to us. There is one resubmission opportunity per NEA assignment.</p> <p>All work submitted (or resubmitted) must be based on the assignment that is live for assessment.</p> <p>For information about feedback see <a href="#">Section 7.3</a>. The final piece of work must be completed solely by the student and teachers must not detail specifically what amendments should be made.</p> |
| Grading                            | Information about unit and qualification grading is in <a href="#">Section 6</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### 3.3 Purpose statement – Certificate



Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate)

Qualification number: 610/3945/7

Overview

#### Who this qualification is for

The Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate) is for students aged 16-19 years old. It will develop knowledge, understanding and skills that will help prepare you for progression to undergraduate study when taken alongside other qualifications and are relevant to the medical science sector.

You might be interested in this qualification if you want a small qualification that builds applied or practical skills, to take alongside and enhance your A Levels or other Level 3 qualifications. You will have the opportunity to apply what you learn to real-life contexts, such as:

- Researching human biology fundamentals.
- Creating and delivering presentations to help patients and health care professionals.
- Planning and performing laboratory investigations involving biomedical techniques.

The qualification will also help you develop independence and confidence in using skills that are relevant to the medical science sector and that prepare you for progressing to university courses where independent study skills are needed. You will develop the following transferable skills that can be used in both higher education and other life and work situations:

- Researching topic areas and recording research sources, then using them to interpret findings and present evidence.
- Problem solving when matching and analysing data.
- Communicating effectively with individuals or groups.

This qualification will complement other learning that you're completing at Key Stage 5. If you are a full-time student, it will be part of your studies along with your A Levels and/or other Level 3 qualifications.

#### What you will study when you take this qualification

Through a combination of theoretical study and hands-on experience, you will develop the necessary knowledge and skills that can support progression to higher education human biology study.

In the examined units, you will study key knowledge and understanding relevant to human biology. In the non examined assessment (NEA) units, you will demonstrate knowledge and skills you learn by completing an applied or practical assignment. More information about the knowledge and skills you will develop is below.

All units in the qualification are mandatory. You must take **all** of these units:

- **F170: Fundamentals of human biology**

This unit is assessed by an exam.

In this unit you will learn about the key topics that are important in human biology. Topics include:

- Topic Area 1 Human cells and tissues
- Topic Area 2 Human physiology, organs and systems
- Topic Area 3 Key concepts in endocrinology, neurobiology and reproduction
- Topic Area 4 Basics of microbiology

- **F172: Genetics**

This unit is assessed by an assignment.

In this unit you will learn about DNA, cell division and inheritance. Topics include:

- Topic Area 1 Fundamentals of genetics
- Topic Area 2 Mode of inheritance
- Topic Area 3 Genetic counselling and genetic testing
- Topic Area 4 Gene therapy and genetic engineering

- **F173: Biomedical techniques**

This unit is assessed by an assignment.

In this unit you will plan and carry out an investigation using a variety of laboratory techniques. Topics include:

- Topic Area 1 What biomedical science is
- Topic Area 2 Diagnostic techniques: cells and microscopy
- Topic Area 3 Diagnostic techniques: biological molecules
- Topic area 4 Planning a clinical investigation
- Topic area 5 Report writing

### **The subjects that complement this qualification**

These subjects might complement this qualification:

- A Level Biology
- A Level Chemistry
- A Level Physical Education
- A Level Psychology
- A Level Sociology

## The types of courses you may progress to

Both the subject-specific knowledge, understanding and skills, and broader transferable skills developed in this qualification will help you progress to further study in related areas such as:

- Biological Sciences degree
- Human Biology degree
- Life Sciences degree
- Biomedical Science degree

## Why you should take the Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate)

There are two qualifications available in human biology these are:

Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate) – this is 180 GLH in size

Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate) – this is 360 GLH in size

You should take this Certificate qualification if you want a small Level 3 qualification that builds some applied knowledge and skills in human biology. This qualification is an Alternative Academic Qualification that is the same size as an AS Level qualification. It is half the size of an A Level. It could be taken alongside A Levels and/or other Level 3 qualifications to enhance your learning, helping you to build broader knowledge and skills that are valued in undergraduate study, and relevant for progression to higher education. You would take this qualification alongside A Levels and/or other Level 3 qualifications as part of your study programme at Key Stage 5.

## More information

More information about this qualification is in these documents:

- Sample Assessment Material (SAM) Question Papers:
  - Unit F170: [Fundamentals of human biology](#)
- Guides to our SAM Question Papers:
  - Unit F170: [Fundamentals of human biology](#)
- SAM Set assignments:
  - Unit F172: [Genetics](#)
  - Unit F173: [Biomedical techniques](#)
- [Student Guide to NEA Assignments: Human Biology](#)

### 3.4 Purpose statement – Extended Certificate



Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate)

Qualification number: 610/3946/9

Overview

#### Who this qualification is for

The Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate) is for students aged 16-19 years old. It will develop knowledge, understanding and skills that will help prepare you for progression to undergraduate study and are relevant to the medical science sector.

You might be interested in this qualification if you want to apply what you learn to practical, real-life contexts, such as:

- Researching health and diseases.
- Planning and performing laboratory investigations involving biomedical techniques.
- Creating and delivering presentations to help patients and health care professionals.
- Creating and delivering nutritional information to individuals with specific needs.

The qualification will also help you develop independence and confidence in using skills that are relevant to the medical science and that prepare you for progressing to university courses where independent study skills are needed. You will develop the following transferable skills that can be used in both higher education and other life and work situations:

- Communicating effectively with individuals or groups.
- Researching topic areas and recording research sources, then using them to interpret findings and present evidence.
- Presenting information, this will involve managing time and identifying aims, purpose, resources, methods.
- Problem solving when matching and analysing data.

The qualification has six units. Each unit has its own assessment and assessment can happen at different points during the year. This unitised, flexible approach to learning and assessment means learning and achievements can be recognised in bite-sized chunks, rather than all at the end of the course. The unitised approach will also be useful preparation if you want to progress to higher education where modular approaches to learning are common.

This qualification will complement other learning that you're completing at Key Stage 5. If you are a full-time student, it will be part of your studies along with A Levels and/or other Level 3 qualifications.



## What you will study when you take this qualification

Through a combination of theoretical study and hands-on experience, you will develop the necessary knowledge and skills that can support progression to higher education human biology study.

In the examined units, you will study key knowledge and understanding relevant to human biology. In the non examined assessment (NEA) units, you will demonstrate knowledge and skills you learn by completing applied or practical assignments. More information about the knowledge and skills you will develop is below.

The qualification has four mandatory units and two optional units.

These are the **mandatory** units – you must take **all** these units:

- F170: Fundamentals of human biology

This unit is assessed by an exam.

In this unit you will learn about the key topics that are important in human biology. Topics include:

- Topic Area 1 Human cells and tissues
- Topic Area 2 Human physiology, organs and systems
- Topic Area 3 Key concepts in endocrinology, neurobiology and reproduction
- Topic Area 4 Basics of microbiology

- F171: Health and disease

This unit is assessed by an exam.

In this unit you will learn about the intriguing and challenging nature of diseases and disorders. Topics include:

- Topic Area 1 Causes and effects of diseases and disorders
- Topic Area 2 Curative management and preventative therapies
- Topic Area 3 The role of immunology
- Topic Area 4 Techniques for diagnosis and monitoring
- Topic Area 5 Reporting, research and confidentiality

- F172: Genetics

This unit is assessed by an assignment.

In this unit you will learn about DNA, cell division and inheritance. Topics include:

- Topic Area 1 Fundamentals of genetics
- Topic Area 2 Mode of inheritance
- Topic Area 3 Genetic counselling and genetic testing
- Topic Area 4 Gene therapy and genetic engineering

- F173: Biomedical techniques

This unit is assessed by an assignment.

In this unit you will plan and carry out an investigation using a variety of laboratory techniques. Topics include:

- Topic Area 1 What biomedical science is
- Topic Area 2 Diagnostic techniques: cells and microscopy
- Topic Area 3 Diagnostic techniques: biological molecules
- Topic area 4 Planning a clinical investigation
- Topic area 5 Report writing

These are **optional** units – you must take **two** of these units:

- F174: Nutrition and metabolism

This unit is assessed by an assignment.

In this unit you will carry out practical investigations involving digestive enzymes and study parts of the digestive system using photomicrographs. Topics include:

- Topic Area 1 Nutrients required for a healthy body
- Topic Area 2 Diets and disorders
- Topic Area 3 Metabolic pathways and control mechanisms
- Topic Area 4 Diagnosis, monitoring and treatment for nutritional/metabolic disorders

- F175: Human reproduction

This unit is assessed by an assignment.

In this unit you will explore the development of the zygote, embryo and foetus and the process of pregnancy and antenatal care. Topics include:

- Topic Area 1 Conception and pregnancy
- Topic Area 2 Pregnancy (antenatal) care
- Topic Area 3 Infertility
- Topic Area 4 Assisted reproduction (AR)

- **F176: The brain**

This unit is assessed by an assignment.

In this unit you will gain a greater insight into the structure and function of the nervous system, including the spinal cord, brain and nerves. Topics include:

- Topic Area 1 Structure and function of the nervous system
- Topic Area 2 Neuron communication and control
- Topic Area 3 Nociception, neurotransmitters and drugs
- Topic Area 4 The diagnosis and treatment of brain disorders/injuries
- Topic Area 5 Monitoring and scanning the brain

- **F177: Drug development**

This unit is assessed by an assignment.

In this unit you will study the stages in the development of a drug and stages in the discovery of a commercial drug/medicine and pre-clinical and clinical trials. Topics include:

- Topic Area 1 Pharmaceutical drugs
- Topic Area 2 Process of drug development
- Topic Area 3 Factors influencing drug development
- Topic area 4 Producing a clinical research proposal

### **The subjects that complement this qualification**

These subjects might complement this qualification:

- A Level Biology
- A Level Chemistry
- A Level Physical Education
- A Level Psychology
- A Level Sociology

### **The types of courses you may progress to**

Both the subject-specific knowledge, understanding and skills, and broader transferable skills developed through these units, will help you progress to further study in related areas such as:

- Biological Sciences degree
- Human Biology degree
- Life Sciences degree
- Biomedical Science degree

## **Why you should take the Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate)**

There are two qualifications available in human biology These are:

Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate) – this is 180 GLH in size

Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate) – this is 360 GLH in size

You should take this Extended Certificate qualification if you want a Level 3 qualification that builds applied knowledge and skills in human biology. This qualification is an Alternative Academic Qualification that is the same size as an A Level. When it is taken alongside other Level 3 qualifications, it will complement them, helping you to build broader knowledge and skills that are valued in undergraduate study, and relevant for progression to higher education. You would take this qualification alongside other Level 3 qualifications as part of your study programme at Key Stage 5.

### **More information**

More information about this qualification is in these documents:

- Sample Assessment Material (SAM) Question Papers:
  - Unit F170: [Fundamentals of human biology](#)
  - Unit F171: [Health and disease](#)
- Guides to our SAM Question Papers:
  - Unit F170: [Fundamentals of human biology](#)
  - Unit F171: [Health and disease](#)
- SAM Set assignments:
  - Unit F172: [Genetics](#)
  - Unit F173: [Biomedical techniques](#)
  - Unit F174: [Nutrition and metabolism](#)
  - Unit F175: [Human reproduction](#)
  - Unit F176: [The brain](#)
  - Unit F177: [Drug development](#)
- [Student Guide to NEA Assignments: Human Biology](#)

## 4 About these qualifications

### 4.1 Qualification size

The size of each qualification is described in terms of Guided Learning Hours (GLH) and Total Qualification Time (TQT).

GLH indicates the approximate time (in hours) you will spend supervising or directing study and assessment activities. We have worked with people who are experienced in delivering related qualifications to determine the content that needs to be taught and how long it will take to deliver.

TQT includes two parts:

- GLH
- an estimate of the number of hours a student will spend on unsupervised learning or assessment activities (including homework) to successfully complete their qualification.

The Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate) is 180 GLH and 225 TQT.

The Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate) is 360 GLH and 450 TQT.

### 4.2 Availability and language

Our Level 3 Cambridge Advanced Nationals are available in England only. They are **not** available in Wales or Northern Ireland.

The qualifications and their assessment materials are available in English only. We will only assess answers written in English.

### 4.3 Prior knowledge and experience

Recognition of prior learning (RPL) is the process for recognising learning that never received formal recognition through a qualification or certification. It includes knowledge and skills gained in school, college or outside of formal learning situations. These may include:

- domestic/family life
- education
- training
- work activities
- voluntary activities.

In most cases RPL will not be appropriate for directly evidencing the requirements of the NEA assignments for the Cambridge Advanced National qualifications. However, if you feel that your student could use RPL to support their evidence, you must follow the guidance provided in our [RPL Policy](#).

## 5 Units

### 5.1 Guidance on unit content

This section describes what must be taught so that students can access all available marks and meet assessment criteria.

#### 5.1.1 Externally assessed units (F170 and F171)

The externally assessed units contain a number of topic areas.

For each topic area, we list the **teaching content** that must be taught and give information on the **breadth and depth** of teaching needed.

#### Teaching content

Questions can be asked about anything in the teaching content or breadth and depth columns.

#### Breadth and depth

The breadth and depth column:

- clarifies the breadth and depth of teaching needed
- indicates the range of knowledge and understanding that can be assessed in the exam
- confirms any aspects that you do not need to teach as 'does not include' statements.

Teaching must cover **both** the **teaching content** and **breadth and depth** columns.

#### Knowledge and understanding

This is what we mean by knowledge and understanding:

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Knowledge     | <ul style="list-style-type: none"> <li>• Be able to identify or recognise an item, for example on a diagram.</li> <li>• Use direct recall to answer a question, for example the definition of a term.</li> </ul>                                                                                                                                                                                                                                                    |
| Understanding | <ul style="list-style-type: none"> <li>• To assess and evidence the perceived meaning of something in greater depth than straight identification or recall.</li> <li>• Understanding will be expressed and presented using terms such as: how; why; when; reasons for; advantages and disadvantages of; benefits and limitations of; purpose of; suitability of; recommendations for improvement; appropriateness of something to/in different contexts.</li> </ul> |

Students will need to **understand** the content unless the breadth and depth column identifies it as knowledge only.

Any item(s) that should be taught as **knowledge** only will start with the word 'know' in the breadth and depth column.

All other content must be taught as understanding.

Opportunities to cover mathematical skills and how science works concepts and skills are exemplified in two columns in the unit content tables. Further information about the requirements for mathematical skills and how science works concepts and skills can be found in [Appendices C and D](#).

### 5.1.2 NEA units (F172-F177)

The NEA units contain a number of topic areas.

For each topic area, we list **teaching content** that must be taught and give **exemplification**. The exemplification shows the teaching expected to equip students to successfully complete their assignments.

### 5.1.3 Command words

[Appendix B](#) gives information about the command words that will be used in the external assessments and the NEA assessment criteria.

### 5.1.4 Performance objectives (POs):

Each Cambridge Advanced National qualification has four Performance Objectives.

|     |                                                                    |
|-----|--------------------------------------------------------------------|
| PO1 | Show knowledge and understanding                                   |
| PO2 | Apply knowledge and understanding                                  |
| PO3 | Analyse and evaluate knowledge, understanding and performance      |
| PO4 | Demonstrate and apply skills and processes relevant to the subject |

PO1 is assessed in the externally assessed unit only.

PO4 is assessed in the NEA units only.

The weightings of the Performance Objectives across the units in the **Certificate** qualification are:

| Performance Objective                   | Externally Assessed unit (range) | NEA units    | Overall weighting |
|-----------------------------------------|----------------------------------|--------------|-------------------|
| PO1                                     | 23.3%-30.0%                      | n/a          | 23.3%-30.0%       |
| PO2                                     | 10.0%-16.7%                      | 17.5%        | 27.5%-34.2%       |
| PO3                                     | 0.0%                             | 20.0%        | 20.0%             |
| PO4                                     | n/a                              | 22.5%        | 22.5%             |
| <b>Overall weighting of assessments</b> | <b>40.0%</b>                     | <b>60.0%</b> | <b>100.0%</b>     |

The weightings of the Performance Objectives across the units in the **Extended Certificate** qualification are:

| Performance Objective                   | Externally Assessed unit (range) | NEA units   | Overall weighting |
|-----------------------------------------|----------------------------------|-------------|-------------------|
| PO1                                     | 13.3%-20.0%                      | n/a         | 13.3%-20.0%       |
| PO2                                     | 15.0%-21.7%                      | 18.8%-19.4% | 33.8%-41.1%       |
| PO3                                     | 5.0%                             | 21.3%-21.9% | 26.3%-26.9%       |
| PO4                                     | n/a                              | 18.8%-20.0% | 18.8%-20.0%       |
| <b>Overall weighting of assessments</b> | <b>40%</b>                       | <b>60%</b>  | <b>100%</b>       |

## 5.2 Externally assessed units

### 5.2.1 Unit F170: Fundamentals of human biology

#### Unit aim

Medical science is constantly advancing at a fast rate. This unit provides some of knowledge and understanding relating to the biology behind these exciting medical advances. These may range from range from diagnosis to therapeutics. Studying human biology at Level 3 with other subjects such as A Levels in Psychology and PE, will give you a solid basis to progress onto degree courses in such areas as healthcare professions, sports science, social care and human physiology.

In this unit you will learn about the structure and function of cells, tissues and organ systems and appreciate the physiological links between such systems in the human body. You will be given the knowledge required to evaluate the impact, detection and treatment of non-functioning systems, with a focus on the endocrine, nervous and reproductive systems. An introduction to the basic features of microbiology will give you the opportunity to consider the key features of beneficial microbes (forming the human biome), pathogens (including viruses) and the immune response.

| Unit F170: Fundamentals of human biology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |                                            |
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| Topic Area 1: Human cells and tissues                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |                                            |
| Teaching content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Breadth and depth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Opportunities to cover: |                                            |
| 1.1 Key features of the cell and methods to observe them                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Maths                   | HSW                                        |
| <b>1.1.1 Generalised human cell and cell specialisation</b> <ul style="list-style-type: none"> <li>□ Definition of the cell</li> <li>□ The structure and function of eukaryotic cells and their components               <ul style="list-style-type: none"> <li>• Cell surface membrane</li> <li>• Cytoplasm</li> <li>• Nucleus</li> <li>• Nucleolus</li> </ul> </li> <li>□ Cell structures and functions:               <ul style="list-style-type: none"> <li>• Cytoplasm</li> <li>• Cell surface membrane</li> <li>• mitochondria</li> <li>• Ribosomes</li> <li>• Smooth and rough endoplasmic reticulum (SER/RER)</li> <li>• Golgi body/apparatus</li> <li>• Vesicles</li> <li>• Lysosomes</li> <li>• Cilium/flagellum</li> <li>• Microvilli</li> </ul> </li> <li>□ Adult stem cell location, function and cell specialisation</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ How these features are found in all specialised cells with the exception of the nucleus in the fully-formed erythrocyte</li> <li>□ The detailed structure and function of cells and all components, including in cell diagrams and photomicrographs</li> <li>□ How ribosomes are located in the cytoplasm and on the surface of the RER and located in the matrix of the mitochondrion</li> <li>□ How vesicles and lysosomes are both formed by the Golgi body/apparatus</li> <li>□ How detailed cell features are seen in electrophotomicrographs using a transmission electron microscope (TEM)</li> <li>□ Why and where stem cells are located in different regions of the adult body</li> </ul> |                         | <b>HSW1</b><br><b>HSW8</b><br><b>HSW11</b> |



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| <ul style="list-style-type: none"> <li>□ Stem cells can remain inactive for many years</li> <br/> <li>□ Structure and function of highly-specialised cells: <ul style="list-style-type: none"> <li>• Sperm cell</li> <li>• Egg cell/ovum</li> <li>• Red blood cell or erythrocyte</li> <li>• White blood cells (neutrophil, lymphocyte, eosinophil and monocyte)</li> <li>• Sensory, relay and motor neurons</li> <li>• Hepatocyte (liver cell)</li> <li>• Renal tubule epithelial cells</li> <li>• Rods and cones in the retina</li> <li>• Ciliated epithelial cells lining the trachea and oviduct</li> <li>• Squamous epithelial cells of alveoli</li> <li>• Skeletal/striated, smooth and cardiac muscle cells</li> <li>• Epithelial cells of gastric pits</li> </ul> </li> <br/> <li>□ Eukaryotic (human) and prokaryotic (bacterial) cells</li> </ul> | <ul style="list-style-type: none"> <li>□ How dormant stem cells are triggered to differentiate by the microenvironment</li> <li>□ How human pluripotent stem cells (PSCs) can be maintained and expanded <i>in vitro</i> for long time periods and then induced to differentiate</li> <li>□ How and why the functions of embryonic and adult stem cells differ</li> <br/> <li>□ How the abundance and features of key organelles differ in relation to the function of highly-specialised cells</li> <br/> <li>□ How eukaryotic (human) and prokaryotic (bacterial) cells compare</li> <li>□ Why the mitochondrion may be considered as a prokaryote existing inside a eukaryotic cell (endosymbiotic theory)</li> <li>□ How ribosomes in eukaryotic and prokaryotic cells differ</li> <br/> <p>Does not include:</p> <li>□ Detailed features of other highly-specialised cells</li> </ul> |  |  |
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| <p><b>1.1.2 Observing cells and organelles</b></p> <ul style="list-style-type: none"> <li>□ Light/optical (LM) microscope</li> <li>□ Preparation of temporary slides</li> <li>□ Use of the stage microtome</li> <li>□ Transmission electron microscope (TEM) and scanning electron microscope (SEM)</li> <li>□ Calculating the magnification and dimensions of cell components</li> <li>□ Units of nm, µm or mm</li> <li>□ Use of differential centrifugation for organelle extraction</li> <li>□ Use of the haemocytometer</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Know the features of the LM microscope</li> <li>□ The advantages and disadvantages of using an LM to study cells</li> <li>□ The steps for preparing a temporary slide for LM observation and the reasons for these steps</li> <li>□ How the features and use of the TEM and SEM can be compared</li> <li>□ The reasons for a TEM or SEM to produce a photomicrograph of a cell or organelle</li> <li>□ How to use the equation:<br/>magnification = <math>\frac{\text{image size}}{\text{actual size}}</math></li> <li>□ How to measure the actual size of an image</li> <li>□ Why different units (nm, µm or mm) for cell/organelle dimensions are used</li> <li>□ Why different organelles or cell fragments are found in the supernatant and pellet</li> <li>□ The advantages and disadvantages of using a haemocytometer or coulter counter</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ The physics of the LM and EM</li> </ul> | <p><b>M1.4</b><br/><b>M1.5</b><br/><b>M2.2</b><br/><b>M4.2</b></p> | <p><b>HSW2</b><br/><b>HSW4</b><br/><b>HSW5</b><br/><b>HSW6</b></p> |
| <p><b>1.1.3 Link between organelle structure and function including:</b></p> <ul style="list-style-type: none"> <li>□ Nucleus</li> <li>□ Nucleolus</li> <li>□ Mitochondrion</li> <li>□ 70S and 80S ribosomes</li> <li>□ SER and RER</li> <li>□ Golgi body/apparatus</li> <li>□ Lysosome</li> </ul>                                                                                                                                                                                                                                     | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How the function of the nucleus and mitochondrion are linked</li> <li>□ Why the functions of the nucleus, ribosome, RER, Golgi body and vesicle/lysosome are linked to complete the process of protein synthesis</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Details of transcription and translation</li> <li>□ Chemistry of cellular respiration</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                    | <p><b>HSW5</b></p>                                                 |

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| <p><b>1.1.4 Structure and function of the cell surface membrane</b></p> <ul style="list-style-type: none"> <li>□ Fluid mosaic model</li> <li>□ Function of each component of the cell surface membrane</li> <li>□ Processes of endocytosis, exocytosis, simple and facilitated diffusion, active transport and osmosis</li> <li>□ Cell-to-cell recognition</li> <li>□ The role of extrinsic proteins as receptor sites</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How the phospholipid bilayer, extrinsic and intrinsic proteins, cholesterol and glycoproteins are arranged in a specific way in the fluid mosaic model</li> <li>□ The advantages and disadvantages of cholesterol in the 'free' cell membranes of endothelial cells of blood vessels</li> <li>□ Why cell-to-cell recognition is the basis of transplant tissue/organ rejection</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Details of charged, gated protein channels</li> <li>□ Calculations of water potential values in osmosis</li> </ul>                                                                                                                                                                 | <p><b>M3.4</b><br/><b>M4.1</b></p>                 | <p><b>HSW9</b></p>                                   |
| <p><b>1.1.5 Mitosis and meiosis</b></p> <ul style="list-style-type: none"> <li>□ Structure of the chromosome, chromatid and centromere</li> <li>□ Molecular structure of DNA and genes</li> <li>□ The cell cycle</li> <li>□ Stages in mitosis, including cytokinesis</li> </ul>                                                                                                                                                   | <p>To include:</p> <ul style="list-style-type: none"> <li>□ The appearance of chromosomes, chromatids and centromeres when viewed by an LM and EM</li> <li>□ Know how bases are paired within the DNA molecule</li> <li>□ How base-pairing is the basis of genetics and inheritance</li> <li>□ The benefits of the genome project</li> <li>□ Why interphase is an active process</li> <li>□ Highly-specialised cells can lose the ability to complete the cell cycle. This is seen in mature red blood cells (erythrocytes versus erythroblasts) when they lose their nucleus.</li> <li>□ The stages of mitosis, including prophase, metaphase, anaphase and telophase</li> <li>□ Know the significance of cell cleavage/cytokinesis</li> <li>□ How nuclear division differs from cell division</li> </ul> | <p><b>M0.1</b><br/><b>M1.3</b><br/><b>M4.1</b></p> | <p><b>HSW9</b><br/><b>HSW10</b><br/><b>HSW11</b></p> |

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| <ul style="list-style-type: none"> <li>□ Stages in meiosis</li> <li>□ Mitosis compared to meiosis</li> <li>□ Basis of inheritance, including monohybrid and dihybrid crosses in the human</li> <li>□ Features of mitochondrial inheritance</li> </ul> | <ul style="list-style-type: none"> <li>□ The stages of meiosis, including prophase I, metaphase I, anaphase I, telophase I and prophase II, metaphase II, anaphase II and telophase II.</li> <li>□ How mitosis differs from meiosis</li> <li>□ Know why crossing-over and random, independent assortment lead to genetic variation</li> <li>□ How to use and interpret the Punnett square</li> <li>□ The advantages and disadvantages of mitochondrial inheritance (via mitochondrial DNA or mtDNA) in the egg cell</li> <li>□ How a baby can have three 'biological parents' due to mitochondrial replacement therapy</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Chromosome and gene mutations</li> </ul> |                            |            |
| <b>1.2 Tissue structure and function</b>                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Maths</b>               | <b>HSW</b> |
| <b>1.2.1 Definition of a tissue</b>                                                                                                                                                                                                                   | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How tissue and organ levels of organisation can be distinguished</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Plant/algal tissues</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |            |
| <b>1.2.2 The link between tissue structure and function</b> <ul style="list-style-type: none"> <li>□ Epithelial</li> <li>□ Muscle</li> <li>□ Bone, cartilage and connective</li> <li>□ Nervous</li> </ul>                                             | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Know the advantages of the basement membrane to epithelial tissue integrity and replacement</li> <li>□ Why the structure of squamous, ciliated and cuboidal epithelial tissues differs in relation to structure</li> <li>□ Muscle tissues can be either skeletal, smooth or cardiac</li> <li>□ Why skeletal, smooth and cardiac muscle tissues have different structures</li> <li>□ Why bone and cartilage tissue can be viewed as special types of connective tissue</li> <li>□ Why the three types of neuron (sensory, relay and motor) differ from each other in relation to their functions</li> </ul>                                                                       | <b>M0.1</b><br><b>M0.2</b> |            |

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| <input type="checkbox"/> Blood                                                                                                                                                                                                                                                                 | <input type="checkbox"/> Know that blood is a special form of tissue<br><input type="checkbox"/> How blood is composed of plasma, white blood cells (WBCs), red blood cells (RBCs) and platelets carried in the watery plasma<br><input type="checkbox"/> Plasma also carries a wide range of molecules and ions<br><br>Does not include:<br><input type="checkbox"/> Sliding filament theory                                                                                                                                                                                                                                                                                                   |                                           |                             |
| <b>1.2.3 Use of tissues in research and development</b><br><input type="checkbox"/> Creating and maintaining <i>in vitro</i> human tissue cultures in a laboratory<br><br><input type="checkbox"/> Applications of stem cell cultures<br><br><input type="checkbox"/> Organoid use in research | To include:<br><input type="checkbox"/> Benefits and limitations of using tissues or organoids for research, rather than using the animal model<br><br><input type="checkbox"/> How tissue cultures are established and maintained in the laboratory<br><input type="checkbox"/> The suitability of tissue culture research to the clinical study of humans<br><br><input type="checkbox"/> Know the characteristic features of an organoid<br><input type="checkbox"/> Benefits and limitations of organoids in research and development<br><br>Does not include:<br><input type="checkbox"/> Details of novel applications not yet approved by the Medicines and Healthcare Regulatory Agency |                                           | <b>HSW4</b><br><b>HSW10</b> |
| <b>Topic Area 2: Human physiology, organs and systems</b>                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                             |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                        | <b>Breadth and depth</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Opportunities to cover:</b>            |                             |
| <b>2.1 Human physiology</b>                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                              | <b>HSW</b>                  |
| <b>2.1.1 The concept of human physiology</b>                                                                                                                                                                                                                                                   | To include:<br><input type="checkbox"/> How human physiology is the applied study of organ system function<br><input type="checkbox"/> Know the role of a physiologist in health and social care, general wellbeing clinics and sports settings                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           | <b>HSW10</b>                |
| <b>2.1.2 The organ</b><br><input type="checkbox"/> Difference between an organ and a system                                                                                                                                                                                                    | To include:<br><input type="checkbox"/> Know that an organ is a group of different tissues sharing a common function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>M0.3</b><br><b>M3.1</b><br><b>M3.5</b> | <b>HSW1</b>                 |

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| <p>□ Structure and functions of the organs in the human body including:</p> <ul style="list-style-type: none"> <li>• Heart</li> <li>• Blood vessels</li> <li>• Muscle</li> <li>• Bone</li> <li>• Liver</li> <li>• Lungs</li> <li>• Stomach</li> <li>• Intestines</li> <li>• Kidney</li> <li>• Pancreas</li> </ul> | <p>□ How the anatomy and histology of the organs relate to their function</p> <p>□ Why all organs have their own blood routes via an artery and vein</p> <p>□ Know that the heart consists of the endocardium, myocardium and pericardium layers, four chambers (right atrium and ventricle and left atrium and ventricle), atrioventricular, pulmonary and cardiac valves and a central septum</p> <p>□ How the cardiac cycle is regulated and maintained</p> <p>□ Know that muscle as an organ consists of muscle tissue, connective tissue, epithelial tissue and is connected to bones by tendons</p> <p>□ Know that bone is both an organ and a tissue, containing calcified matrix, fibrocytes, collagen/fibres, and different stages of osteocyte development</p> <p>□ How damaged bone has the ability to regrow, involving the migration and activity of fibrocytes and osteocytes and a supply of calcium ions and energy (via glucose molecules)</p> <p>□ How the liver is formed from hepatocytes surrounding blood sinuses and canaliculi</p> <p>□ Why the liver has a double blood supply (hepatic artery and hepatic portal vein)</p> <p>□ Know that the lungs present a large surface area for gaseous exchange via many alveoli, form right and left lobes and are connected to the external environment along the trachea, bronchi and bronchioles</p> <p>□ How gas exchange occurs at the alveoli</p> <p>□ How the stomach can be sealed using the cardiac and pyloric sphincters, is the site of digestion and absorption and how the gastric wall contains gastric pits for the secretion of hydrochloric acid, enzymes and mucus</p> <p>□ That the small intestine consists of the duodenum and ileum</p> <p>□ How the small intestine carries out digestion and absorption</p> |  |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li><input type="checkbox"/> Know that the large intestine consists of the caecum, appendix, colon, rectum and anus</li> <li><input type="checkbox"/> How the large intestine is involved in digestion, including water reabsorption and faeces formation</li> <li><input type="checkbox"/> How the biome within the large intestine is responsible for different functions</li> <li><input type="checkbox"/> Know that each kidney is formed from an outer capsule, cortex, medulla, renal pyramids, calyx, ureter and is the site of ultrafiltration, reabsorption and urine formation</li> <li><input type="checkbox"/> Why the pancreas has both an exocrine and endocrine function</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Brain</li> <li><input type="checkbox"/> Nerve</li> <li><input type="checkbox"/> Gonads</li> </ul> |  |                      |
| <b>2.1.3 Biological basis of disease/failure of organs</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Causes of disease and failure in organs:             <ul style="list-style-type: none"> <li>• Heart defects                 <ul style="list-style-type: none"> <li>○ Ventral septal defect (VSD)</li> <li>○ Atrial septal defect (ASD)</li> <li>○ Valve malfunction</li> </ul> </li> <li>• Atherosclerosis</li> <li>• Aortic/pulmonary aneurism</li> <li>• Muscle deterioration</li> <li>• Osteoporosis</li> <li>• Liver cirrhosis</li> <li>• Asthma, emphysema, chronic obstructive pulmonary disease (COPD) and lung cancer</li> <li>• Stomach ulcers and cancer</li> <li>• Cancer of the colon and inflammatory bowel disease (IBD)</li> <li>• Kidney failure</li> <li>• Islets of Langerhans/diabetes and pancreatic cancer</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How the symptoms of disease and organ failure are linked to changes in the structure and function of cells/tissues</li> <li><input type="checkbox"/> How the appearance of healthy and diseased heart and lung tissues differs</li> <li><input type="checkbox"/> How osteoporosis can be monitored via DEXA (dual energy X-ray absorptiometry)</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Brain disease, malfunctioning reproductive systems</li> </ul>                                                                                                                                                                                                                                                                                                                              |  | <b>HSW3<br/>HSW5</b> |
| <b>2.1.4 Transplanted and artificial organs</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Transplants/corrective surgery:             <ul style="list-style-type: none"> <li>• Heart</li> <li>• Liver</li> <li>• Lungs</li> <li>• Stomach</li> <li>• Intestines</li> <li>• Kidney</li> <li>• Bone</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Why transplanted organs are rejected</li> <li><input type="checkbox"/> The advantages and disadvantages of artificial organs</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Mechanical details of a dialysis machine</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  | <b>HSW11</b>         |

| <b>2.2 Systems in the human body</b>                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Maths</b> | <b>HSW</b> |
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| <b>2.2.1 The system</b><br><input type="checkbox"/> The definition of a system                                                                                                                                                                                                                                                                              | <p>To include:</p> <input type="checkbox"/> Know that a system is a group of different organs sharing a common function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |            |
|                                                                                                                                                                                                                                                                                                                                                             | <p>Does not include:</p> <input type="checkbox"/> Plant/algal systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |            |
| <b>2.2.2 Structure and function of different systems</b><br><input type="checkbox"/> Blood circulatory<br><input type="checkbox"/> Lymphatic<br><input type="checkbox"/> Musculoskeletal<br><input type="checkbox"/> Homeostatic<br><input type="checkbox"/> Gastrointestinal<br><input type="checkbox"/> Excretory<br><input type="checkbox"/> Respiratory | <p>To include:</p> <input type="checkbox"/> How the blood circulatory and nervous systems support the functioning of the other systems<br><input type="checkbox"/> That the blood circulatory system is responsible for the transport of blood, circulation of oxygen/carbon dioxide, antibodies, red and white blood cells, molecules including glucose and hormones and for thermoregulation<br><input type="checkbox"/> How the structure and function of the lymphatic system differs from that of the blood circulatory system<br><input type="checkbox"/> Know that the musculoskeletal system supports movement and balance and that the bones also act as a calcium storage site and produce blood cells<br><input type="checkbox"/> How the homeostatic system is responsible for the processes of thermoregulation, plasma glucose regulation and osmoregulation<br><input type="checkbox"/> Know that the gastrointestinal system consists of the buccal cavity, oesophagus, stomach and small and large intestines<br><input type="checkbox"/> How the excretory system includes the sweat glands in the skin but also the kidneys for the excretion of urea<br><input type="checkbox"/> Know that the respiratory system consists of the trachea, bronchi, bronchioles, lungs, rib cage and intercostal/diaphragm muscles and carries out inspiration and expiration | <b>M3.6</b>  |            |
|                                                                                                                                                                                                                                                                                                                                                             | <p>Does not include:</p> <input type="checkbox"/> Nervous and reproductive systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |            |



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| <b>2.2.3 Measuring the activity of systems, including:</b> <ul style="list-style-type: none"> <li>□ Sphygmomanometer</li> <li>□ Radial pulse readings</li> <li>□ Electrocardiogram (ECG) readings</li> <li>□ Ultrasound scans</li> <li>□ Colonoscopy</li> <li>□ Urinalysis</li> <li>□ Blood glucose levels</li> <li>□ Thermometer</li> <br/> <li>□ Spirometry</li> <br/> <li>□ Peak flow readings</li> <li>□ Fractional exhaled nitric oxide (FeNO) test</li> </ul>    | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to use each measurement tool</li> <li>□ How each type of measurement tool contributes to the diagnosis of a condition or disease</li> <li>□ The benefits and limitations of using each form of measurement tool</li> <li>□ How to interpret blood glucose levels via the glucose tolerance test</li> <br/> <li>□ How to calculate the pulmonary ventilation rate using<br/> <math>PVR = \text{breathing rate (breaths min}^{-1}) \times \text{tidal volume (cm}^3)</math></li> <li>□ The reasons for a change in the pulmonary ventilation rate when undergoing exercise or in response to a heart defect or disease</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ The physics or mechanics of the tools used</li> </ul> | <b>M0.4</b><br><b>M1.6</b><br><b>M2.3</b><br><b>M2.4</b><br><b>M3.2</b> | <b>HSW4</b><br><b>HSW10</b> |
| <b>Topic Area 3: Key concepts in endocrinology, neurobiology and reproduction</b>                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                         |                             |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Breadth and depth</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Opportunities to cover:</b>                                          |                             |
| <b>3.1 Key concepts of endocrinology</b>                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Maths</b>                                                            | <b>HSW</b>                  |
| <b>3.1.1 The endocrine system and homeostasis</b> <ul style="list-style-type: none"> <li>□ Key features of the endocrine system and hormones: <ul style="list-style-type: none"> <li>• Adrenaline</li> <li>• Thyroxine</li> <li>• Somatostatin</li> <li>• Erythropoietin</li> <li>• Calcitonin</li> <li>• Insulin</li> <li>• ADH (anti-diuretic hormone)</li> </ul> </li> <li>□ Definition and significance of homeostasis</li> <li>□ The homeostasis model</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Why the endocrine system is generally slower to respond to stimuli but the response is longer lasting than the nervous system</li> <li>□ The endocrine glands/tissues responsible for producing the hormones listed and the action of each hormone</li> <li>□ How synthetic hormones can be used as a form of therapy</li> <br/> <li>□ Know that homeostasis is the maintenance of a constant internal body environment</li> <li>□ The steps of the homeostasis model, including receptors, monitoring centre, effectors and negative feedback</li> </ul>                                                                                                                                                                                                   | <b>M0.4</b>                                                             | <b>HSW11</b>                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |              |
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| <input type="checkbox"/> Principles of <ul style="list-style-type: none"> <li>• Thermoregulation</li> <li>• Osmoregulation</li> <li>• Glucose regulation</li> </ul>                                                                                                                                                                                                                                                                                                                                     | <input type="checkbox"/> The principles of hormonal and/or nervous control in relation to thermoregulation, osmoregulation and glucose regulation (avoiding hypoglycaemia and hyperglycaemia) <p>Does not include:</p> <input type="checkbox"/> Sex hormones and neurotransmitters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |              |
| <b>3.1.2 Monitoring homeostasis</b> <input type="checkbox"/> Symptoms of malfunctioning endocrine systems: <ul style="list-style-type: none"> <li>• Thermoregulation</li> <li>• Osmoregulation</li> <li>• Glucose regulation</li> </ul><br><input type="checkbox"/> Physiological tests used to monitor homeostatic systems: <ul style="list-style-type: none"> <li>• Core body temperature testing</li> <li>• Blood osmotic potential and pressure testing</li> <li>• Blood-glucose testing</li> </ul> | <p>To include:</p> <input type="checkbox"/> The reason for the differences in symptoms of hypothermia and hyperthermia <input type="checkbox"/> How malfunctioning osmoregulation can be offset by adequate body hydration (drinking an appropriate supply of water on a daily basis) <input type="checkbox"/> The causes and symptoms of type 1 and type 2 diabetes <input type="checkbox"/> Know the characteristic features of hypoglycaemia and hyperglycaemia <input type="checkbox"/> The impact of changes in lifestyle to reduce long term effects of type 1 and type 2 diabetes<br><input type="checkbox"/> The advantages and disadvantages of each physiological test <input type="checkbox"/> Why a fasting period is needed for the glucose tolerance test <input type="checkbox"/> The advantages and disadvantages of non-invasive blood glucose testing technology to monitor and regulate diabetes <p>Does not include:</p> <input type="checkbox"/> Sex hormones | <b>M1.5</b><br><b>M3.1</b><br><b>M3.2</b> | <b>HSW10</b> |
| <b>3.2 Key concepts of neurobiology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Maths</b>                              | <b>HSW</b>   |
| <b>3.2.1 The structure and function of the nervous system</b> <input type="checkbox"/> Central nervous system (CNS) versus autonomic nervous system (ANS) <input type="checkbox"/> The structure and function of neurons, including the myelin sheath and nodes of Ranvier <input type="checkbox"/> Key features of nerve impulse transmission                                                                                                                                                          | <p>To include:</p> <input type="checkbox"/> The functional links between the CNS and ANS <input type="checkbox"/> How receptors, sensory, relay and motor neurons and effectors function in the spinal reflex arc <input type="checkbox"/> Know the stages of resting and action potentials and the significance of polarisation, depolarisation and hyperpolarisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>M3.6</b>                               | <b>HSW7</b>  |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>□ The causes and symptoms of multiple sclerosis and the impact of the disease on impulse transmission via a changed saltatory response</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Details of ionic exchange during nerve impulse transmission</li> <li>□ Nervous control of metabolism</li> </ul>                                                                                                                                                                                                                                                                                                                  |              |             |
| <b>3.2.2 Basic features of the brain and spinal cord</b> <ul style="list-style-type: none"> <li>□ Structure and function of the brain</li> <li>□ Structure and function of the spinal cord</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to interpret vertical section (VS) and transverse section (TS) images of the brain and spinal cord</li> <li>□ Know that the brain consists of defined regions including the cerebral hemispheres/cerebrum cerebellum, hypothalamus, pituitary gland and medulla</li> <li>□ Know the location and importance of the meninges and ventricles in the brain</li> <li>□ The reasons for taking samples of cerebrospinal fluid</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Details of different parts of the brain and spinal cord</li> <li>□ Detailed histology of structures listed</li> </ul> | <b>M1.4</b>  | <b>HSW1</b> |
| <b>3.3 Key concepts of reproduction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Maths</b> | <b>HSW</b>  |
| <b>3.3.1 Structure and function of the reproductive system</b> <ul style="list-style-type: none"> <li>□ Common features of the female system including:               <ul style="list-style-type: none"> <li>• Cervix</li> <li>• Ovaries</li> <li>• Oviducts</li> <li>• Uterus</li> <li>• Vagina</li> <li>• Vulva</li> </ul> </li> <li>□ Common features of the male system including:               <ul style="list-style-type: none"> <li>• Testes</li> <li>• Epididymis</li> <li>• Vas deferens</li> <li>• Prostate gland</li> <li>• Seminal vesicle</li> <li>• Urethra</li> <li>• Penis</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How the different structures work together within the systems</li> <li>□ Why individuals with variations in sex traits (intersex) may have reproductive systems outside of the common male/female systems</li> <li>□ How to interpret photomicrographs of structures in the reproductive systems</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Details of the menstrual cycle</li> </ul>                                                                                                                                                                                                         |              | <b>HSW2</b> |

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| <b>3.3.2 Hormonal control of gametogenesis</b><br><input type="checkbox"/> Role of hormones in reproductive systems <ul style="list-style-type: none"> <li>• Follicle-stimulating hormone (FSH)</li> <li>• Progesterone</li> <li>• Oestrogen</li> <li>• Luteinising hormone (LH)</li> <li>• Testosterone</li> </ul>                                                                                                            | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Why ovulation has evolved to become periodic but sperm production is continuous</li> <li><input type="checkbox"/> The roles of the hormones in relation to the development of secondary sexual characteristics, gametogenesis, fertilisation, pregnancy and birth</li> <li><input type="checkbox"/> How variations in sex traits (intersex) can be caused by differences in hormone levels and the impact on secondary sexual characteristics and gametogenesis</li> </ul> Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Detailed structure of the hypothalamus and pituitary gland</li> </ul> | <b>M1.6</b>                    |             |
| <b>3.3.3 Reproductive changes during ageing</b><br><input type="checkbox"/> Onset of menopause<br><input type="checkbox"/> Use of hormones and surgery to delay or reduce the impact of menopause<br><input type="checkbox"/> Causes and symptoms of structural and functional changes in the male reproductive system                                                                                                         | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> The advantages and disadvantages of pregnancy in later life</li> <li><input type="checkbox"/> The cause and symptoms of menopause, including the effect of different therapies</li> <li><input type="checkbox"/> Why hypertrophy of the prostate gland affects urination and sperm discharge</li> </ul> Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Detailed histological changes in the reproductive systems during ageing</li> </ul>                                                                                                                                                       | <b>M4.2</b>                    | <b>HSW9</b> |
| <b>Topic Area 4: Basics of microbiology</b>                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |             |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Breadth and depth</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Opportunities to cover:</b> |             |
| <b>4.1 Key features of microbes</b>                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>                   | <b>HSW</b>  |
| <b>4.1.1 Features of bacteria found in humans</b><br><input type="checkbox"/> Structure and function of components of bacterial cells: <ul style="list-style-type: none"> <li>• Capsule/slime layer</li> <li>• Peptidoglycan cell wall</li> <li>• Cell surface membrane</li> <li>• Cytoplasm</li> <li>• 70S ribosomes</li> <li>• DNA loop</li> <li>• Plasmids</li> <li>• Mesosomes</li> <li>• Rotary-like flagellum</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Know the functions of the different structures listed for bacterial cells</li> <li><input type="checkbox"/> How gram positive and gram-negative bacteria differ</li> <li><input type="checkbox"/> Classification of bacterial cells as coccus, bacillus and spiral</li> </ul> Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Detailed structure of the cell wall</li> </ul>                                                                                                                                                                                                                     |                                | <b>HSW5</b> |

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| <p><b>4.1.2 Features of fungi found in humans</b></p> <ul style="list-style-type: none"> <li>□ Structure and function of fungal components             <ul style="list-style-type: none"> <li>• Cytoplasm</li> <li>• Chitin cell wall</li> <li>• Septum</li> <li>• Hypha</li> <li>• Mycelium</li> <li>• Spores</li> </ul> </li> <li>□ Fungi are parasitic or saprophytic             <ul style="list-style-type: none"> <li>• Endoparasitic</li> <li>• Ectoparasitic</li> <li>• Saprophytic fungi</li> </ul> </li> <li>□ Characteristics of common fungal diseases in humans             <ul style="list-style-type: none"> <li>• Aspergillosis (<i>Aspergillus sp.</i>)</li> <li>• Vaginal candidiasis (<i>Candida sp.</i>)</li> <li>• Athlete's foot (<i>Tenia sp.</i>)</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Recognise the key structures of fungi in photomicrographs and drawings</li> <li>□ Link the structure of each component to its function, to include:             <ul style="list-style-type: none"> <li>• the cytoplasm for cell shape and site of reactions</li> <li>• Chitin cell wall for cell shape and protection</li> <li>• Hypha for extracellular digestion and colonisation of substrate</li> <li>• Septum (containing perforations) to enable movement of molecules and ions from 'cell' to 'cell' and isolation of diseased or non-functioning 'cells'</li> <li>• Mycelium as the collection of branching hyphae and spores for reproduction and dispersal</li> </ul> </li> <li>□ Know the role and impact of extracellular, hydrolytic enzymes secreted by fungal hyphae</li> <li>□ The characteristics of endoparasitic, ectoparasitic and saprophytic fungi living on or inside the human body</li> <li>□ The differences between parasitic and saprophytic fungi in relation to their lifestyles and impact on the human body</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Detailed process of sporulation and sexual reproduction</li> </ul> | <p><b>M0.1</b><br/><b>M1.1</b></p> |                    |
| <p><b>4.1.3 Location of bacteria in the human body and external environment</b></p> <ul style="list-style-type: none"> <li>□ Locations of bacteria in the human body:             <ul style="list-style-type: none"> <li>• Skin surface</li> <li>• Conjunctiva</li> <li>• Mucous membranes</li> <li>• Teeth</li> <li>• Gastrointestinal tract (colon)</li> <li>• Reproductive tract</li> <li>• Renal tract</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to collect samples, using the aseptic technique</li> <li>□ Advantages and disadvantages of taking bacterial samples from the external environment</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Collection of clinical samples from diseased tissue</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p><b>M1.7</b></p>                 | <p><b>HSW4</b></p> |

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| <input type="checkbox"/> Locations of bacteria in the external environment: <ul style="list-style-type: none"> <li>• Air</li> <li>• Water</li> <li>• Soil</li> <li>• Surface of plants</li> <li>• Surface of other animals</li> </ul>                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |             |
| <b>4.1.4 Reproduction and culture of bacteria</b> <input type="checkbox"/> Binary fission (asexual)<br><input type="checkbox"/> Sex pili<br><br><input type="checkbox"/> Use of agar plates and nutrient broths to culture bacteria<br><br><input type="checkbox"/> The aseptic technique | <p>To include:</p> <input type="checkbox"/> Know that binary fission is a form of asexual reproduction and that the products are identical unless a mutation occurs during the process<br><input type="checkbox"/> How to interpret data via graphs showing growth of bacterial population<br><input type="checkbox"/> How to calculate bacterial population growth using,<br>Estimate of bacterial population<br>$= 1 \times 2^{\text{number of divisions}}$<br><input type="checkbox"/> The features of lag, exponential, stationary and death stages of bacterial populations, in the context of environmental factors and natality/mortality of bacterial cells<br><input type="checkbox"/> Know the factors promoting reproduction and death of bacterial cells within culture vessels<br><br><input type="checkbox"/> How agar is suitable as a growth medium for bacteria in the laboratory<br><input type="checkbox"/> Know the key features of bacterial cultures when grown in agar dishes or nutrient broth<br><br><input type="checkbox"/> The steps of the aseptic technique when obtaining bacterial samples, creating bacterial suspensions in nutrient broth/agar and streaking the surface of an agar plate<br><input type="checkbox"/> The benefits and limitations of the aseptic technique in the context of personal safety and contamination of cultures<br><input type="checkbox"/> How to create a health and safety record for carrying out the aseptic technique<br><br><p>Does not include:</p> <input type="checkbox"/> Identification of bacteria via colony colour and morphology | <b>M0.5</b><br><b>M2.5</b><br><b>M3.6</b> | <b>HSW4</b> |

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## Assessment guidance

This unit is assessed by an exam. The exam is 1 hour and 15 minutes and has **60** marks in total. All the questions in the exam are compulsory.

A range of question types will be used in this assessment including:

- Forced choice/controlled response questions – including MCQ
- Short answer, closed response questions (with or without diagrams)
- Short answer with calculation/working
- Extended constructed response with points-based mark scheme

Content will be sampled from all topic areas, with at least one question or part question relating to each topic area.

Content in this exam will have links to the 'How Science Works Concepts and Skills' and 'Mathematical skills for Human Biology'.

This will be conducted under examination conditions. For more details refer to the [Administration area](#).

A range of question types will be used in the exam.

The [guide to our Sample Assessment Material for this unit](#) gives more information about the layout and expectations of the exam.

The exam for this unit assesses the following Performance Objectives:

- PO1 – Show knowledge and understanding
- PO2 – Apply knowledge and understanding

### **Synoptic assessment**

This unit allows students to gain underpinning knowledge and understanding relevant to the qualification and sector. The NEA units draw on and strengthen this learning with students applying their learning in an applied or practical way.

The following NEA units have synoptic links with this unit. The synoptic grids at the end of these NEA units show these synoptic links.

- Unit F172: Genetics
- Unit F173: Biomedical techniques
- Unit F174: Nutrition and metabolism
- Unit F175: Human reproduction
- Unit F176: The brain

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).



## 5.2.2 Unit F171: Health and disease

### Unit aim

The nature of diseases and disorders is always a challenging and intriguing topic. The therapies involved in treating diseases and disorders is ever evolving, aided by ongoing medical research. This unit considers these things, along with the role played by immunology, diagnosis and monitoring in today's healthcare system.

In this unit you will learn about physiological disorders and communicable diseases that can impact on the health of individuals in terms of their causes and effects. You will be given the opportunity to understand the skills needed to review, measure and research this aspect of human biology. You will review the present and future role of immunology in fighting disease. By studying diagnostic and monitoring techniques you will gain an understanding of how disease change can be measured. Finally, you will examine how research is reported with consideration given to patient confidentiality.

| Unit F171: Health and disease                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |              |
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| Topic Area 1: Causes and effects of diseases and disorders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |              |
| Teaching content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Breadth and depth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Opportunities to cover: |              |
| <b>1.1 Definitions of health and disease</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>            | <b>HSW</b>   |
| <b>Definitions of health, wellbeing and disease</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Physical health</li> <li><input type="checkbox"/> Mental health</li> <li><input type="checkbox"/> Social health</li> <li><input type="checkbox"/> Disease</li> <li><input type="checkbox"/> Medical disorder</li> <li><input type="checkbox"/> Medical sign</li> <li><input type="checkbox"/> Medical symptom</li> <li><input type="checkbox"/> Medical syndrome</li> <li><input type="checkbox"/> Medical condition</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Know the World Health Organization definition of health</li> <li><input type="checkbox"/> Know definitions of the list in 1.1</li> <li><input type="checkbox"/> How physical, mental and social health are a measurement of overall health</li> <li><input type="checkbox"/> How the terms disease and disorder are used interchangeably</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         | <b>HSW12</b> |
| <b>1.2 The nature of physiological disorders/diseases</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>            | <b>HSW</b>   |
| <b>1.2.1 Physiological disorders/diseases and their effects</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Disorders of the nervous system               <ul style="list-style-type: none"> <li>• Motor neurone disease (MND)</li> <li>• Parkinson's</li> </ul> </li> <li><input type="checkbox"/> Disorders of the circulatory system               <ul style="list-style-type: none"> <li>• Abdominal aortic aneurysm</li> <li>• Hypertension</li> </ul> </li> <li><input type="checkbox"/> Disorders of the respiratory system               <ul style="list-style-type: none"> <li>• Asthma</li> <li>• Chronic Obstructive Pulmonary Disease (COPD)</li> </ul> </li> <li><input type="checkbox"/> Disorders of the digestive system               <ul style="list-style-type: none"> <li>• Crohn's disease</li> <li>• Hiatus hernia</li> </ul> </li> <li><input type="checkbox"/> Disorders of the urinary system               <ul style="list-style-type: none"> <li>• Nephritis</li> <li>• Polycystic Kidney Disease (PKD)</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> The main changes to the relevant physiology of the body systems caused by each disorder/disease</li> <li><input type="checkbox"/> The main changes to overall body functions caused by each disorder/disease</li> <li><input type="checkbox"/> Know the main observable signs of each disorder/disease</li> <li><input type="checkbox"/> Know the main symptoms felt and experienced by individuals with each disease</li> <li><input type="checkbox"/> How the disorder/disease impacts on the individual, family and society in general</li> </ul><br>Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Changes at the cellular level</li> <li><input type="checkbox"/> Diseases/disorders other than those specified in the teaching content</li> </ul> | <b>M1.2</b>             | <b>HSW10</b> |

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| <ul style="list-style-type: none"> <li><input type="checkbox"/> Disorders of the musculoskeletal system <ul style="list-style-type: none"> <li>• Multiple sclerosis</li> <li>• Rheumatoid arthritis</li> </ul> </li> <li><input type="checkbox"/> Cancer in various organ systems <ul style="list-style-type: none"> <li>• Hodgkin's lymphoma</li> <li>• Melanoma</li> </ul> </li> <li><input type="checkbox"/> Deficiency diseases <ul style="list-style-type: none"> <li>• Iron deficiency anaemia</li> <li>• Vitamin D deficiency and rickets</li> </ul> </li> <li><input type="checkbox"/> Genetic disorders <ul style="list-style-type: none"> <li>• Cystic fibrosis</li> <li>• Sickle cell anaemia</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                    |
| <p><b>1.2.2 Causes of physiological disorders/diseases</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Autoimmunity</li> <li><input type="checkbox"/> Diet and exercise</li> <li><input type="checkbox"/> Environmental</li> <li><input type="checkbox"/> Infection</li> <li><input type="checkbox"/> Inherited traits</li> <li><input type="checkbox"/> Lifestyle choices</li> <li><input type="checkbox"/> Occupation</li> <li><input type="checkbox"/> Treatment for other illnesses (polypharmacy)</li> </ul> <p>Specified disorders/diseases:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Air pollution and asthma</li> <li><input type="checkbox"/> COPD and smoking</li> <li><input type="checkbox"/> COVID-19 and the pandemic</li> <li><input type="checkbox"/> Cystic fibrosis and inherited traits</li> <li><input type="checkbox"/> Hypertension and obesity</li> <li><input type="checkbox"/> Rheumatoid arthritis and autoimmunity</li> <li><input type="checkbox"/> Polypharmacy and Adverse Drug Reactions (ADR)</li> <li><input type="checkbox"/> Sheep farmers and hydatid disease</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Know that disorders/diseases may be caused by multiple factors</li> <li><input type="checkbox"/> How the factor(s) may influence the development of the specified disorders/diseases</li> <li><input type="checkbox"/> What is meant by autoimmunity</li> <li><input type="checkbox"/> The role of diet and exercise in health and wellbeing</li> <li><input type="checkbox"/> How environment can affect health and wellbeing</li> <li><input type="checkbox"/> Why inherited traits influence health and wellbeing</li> <li><input type="checkbox"/> How occupation is a major contributor to injuries and disease as well as economic loss</li> <li><input type="checkbox"/> What is meant by polypharmacy</li> <li><input type="checkbox"/> Benefits and limitations of polypharmacy</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Diseases/disorders other than those specified</li> </ul> | <p><b>M1.3</b><br/><b>M1.6</b><br/><b>M3.6</b></p> | <p><b>HSW5</b></p> |
| <b>1.3 The nature of communicable diseases</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Maths</b>                                       | <b>HSW</b>         |
| <p><b>1.3.1 Causes of communicable diseases</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Viruses: <ul style="list-style-type: none"> <li>• COVID-19</li> <li>• HIV and AIDS</li> </ul> </li> <li><input type="checkbox"/> Bacteria: <ul style="list-style-type: none"> <li>• Lyme disease</li> <li>• Methicillin-resistant Staphylococcus aureus (MRSA)</li> <li>• Tuberculosis</li> </ul> </li> <li><input type="checkbox"/> Fungi: <ul style="list-style-type: none"> <li>• Candidiasis</li> <li>• Histoplasmosis</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How parasitic adaptations of these groups of organisms, allow transmission and entry into the body</li> <li><input type="checkbox"/> How preventative measures may reduce the risk of causes and spread of communicable diseases</li> <li><input type="checkbox"/> Suitability of the role of the following as modes of transmission: <ul style="list-style-type: none"> <li>• Air</li> <li>• Water</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>M1.6</b></p>                                 | <p><b>HSW9</b></p> |

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| <ul style="list-style-type: none"> <li>□ Protozoans: <ul style="list-style-type: none"> <li>• Malaria</li> <li>• Toxoplasmosis</li> </ul> </li> <li>□ Multicellular parasites: <ul style="list-style-type: none"> <li>• Fasciolosis (liver fluke)</li> <li>• Hydatid disease (tapeworm)</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• Food</li> <li>• Touch</li> <li>• Saliva</li> <li>• Sexual organs</li> <li>• Placenta</li> <li>• Birth canal</li> <li>• Contaminated blood products</li> <li>• Contaminated body fluids</li> <li>• Insects</li> <li>• Flatworms</li> <li>• Roundworms</li> <li>• Ticks and mites</li> </ul> <ul style="list-style-type: none"> <li>□ Appropriateness of respiratory tract; gastrointestinal tract; urinogenital openings; broken skin; as portals of entry</li> <li>□ Know multicellular parasites are usually defined as helminths and ectoparasites</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Prion diseases</li> <li>□ Diseases/disorders other than those specified in the teaching content</li> </ul> |             |             |
| <p><b>1.3.2 Effects of communicable diseases</b></p> <ul style="list-style-type: none"> <li>□ Viral diseases: <ul style="list-style-type: none"> <li>• COVID-19</li> <li>• HIV and AIDS</li> </ul> </li> <li>□ Bacterial diseases: <ul style="list-style-type: none"> <li>• Lyme disease</li> <li>• Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA)</li> <li>• Tuberculosis</li> </ul> </li> <li>□ Fungal diseases: <ul style="list-style-type: none"> <li>• Candidiasis (vaginal thrush)</li> <li>• Histoplasmosis</li> </ul> </li> <li>□ Protozoan diseases: <ul style="list-style-type: none"> <li>• Malaria</li> <li>• Toxoplasmosis</li> </ul> </li> <li>□ Multicellular parasite diseases: <ul style="list-style-type: none"> <li>• Fasciolosis (liver fluke)</li> <li>• Hydatid disease (tapeworm)</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Know observable signs of diseases at macroscopic and microscopic level</li> <li>□ Know symptoms felt and experienced</li> <li>□ The advantages and disadvantages of identifying diseases by signs and symptoms</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Prion diseases</li> <li>□ Diseases/disorders other than those specified in the teaching content</li> </ul>                                                                                                                                                                                                                                                                                                                    | <b>M2.2</b> | <b>HSW8</b> |

| <b>Topic Area 2: Curative management and preventative therapies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |                                           |
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| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Breadth and depth</b>                  |                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Opportunities to cover:</b>            |                                           |
| <b>2.1 Curative therapies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b>                              | <b>HSW</b>                                |
| <ul style="list-style-type: none"> <li>□ Antimicrobials               <ul style="list-style-type: none"> <li>• Effect of different antibiotics on the growth of bacteria on agar plates</li> <li>• Koch's postulates</li> </ul> </li> <li>□ Casts               <ul style="list-style-type: none"> <li>• Fibreglass</li> <li>• Plaster</li> </ul> </li> <li>□ Chemotherapy</li> <li>□ Dietary programmes</li> <li>□ Surgery</li> <li>□ Transplants               <ul style="list-style-type: none"> <li>• Gene</li> <li>• Cell</li> <li>• Organ</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How antibiotic discs/wells can be used to investigate bacterial growth on agar plates, including the use of control discs</li> <li>□ How pathogens are destroyed by antimicrobials</li> <li>□ How misuse of antibiotics may result in them becoming ineffective and lead to resistance</li> <li>□ How Koch's postulates establish whether a particular microorganism causes a particular disease</li> <li>□ How a control in antibiotic investigations helps validate experimental performance</li> <li>□ How antibiotic and bacterial investigations can be made more valid</li> <li>□ How these curative therapies may lead to a cure if the treatment period is completed</li> <li>□ How the use of these curative therapies may be influenced by the health status of the patient and various external factors</li> <li>□ Advantages and disadvantages of different ways to manage diseases/disorders</li> <li>□ Benefits and limitations of different types of plaster casts</li> <li>□ Reasons for introducing dietary programmes</li> <li>□ Suitability of the role of curative surgery and chemotherapy in cancer treatment</li> <li>□ Suitability of the role of organ, cellular and molecular therapies</li> <li>□ Appropriateness of the role of transplants in disease treatment</li> </ul> | <b>M0.4</b><br><b>M2.1</b><br><b>M3.6</b> | <b>HSW1</b><br><b>HSW4</b><br><b>HSW5</b> |
| <b>2.2 Management Therapies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b>                              | <b>HSW</b>                                |
| <p><b>Types of management therapies</b></p> <p>Palliative care</p> <ul style="list-style-type: none"> <li>□ Types of diseases/disorders that can be managed               <ul style="list-style-type: none"> <li>• Renal disease                   <ul style="list-style-type: none"> <li>○ Nephritis</li> <li>○ Polycystic Kidney Disease (PKD)</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                                                    | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Purpose of the role of palliative care at the end of life</li> <li>□ Why some diseases cannot be cured</li> <li>□ How management may relieve symptoms, improve quality of, and extend life</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>M1.6</b><br><b>M3.1</b><br><b>M3.3</b> | <b>HSW10</b>                              |

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| <ul style="list-style-type: none"> <li>• Autoimmune diseases <ul style="list-style-type: none"> <li>○ Multiple sclerosis</li> <li>○ Rheumatoid arthritis</li> </ul> </li> <li>• Retinal diseases <ul style="list-style-type: none"> <li>○ Diabetic retinopathy</li> <li>○ Macular degeneration</li> </ul> </li> <li>• Neurodegenerative diseases <ul style="list-style-type: none"> <li>○ Motor neurone disease (MND)</li> <li>○ Parkinson's</li> </ul> </li> <li>• Digestive diseases <ul style="list-style-type: none"> <li>○ Crohn's disease</li> <li>○ Hiatus hernia</li> </ul> </li> </ul> <p>□ Ways of managing diseases/disorders</p> <ul style="list-style-type: none"> <li>• Medication</li> <li>• Supportive therapies <ul style="list-style-type: none"> <li>○ Dialysis</li> <li>○ Occupational therapy</li> <li>○ Physiotherapy</li> <li>○ Speech therapy</li> </ul> </li> <li>• Exercise</li> <li>• Chemotherapy</li> <li>• Cognitive therapy</li> <li>• Surgery</li> </ul> | <p>□ Why some diseases may go into remission</p> <p>□ The potential that some diseases may be cured in the future</p>                                                                                                                                                                                                                                                                                  |                                |              |
| <b>2.3 Preventative therapies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <b>Types of preventative therapy strategies</b> <ul style="list-style-type: none"> <li>□ Allergy and food intolerance testing</li> <li>□ Check-ups</li> <li>□ Health promotion/education programmes</li> <li>□ Meal plans</li> <li>□ Patient counselling</li> <li>□ Screenings</li> <li>□ Vaccinations</li> <li>□ Well baby/well child visits</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How preventive health care aims to improve patient well-being, prevent disease, disability, and death</li> <li>□ Why the detection of pre or early stages of chronic diseases lead to more successful outcomes</li> <li>□ The difference between allergy and intolerance</li> <li>□ Reasons for preventative therapy strategies</li> </ul> | <b>M1.2</b><br><b>M1.7</b>     | <b>HSW12</b> |
| <b>Topic Area 3: The role of immunology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                        |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Breadth and depth</b>                                                                                                                                                                                                                                                                                                                                                                               | <b>Opportunities to cover:</b> |              |
| <b>3.1 The immune system</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <b>Lines of Defence</b> <ul style="list-style-type: none"> <li>□ Innate immunity – first line of defence and non-specific <ul style="list-style-type: none"> <li>• Physical barriers</li> <li>• Chemical barriers</li> <li>• Cells</li> </ul> </li> <li>□ Adaptive immunity - second line of defence and specific <ul style="list-style-type: none"> <li>• Antibodies</li> <li>• Specialised cells</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How physical and chemical barriers – skin, mucous membranes and their secretions assist in defence</li> <li>□ Know the role of macrophages, neutrophils, basophils, mast cells</li> <li>□ Know the role of specialised B and T cells</li> <li>□ Know the gamma globulin structure</li> <li>□ The function of antibodies</li> </ul>         | <b>M0.1</b><br><b>M4.1</b>     | <b>HSW1</b>  |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <input type="checkbox"/> The formation of the antigen-antibody complex and its role in the immune response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |                       |
| <b>3.2 Immune dysfunction and clinical immunology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Maths</b>                   | <b>HSW</b>            |
| <b>3.2.1 Clinical immunology as the study of disease caused by immune system dysfunction</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Immunodeficiency               <ul style="list-style-type: none"> <li>• Primary</li> <li>• Acquired</li> </ul> </li> <li><input type="checkbox"/> Allergies reaction to allergens</li> <li><input type="checkbox"/> Asthma</li> <li><input type="checkbox"/> Autoimmune disease</li> <li><input type="checkbox"/> Cancer</li> <li><input type="checkbox"/> Transplants</li> </ul>                                                                                                           | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How clinical immunology contributes to identifying immune dysfunction, its pathways and origins</li> <li><input type="checkbox"/> How types of problems with the immune system impair its ability to defend against allergens, infections or against 'self' and the resulting consequences</li> <li><input type="checkbox"/> How clinical immunology contributes to improvements in healthcare</li> </ul>                                                                                                                                             | <b>M3.6</b>                    | <b>HSW9<br/>HSW10</b> |
| <b>3.2.2 Vaccines</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Inactivated vaccines</li> <li><input type="checkbox"/> Live attenuated vaccine</li> <li><input type="checkbox"/> Messenger RNA (mRNA)</li> <li><input type="checkbox"/> Subunit               <ul style="list-style-type: none"> <li>• Protein</li> <li>• Polysaccharide</li> <li>• Conjugate</li> </ul> </li> <li><input type="checkbox"/> Toxoid vaccines</li> <li><input type="checkbox"/> Viral vector vaccines</li> </ul>                                                                                                                                     | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How new therapies and treatments can manage or cure a condition by altering the way the immune system works</li> <li><input type="checkbox"/> How vaccine types differ from each other</li> <li><input type="checkbox"/> The role of vaccines in priming the immune system and boosting the immune reaction to specific pathogens</li> <li><input type="checkbox"/> Advantages and disadvantages of vaccine types</li> </ul> Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> The manufacture of vaccines</li> </ul> | <b>M3.5</b>                    | <b>HSW9<br/>HSW10</b> |
| <b>Topic Area 4: Techniques for diagnosis and monitoring</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |                       |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Breadth and depth</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Opportunities to cover:</b> |                       |
| <b>4.1 Diagnostic techniques</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Maths</b>                   | <b>HSW</b>            |
| <b>Stages in medical diagnosis</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Medical history</li> <li><input type="checkbox"/> Physical examination               <ul style="list-style-type: none"> <li>• Auscultation</li> <li>• Inspection</li> <li>• Palpation</li> <li>• Percussion</li> </ul> </li> <li><input type="checkbox"/> Initial tests and measurements               <ul style="list-style-type: none"> <li>• Blood pressure values</li> <li>• Body mass index (BMI)</li> <li>• Lung volumes values</li> <li>• Oxygen levels values</li> <li>• Peak flow values</li> <li>• Temperature value</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How interpersonal skills and general approach of the medical practitioner in establishing the medical history may improve the diagnosis outcome</li> <li><input type="checkbox"/> How consultation room design may improve the diagnosis outcome</li> </ul>                                                                                                                                                                                                                                                                                           | <b>M0.4<br/>M1.1</b>           | <b>HSW2</b>           |

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| <ul style="list-style-type: none"> <li>□ Further diagnostic investigations <ul style="list-style-type: none"> <li>• Biopsies</li> <li>• Blood</li> <li>• Cognitive</li> <li>• Mammogram</li> <li>• Urine</li> </ul> </li> <li>□ Medical practitioners and the use of interpersonal skills</li> </ul>                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>□ How good practice is achieved in different stages of medical diagnosis</li> <li>□ How to interpret the results of diagnostic techniques</li> <li>□ How to calculate BMI and what the results mean</li> <li>□ Reasons for the different stages of medical diagnosis being performed</li> <li>□ Advantages and disadvantages of different stages of medical diagnosis</li> <li>□ Roles of the medical practitioners in different stages of medical diagnosis</li> </ul> <p>Examples of <b>medical practitioners</b> may include:</p> <ul style="list-style-type: none"> <li>□ General Practitioner (GP)</li> <li>□ Nurse</li> <li>□ Pathologist</li> <li>□ Radiologist</li> <li>□ Dermatologist</li> </ul> |                            |              |
| <b>4.2 Monitoring techniques</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Maths</b>               | <b>HSW</b>   |
| <b>4.2.1 Groups requiring monitoring</b> <ul style="list-style-type: none"> <li>□ Acute conditions</li> <li>□ Child development</li> <li>□ Chronic conditions</li> <li>□ Employees requiring statutory medicals <ul style="list-style-type: none"> <li>• Contractual requirements</li> <li>• HSE requirements</li> </ul> </li> <li>□ Specialist clinics <ul style="list-style-type: none"> <li>• Asthma</li> <li>• Diabetes</li> </ul> </li> <li>□ Specific group screening <ul style="list-style-type: none"> <li>• Abdominal aortic aneurysm</li> <li>• Breast</li> <li>• Cervical</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Reasons for screening particular cohorts</li> <li>□ Appropriateness of techniques for the individual/group/situation</li> <li>□ Why some employees require statutory medicals</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>M1.2</b>                | <b>HSW11</b> |
| <b>4.2.2 Methods of monitoring</b> <ul style="list-style-type: none"> <li>□ Repeat of relevant initial diagnostic tests and measurements</li> <li>□ Clinical scoring systems <ul style="list-style-type: none"> <li>• Disease Activity Scores (DAS28)</li> <li>• Unified Parkinson's Disease Rating Scale (UPDRS)</li> </ul> </li> <li>□ Electronic monitoring</li> <li>□ Reagent strips</li> </ul>                                                                                                                                                                                                         | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Advantages and disadvantages of monitoring methods</li> <li>□ How regular monitoring and screening improves the health of an individual/cohort</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ How the diagnostic tests or electronic devices work</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | <b>M1.6</b><br><b>M1.7</b> | <b>HSW3</b>  |

| <b>Topic Area 5: Reporting, research and confidentiality</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                           |                             |
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| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Breadth and depth</b>                  |                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Opportunities to cover:</b>            |                             |
| <b>5.1 Reporting</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                              | <b>HSW</b>                  |
| <b>5.1.1 Types of health data gathered by</b> <ul style="list-style-type: none"> <li>□ Healthcare professionals <ul style="list-style-type: none"> <li>• Clinical trials</li> <li>• Electronic records</li> <li>• Health surveys</li> <li>• Manual records</li> <li>• National databases</li> <li>• Patient disease registries</li> </ul> </li> <li>□ Patients <ul style="list-style-type: none"> <li>• Mobile Apps</li> <li>• Screening tests and dietary monitoring</li> <li>• Social media posts</li> <li>• Wearable devices</li> </ul> </li> <li>□ Wider information <ul style="list-style-type: none"> <li>• Climate and pollution monitoring</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li>□ Benefits of completing health data research</li> <li>□ Benefits and limitations of manual and electronic record gathering</li> <li>□ Reasons for accessing different types of health data</li> <li>□ Advantages and disadvantages of screening tests and dietary monitoring</li> <li>□ How social media may influence people's attitude to health data</li> <li>□ Advantages and disadvantages of apps and wearable devices</li> <li>□ How some wearable devices work in conjunction with mobile apps</li> <li>□ Why climate and pollution monitoring are important from a public health perspective</li> </ul> | <b>M1.3</b><br><b>M1.7</b>                | <b>HSW2</b><br><b>HSW10</b> |
| <b>5.1.2 The process of analytics</b> <ul style="list-style-type: none"> <li>□ Data collection</li> <li>□ Interpretation</li> <li>□ Reporting</li> <li>□ Extraction</li> <li>□ Transformation</li> <li>□ Analysis</li> <li>□ Types of analytics <ul style="list-style-type: none"> <li>• Descriptive - What happened?</li> <li>• Diagnostic - Why did it happen?</li> <li>• Predictive - What may happen?</li> <li>• Prescriptive - Make it happen?</li> </ul> </li> </ul>                                                                                                                                                                                                | To include: <ul style="list-style-type: none"> <li>□ How analytics discover meaningful patterns in data</li> <li>□ The specific order of the process of analytics and what happens at each stage of the process</li> <li>□ Advantages and disadvantages of different types of data analytics in health care</li> </ul> Does not include: <ul style="list-style-type: none"> <li>□ Detailed explanations of the different types of analytics</li> </ul>                                                                                                                                                                                                               | <b>M0.3</b><br><b>M0.4</b><br><b>M3.1</b> |                             |
| <b>5.2 Research</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                              | <b>HSW</b>                  |
| <b>Approach to research</b> <ul style="list-style-type: none"> <li>□ Types of research <ul style="list-style-type: none"> <li>• Qualitative</li> <li>• Quantitative</li> </ul> </li> <li>□ Dependent upon <ul style="list-style-type: none"> <li>• Finance</li> <li>• Practical feasibility</li> <li>• Staffing</li> <li>• Scientific basis</li> </ul> </li> <li>□ Research methodology <ul style="list-style-type: none"> <li>• Clinical</li> <li>• Epidemiological</li> <li>• Experimental</li> </ul> </li> </ul>                                                                                                                                                       | To include: <ul style="list-style-type: none"> <li>□ How the type of research will determine what methodology and study are used</li> <li>□ The difference between qualitative and quantitative research</li> <li>□ The difference between the stated research methodologies</li> </ul>                                                                                                                                                                                                                                                                                                                                                                              | <b>M1.6</b>                               | <b>HSW3</b><br><b>HSW6</b>  |



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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| <input type="checkbox"/> Types of study <ul style="list-style-type: none"> <li>• Case-controlled studies</li> <li>• Cohort studies</li> <li>• Randomised control trials (RCTs)</li> </ul>                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |
| <b>5.3 Confidentiality</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b> | <b>HSW</b>   |
| <b>Confidentiality is maintained through</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Data sharing agreements</li> <li><input type="checkbox"/> Health professional contracts</li> <li><input type="checkbox"/> Government legislation or case law <ul style="list-style-type: none"> <li>• Data Protection Act 2018 (DPA)</li> <li>• Common Law Duty of Confidentiality (CLDC)</li> </ul> </li> <li><input type="checkbox"/> Professional codes of conduct or best practice</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How health professionals can ensure patient confidentiality</li> <li><input type="checkbox"/> Reasons for and against disclosing health data to a third party</li> <li><input type="checkbox"/> Know the DPA 2018 covers personal data (Article 6) and health data (Article 9)</li> <li><input type="checkbox"/> How general disclosure to a third party can be made under CLDC in order to avoid a breach of confidentiality</li> </ul> Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Details of the above Act, Articles and Common Law</li> </ul> |              | <b>HSW11</b> |

### Assessment guidance

This unit is assessed by an exam. The exam is 1 hour and 15 minutes and has **60** marks in total. All the questions in the exam are compulsory.

A range of question types will be used in this assessment including:

- Forced choice/controlled response questions – including MCQs
- Short answer, closed response questions (with or without diagrams)
- Short answer with calculation/working
- Extended constructed response with points-based mark scheme
- Extended constructed response with levels of response mark scheme

Content will be sampled from all topic areas, with at least one question or part question relating to each topic area.

Content in this exam will have links to the 'How Science Works Concepts and Skills' and 'Mathematical skills for Human Biology'.

This will be conducted under examination conditions. For more details refer to the [Administration area](#).

A range of question types will be used in the exam.

The [guide to our Sample Assessment Material for this unit](#) gives more information about the layout and expectations of the exam.

The exam for this unit assesses the following Performance Objectives:

- PO1 – Show knowledge and understanding
- PO2 – Apply knowledge and understanding
- PO3 – Analyse and evaluate knowledge, understanding and performance.

### **Synoptic assessment**

This unit allows students to gain underpinning knowledge and understanding relevant to the qualification and sector. The NEA units draw on and strengthen this learning as students will apply their learning to practical or applied tasks.

The following NEA units have synoptic links with this unit. The synoptic grids at the end of these NEA units show these synoptic links.

- Unit F172: Genetics
- Unit F174: Nutrition and metabolism
- Unit F175: Human reproduction
- Unit F176: The brain
- Unit F177: Drug development

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

## 5.3 NEA Units

### 5.3.1 Unit F172: Genetics

#### Unit Aim

Genetics has a central role in the study of Human Biology. Genetics helps to explain what makes us all unique, why family members look alike, why some diseases run in families and how human evolution occurs. By studying the structure and function of our genes, scientists are able to understand how the body works and how we can use this knowledge to benefit individuals and society as a whole. This unit explores the main principles of genetics and inheritance, particularly in relation to genetic disorders. It looks at the emerging roles of genetic testing and the modification of genes to prevent or cure inherited disorders.

In this unit you will build on knowledge of DNA, cell division and inheritance from Unit F170 Fundamentals of human biology. You will learn how to apply and use mathematical techniques to determine probability of inheritance in human genetic disorders. You will also learn the principles of genetic testing how it is used and the importance of genetic counsellors. Finally, you will learn how to investigate recent advances in gene therapy and genetic engineering, and the potential importance of these technologies in the future.

| <b>Unit F172: Genetics</b>                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------|
| <b>Topic Area 1: Fundamentals of genetics</b>                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                    | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Opportunities to cover:</b> |              |
| <b>1.1 DNA</b>                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <input type="checkbox"/> Function of DNA: <ul style="list-style-type: none"> <li>• Replication</li> <li>• Protein synthesis:               <ul style="list-style-type: none"> <li>○ Transcription</li> <li>○ Translation</li> </ul> </li> </ul> <input type="checkbox"/> Role of telomeres | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Key features of each process</li> <li><input type="checkbox"/> The importance of each process to the cell</li> </ul><br><input type="checkbox"/> The importance of the telomeres <input type="checkbox"/> The effect of ageing on telomeres<br><br>Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Structure of ribosomes</li> <li><input type="checkbox"/> Structure of mRNA</li> </ul> |                                | <b>HSW1</b>  |
| <b>1.2 Gene expression</b>                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <input type="checkbox"/> Gene expression<br><br><br><br><br><br><br><br><br><br><input type="checkbox"/> Gene regulation                                                                                                                                                                   | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Meaning of gene expression</li> <li><input type="checkbox"/> How gene expression is measured</li> <li><input type="checkbox"/> What factors can influence gene expression</li> </ul><br><input type="checkbox"/> Meaning of gene regulation <input type="checkbox"/> Reasons why gene expression and gene regulation are important                                                                                         |                                | <b>HSW9</b>  |
| <b>1.3 Diversity and variation</b>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <b>1.3.1</b>                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>M1.3</b>                    | <b>HSW10</b> |
| <input type="checkbox"/> Phenotypic variation can be caused by: <ul style="list-style-type: none"> <li>• Genotypic variation</li> <li>• Environmental variation</li> </ul>                                                                                                                 | To include:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>M1.6</b>                    |              |

|                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |             |
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| <ul style="list-style-type: none"> <li>□ Genotypic variation occurs because of: <ul style="list-style-type: none"> <li>• Genetic recombination</li> <li>• Gene variants</li> </ul> </li> <li>□ Environmental variation</li> </ul>                                                  | <ul style="list-style-type: none"> <li>□ The process and key features of recombination</li> <li>□ Why recombination is important</li> <li>□ How the process of recombination has been used to map human genes</li> <li>□ Meaning of the term variant</li> <li>□ How recombination and variants contribute to evolution</li> <li>□ What environmental factors can contribute to phenotypic variation in humans</li> <li>□ How environmental factors can alter genes or gene expression</li> </ul>     |                                |             |
| <b>1.3.2</b> <ul style="list-style-type: none"> <li>□ Investigating phenotypic variation in a discrete population</li> </ul>                                                                                                                                                       | To include: <ul style="list-style-type: none"> <li>□ How investigations of phenotypic characteristics in a discrete population are carried out</li> <li>□ Why it is important to compare data from investigations with national statistics</li> <li>□ Limitations of comparing data with national statistics: <ul style="list-style-type: none"> <li>• Effects of age and sex on values</li> <li>• Effects of ethnicity on values</li> <li>• Effects of environment on values</li> </ul> </li> </ul> | <b>M3.1</b>                    | <b>HSW2</b> |
| <b>Topic Area 2: Mode of inheritance</b>                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |             |
| <b>Teaching content</b>                                                                                                                                                                                                                                                            | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Opportunities to cover:</b> |             |
| <b>2.1 Mendelian inheritance</b>                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                   | <b>HSW</b>  |
| <b>2.1.1</b> <ul style="list-style-type: none"> <li>□ Monohybrid inheritance of: <ul style="list-style-type: none"> <li>• Normal trait</li> <li>• Single gene disorder</li> <li>• Codominance</li> <li>• Incomplete inheritance</li> <li>• Sex-linked trait</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li>□ Monohybrid crosses giving genotypes and phenotypes</li> <li>□ Punnett squares</li> </ul>                                                                                                                                                                                                                                                                                                                                                        | <b>M1.3</b>                    |             |
| <b>2.1.2</b> <ul style="list-style-type: none"> <li>□ Dihybrid inheritance of two non-linked autosomal genes</li> <li>□ Predicting genotypic and phenotypic ratios</li> </ul>                                                                                                      | To include: <ul style="list-style-type: none"> <li>□ Dihybrid crosses giving genotypes and phenotypes</li> <li>□ How two-trait Punnett squares are used</li> <li>□ How chi-squared tests use expected and observed data</li> <li>□ The statistical significance of differences in data and probabilities</li> </ul>                                                                                                                                                                                  | <b>M1.1</b><br><b>M1.3</b>     |             |

| <b>2.2 DNA mutations</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>               | <b>HSW</b>                 |
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| <b>2.2.1 Genetic mutations caused by changes in the sequence of DNA:</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Deletion</li> <li><input type="checkbox"/> Inversion</li> <li><input type="checkbox"/> Substitution</li> <li><input type="checkbox"/> Duplication</li> </ul>                                                                                                                                                                                                                                                                                              | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Key features of each way that mutations can occur in DNA</li> <li><input type="checkbox"/> Representation of each way that the DNA sequence can change using diagrams</li> <li><input type="checkbox"/> The effect of changes in DNA to which amino acid is expressed, and therefore to proteins that are produced</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            | <b>HSW1</b>                |
| <b>2.2.2 Genetic mutations:</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Acquired mutations</li> <li><input type="checkbox"/> Hereditary mutations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                               | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Comparison of key features of both types of genetic mutations</li> <li><input type="checkbox"/> Factors that can cause acquired mutations</li> <li><input type="checkbox"/> Consequences of genetic mutations, including the effect the mutation can have on:               <ul style="list-style-type: none"> <li>• Genes or Gene expression</li> <li>• Protein production</li> <li>• Physiological processes in the body</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                              | <b>M3.6</b>                | <b>HSW11</b>               |
| <b>2.3 Genetic disorders</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>               | <b>HSW</b>                 |
| <b>2.3.1</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Types of genetic disorders:               <ul style="list-style-type: none"> <li>• Single gene</li> <li>• Chromosomal</li> <li>• Complex (polygenic)</li> </ul> </li> <li><input type="checkbox"/> Types of single gene disorders:               <ul style="list-style-type: none"> <li>• Autosomal dominant</li> <li>• Autosomal recessive gene</li> <li>• X-linked dominant</li> <li>• X-linked recessive</li> </ul> </li> <li><input type="checkbox"/> Human pedigree analysis in single gene disorders</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Meaning of the term genetic disorder</li> <li><input type="checkbox"/> Key features of each type of single gene disorder</li> <li><input type="checkbox"/> Patterns of inheritance of single gene disorders using genetic crosses and Punnett squares</li> <li><input type="checkbox"/> How human pedigree analysis is used to identify the type of single gene disorder</li> <li><input type="checkbox"/> How single gene disorders can be tracked through families and risks to future generations predicted</li> </ul> <p>Examples of <b>single gene disorders</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Cystic Fibrosis</li> <li><input type="checkbox"/> Sickle cell anaemia</li> <li><input type="checkbox"/> Huntington's disease</li> </ul> | <b>M3.1</b>                | <b>HSW7</b><br><b>HSW9</b> |
| <b>2.3.2</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Chromosomal disorders can be caused by changes in:               <ul style="list-style-type: none"> <li>• The number of chromosomes</li> <li>• The structure of chromosomes</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                    | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How changes in the number and structure of chromosomes can occur</li> <li><input type="checkbox"/> Identification of chromosome disorders from diagrams</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>M1.3</b><br><b>M1.4</b> | <b>HSW1</b>                |

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|                                                                                                                                                                                                                                                                                                                                | <p>Examples of <b>chromosomal disorders</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Down syndrome</li> <li><input type="checkbox"/> Klinefelter syndrome</li> <li><input type="checkbox"/> Turner syndrome</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |              |
| <p><b>2.3.3</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Complex genetic disorders (polygenic) caused by a combination of: <ul style="list-style-type: none"> <li>• Many genes</li> <li>• Lifestyle and environmental factors</li> </ul> </li> </ul>                                                | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Why it is harder to track patterns of inheritance for complex genetic disorders</li> <li><input type="checkbox"/> Meaning of the term genetic predisposition</li> <li><input type="checkbox"/> How people with a genetic predisposition may be able to reduce their risk</li> </ul> <p>Examples of <b>complex genetic disorders</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Type 2 diabetes</li> <li><input type="checkbox"/> Coronary heart disease</li> <li><input type="checkbox"/> Atherosclerosis</li> </ul>                                                                                                                                                                                                                 | <b>M3.1</b>                    | <b>HSW12</b> |
| <b>Topic Area 3: Genetic counselling and genetic testing</b>                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                        | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Opportunities to cover:</b> |              |
| <b>3.1 Genetic counselling</b>                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>                   | <b>HSW</b>   |
| <p><b>3.1.1</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> What genetic counselling is</li> <li><input type="checkbox"/> The role of a genetic counsellor: <ul style="list-style-type: none"> <li>• Providing information and support</li> <li>• Assessing risk of inheritance</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Why different individuals might have genetic counselling</li> <li><input type="checkbox"/> Why individuals might have genetic counselling before or after genetic testing</li> </ul> <p>Examples of <b>the role of a genetic counsellor</b>:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Providing information and support about: <ul style="list-style-type: none"> <li>• Different genetic tests</li> <li>• How to arrange tests</li> <li>• How to understand test results</li> <li>• Support groups for a patient or for a family</li> </ul> </li> <li><input type="checkbox"/> Assessing risk of inheritance: <ul style="list-style-type: none"> <li>• Looking at family medical history</li> <li>• Using a family tree</li> </ul> </li> </ul> | <b>M3.1</b>                    | <b>HSW9</b>  |
| <p><b>3.1.2</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Genetic tests: <ul style="list-style-type: none"> <li>• Molecular tests</li> <li>• Chromosomal tests</li> <li>• Gene expression tests</li> <li>• Biochemical tests</li> </ul> </li> </ul>                                                  | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How genetic tests are taken</li> <li><input type="checkbox"/> Key features of each test</li> <li><input type="checkbox"/> Similarities and differences between the tests</li> <li><input type="checkbox"/> Reasons for selecting one type of test over another</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>M1.3</b>                    | <b>HSW8</b>  |

| <b>3.2 Different types of genetic tests</b>                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b> | <b>HSW</b>            |
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| <b>3.2.1</b><br><input type="checkbox"/> Genetic tests in adults: <ul style="list-style-type: none"> <li>• Diagnostic tests</li> <li>• Assessing risk of genetic disorder</li> <li>• Ancestry genetic tests</li> </ul>                                                                                                                                                | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Key features of each type of genetic test in adults</li> <li><input type="checkbox"/> What information each test provides</li> <li><input type="checkbox"/> How tests differ from each other</li> </ul>                                                                                                                                                                                                                                                                                                        |              | <b>HSW11</b>          |
| <b>3.2.2</b><br><input type="checkbox"/> Genetic tests in embryos and babies: <ul style="list-style-type: none"> <li>• Prenatal tests</li> <li>• New-born screening</li> </ul>                                                                                                                                                                                        | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Why and how tests are carried out</li> <li><input type="checkbox"/> The advantages and disadvantages of tests</li> <li><input type="checkbox"/> Which disorders are targeted by both types of test, and why</li> <li><input type="checkbox"/> Importance of tests</li> <li><input type="checkbox"/> Reasons why new-born screening is the most common type of genetic testing</li> <li><input type="checkbox"/> Reasons why there are regional differences in prenatal tests and new-born screening</li> </ul> | <b>M1.6</b>  | <b>HSW1<br/>HSW4</b>  |
| <b>3.2.3</b><br><input type="checkbox"/> Preimplantation tests used in the process of <i>in vitro</i> fertilisation (IVF)<br><input type="checkbox"/> Basic outline of the process of IVF                                                                                                                                                                             | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How preimplantation tests are used in IVF</li> <li><input type="checkbox"/> Advantages and disadvantages of preimplantation testing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 | <b>M1.3</b>  | <b>HSW4<br/>HSW12</b> |
| <b>3.3 Privacy and ethics</b>                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b> | <b>HSW</b>            |
| <b>3.3.1</b><br><input type="checkbox"/> Privacy and ethical issues in genetic testing: <ul style="list-style-type: none"> <li>• Confidentiality of personal information</li> <li>• Sharing of information</li> <li>• Storage of DNA information</li> <li>• Consequences of positive genetic test results</li> <li>• Accuracy of results and false results</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How each issue arises in genetic testing</li> <li><input type="checkbox"/> Why each ethical issue is important</li> <li><input type="checkbox"/> How the issues can be solved or minimised</li> </ul>                                                                                                                                                                                                                                                                                                          | <b>M1.5</b>  | <b>HSW9<br/>HSW10</b> |
| <b>3.3.2</b><br><input type="checkbox"/> Concerns about storage of DNA information on a DNA database: <ul style="list-style-type: none"> <li>• Surveillance</li> <li>• Discrimination</li> <li>• DNA evidence is not always 100% accurate</li> </ul>                                                                                                                  | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How each concern arises</li> <li><input type="checkbox"/> Why each concern is important</li> <li><input type="checkbox"/> How the concerns could be addressed</li> </ul>                                                                                                                                                                                                                                                                                                                                       | <b>M0.4</b>  | <b>HSW8<br/>HSW9</b>  |

| Topic Area 4: Gene therapy and genetic engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                           |                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|
| Teaching content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Exemplification                           |                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Opportunities to cover:                   |                                            |
| <b>4.1 Gene therapy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                              | <b>HSW</b>                                 |
| <ul style="list-style-type: none"> <li>□ Gene therapy corrects genetic defects by:               <ul style="list-style-type: none"> <li>• Replacing defective genes</li> <li>• Turning off defective genes</li> <li>• Turning on healthy genes</li> <li>• Training the immune system to recognise diseased cells</li> </ul> </li> <li>□ Genes can be altered in:               <ul style="list-style-type: none"> <li>• Somatic cells</li> <li>• Germline cells</li> </ul> </li> <li>□ Methods of delivery of gene therapy:               <ul style="list-style-type: none"> <li>• <i>ex vivo</i> (<i>in vitro</i>)</li> <li>• <i>in vivo</i></li> <li>• <i>in situ</i> gene therapies</li> </ul> </li> <li>□ The use of vectors in gene therapy</li> </ul> | <ul style="list-style-type: none"> <li>To include:               <ul style="list-style-type: none"> <li>□ Key features of the different ways that genes can be altered in gene therapy</li> <li>□ Benefits of gene therapy</li> <li>□ Risks and challenges involved in gene therapy</li> <li>□ Examples of the use of gene therapy</li> </ul> </li> <li>□ Key differences between somatic and germline cells and their use in gene therapies</li> <li>□ Key features and differences between <i>ex vivo</i>, <i>in vivo</i> and <i>in situ</i> gene therapies</li> <li>□ The advantages and disadvantages of each method of delivery</li> <li>□ Why vectors are used in gene therapy</li> <li>□ Best vectors to use</li> <li>□ The advantages of using viruses as vectors in gene therapy</li> </ul> | <b>M0.2</b><br><b>M0.5</b><br><b>M1.1</b> | <b>HSW9</b>                                |
| <b>4.2 Genetic engineering</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                              | <b>HSW</b>                                 |
| <b>4.2.1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                           | <b>HSW9</b>                                |
| <ul style="list-style-type: none"> <li>□ Genetic engineering and recombinant DNA technologies</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>To include:               <ul style="list-style-type: none"> <li>□ Key features of genetic engineering</li> <li>□ Purpose of genetic engineering</li> <li>□ Comparison of genetic engineering to gene therapy in terms of techniques, purpose and ethics</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                           |                                            |
| <b>4.2.2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>M3.1</b>                               | <b>HSW1</b><br><b>HSW9</b><br><b>HSW11</b> |
| <ul style="list-style-type: none"> <li>□ Genetic engineering in humans</li> <li>□ CRISPR technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>To include:               <ul style="list-style-type: none"> <li>□ Reasons why genetic engineering might be used in humans</li> <li>□ Advantages and disadvantages of genetic engineering in humans</li> <li>□ Ethics of genetic engineering in humans</li> </ul> </li> <li>□ Key features of CRISPR technology</li> <li>□ Potential uses of CRISPR technology in humans</li> <li>□ Benefits, limitations, and ethics of CRISPR technology</li> </ul>                                                                                                                                                                                                                                                                                                         |                                           |                                            |



## Assessment criteria

The table below gives the assessment criteria for the tasks in the set assignment for this unit. The assessment criteria indicate what is required in these tasks.

This qualification has a compensatory approach. This means that the unit grade awarded is based on the **total** number of achieved criteria for the unit (see [Section 6.4](#)). Students do **not** have to achieve **all** criteria for a specific grade to achieve that unit grade (e.g. achieve all pass criteria to achieve a pass grade).

[Section 7.4](#) provides full information on how to assess the NEA units and apply the assessment criteria. Students' work must show that all aspects of a criterion have been met in sufficient detail for it to be **successfully achieved** (see [Section 7.4.1](#)). If a student's work does not fully meet a criterion, you must not award that criterion.

The command words used in the assessment criteria are defined in [Appendix B](#).

| Pass                                                                                                                               | Merit                                                                                                                                     | Distinction                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1:</b> Use research to <b>summarise</b> DNA function for someone with the genetic disorder.                                    | <b>M1:</b> Use research to <b>compare</b> the functioning gene/chromosome to the malfunctioning gene/chromosome for the genetic disorder. | <b>D1: Assess</b> how physiological processes are affected by the genetic disorder.                                                    |
| <b>P2:</b> Use research to <b>explain</b> how genes determine the signs and symptoms of the genetic disorder.                      | <b>M2:</b> Use research to <b>describe</b> how gene expression and gene regulation contribute to the genetic disorder.                    |                                                                                                                                        |
| <b>P3:</b> Use research to <b>describe</b> how the genetic disorder is caused by type(s) of variation.                             |                                                                                                                                           |                                                                                                                                        |
| <b>P4:</b> Use research to <b>describe</b> the mode of inheritance of the genetic disorder.                                        |                                                                                                                                           |                                                                                                                                        |
| <b>P5:</b> Use research to <b>describe</b> how relevant gene therapies are for the genetic disorder.                               | <b>M3:</b> Use research to <b>describe</b> the medical benefits and risks of gene therapy for the genetic disorder.                       | <b>D2: Discuss three</b> advantages and <b>three</b> disadvantages of the potential for genetic engineering for this genetic disorder. |
| <b>P6:</b> Use research to <b>describe</b> how genes are altered through the most relevant gene therapy for this genetic disorder. | <b>M4: Analyse</b> the challenges involved with gene therapy for the genetic disorder.                                                    |                                                                                                                                        |
| <b>P7: Explain</b> the method of delivery for the most relevant gene therapy for this genetic disorder.                            |                                                                                                                                           |                                                                                                                                        |

| Pass                                                                                                                                                                                                                                                                                                                                 | Merit                                                                                                   | Distinction                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>P8:</b> Use research to <b>summarise</b> how a genetic counsellor may be able to assist the patient.</p> <p><b>P9:</b> <b>Explain</b> the potential impact of the genetic disorder on the mental health of the patient.</p> <p><b>P10:</b> <b>Explain</b> how privacy and ethical issues can be addressed for the patient.</p> | <p><b>M5:</b> <b>Explain</b> how genetic counselling would be beneficial in the case study context.</p> | <p><b>D3:</b> <b>Discuss</b> the relevance of gene therapies in the case study context.</p>                                                            |
| <p><b>P11:</b> <b>Create</b> diagrammatic representation(s) to show the inheritance of the genetic disorder in the case study context.</p>                                                                                                                                                                                           |                                                                                                         |                                                                                                                                                        |
| <p><b>P12:</b> <b>Explain</b> the type of genetic test(s) that is appropriate to diagnose the genetic disorder.</p>                                                                                                                                                                                                                  |                                                                                                         |                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                      | <p><b>M6:</b> <b>Explain</b> what the diagrammatic representation(s) means for the patient.</p>         | <p><b>D4:</b> <b>Discuss</b> what the diagrammatic representation(s) show about the inheritance of the genetic disorder in the case study context.</p> |
|                                                                                                                                                                                                                                                                                                                                      | <p><b>M7:</b> <b>Analyse</b> the role of genetic test(s) in the case study context.</p>                 | <p><b>D5:</b> <b>Assess three</b> available options for managing the outcomes of the genetic disorder in the case study context.</p>                   |

### Assessment guidance

This assessment guidance gives you information relating to the assessment criteria. There might not be additional assessment guidance for each assessment criterion. It is included only where it is needed.

| Assessment Criteria | Assessment guidance                                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task 1              | <ul style="list-style-type: none"> <li>The research element of the criteria in this Task does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criteria.</li> </ul>                                     |
| P1                  | <ul style="list-style-type: none"> <li>Students must use research to summarise DNA function for someone with the genetic disorder. Students must consider the impact on different sexes and at different life stages.</li> </ul>                                                |
| M1                  | <ul style="list-style-type: none"> <li>Students need to compare the functioning gene or chromosome to the malfunctioning gene or chromosome for the genetic disorder. Whether the focus is on 'gene' or 'chromosome' will depend on the genetic disorder.</li> </ul>            |
| Task 2              | <ul style="list-style-type: none"> <li>The research element of the criteria in this Task does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criteria.</li> </ul>                                     |
| P5                  | <ul style="list-style-type: none"> <li>Students must use research to describe how relevant <b>at least two</b> gene therapies are for the genetic disorder.</li> <li>If at least two gene therapies are <b>not</b> relevant then there must be a description of why.</li> </ul> |
| M3                  | <ul style="list-style-type: none"> <li><b>M3</b> is an extension of <b>P5</b>.</li> </ul>                                                                                                                                                                                       |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M4          | <ul style="list-style-type: none"> <li>Students must analyse the challenges involved with gene therapy for the genetic disorder.</li> <li>The challenges might be holistic, like financial, practical or ethical considerations, or specific, like the number of genes affecting the genetic disorder, the countries the gene therapy is offered or people's understanding of the gene therapy.</li> </ul>                                                      |
| D2          | <ul style="list-style-type: none"> <li>Students must discuss <b>three</b> advantages and <b>three</b> disadvantages of the potential for genetic engineering for this genetic disorder.</li> <li>This discussion might include, for example, exploring whether genetic engineering would be financially viable, ethical concerns, the complications of research, the impact on those who have the genetic disorder, improvements to quality of life.</li> </ul> |
| Task 3      | <ul style="list-style-type: none"> <li>In Task 3, where a criterion focuses on 'the patient' then students must focus on the patient. There is no expectation that they discuss the rest of the case study context.</li> <li>In Task 3, where a criterion focuses on 'in the case study context' then students must include the whole case study context, for example, other family members, potential children, partners.</li> </ul>                           |
| P8          | <ul style="list-style-type: none"> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul>                                                                                                                                                                                                                               |
| P10         | <ul style="list-style-type: none"> <li>Students explain how at least <b>two</b> privacy issues and at least <b>two</b> ethical issues can be addressed for the patient.</li> <li>If at least <b>two</b> privacy issues and/or ethical issues are not relevant then there must be an explanation of why.</li> </ul>                                                                                                                                              |
| P11, M6, D4 | <ul style="list-style-type: none"> <li>For <b>P11</b>, <b>M6</b> and <b>D4</b>, students should include all relevant diagrammatic representations from Topic Area 2.2 DNA mutations as appropriate for the genetic disorder.</li> </ul>                                                                                                                                                                                                                         |
| M5          | <ul style="list-style-type: none"> <li><b>M5</b> is an extension of <b>P8</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |
| M6          | <ul style="list-style-type: none"> <li><b>M6</b> is an extension of <b>P11</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      |
| M7          | <ul style="list-style-type: none"> <li><b>M7</b> is an extension of <b>P12</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      |
| D3          | <ul style="list-style-type: none"> <li>Students must discuss the relevance of gene therapies in the case study context, with part of the discussion potentially being whether gene therapy is the most appropriate option or if there are other treatments available.</li> </ul>                                                                                                                                                                                |
| D4          | <ul style="list-style-type: none"> <li><b>D4</b> is an extension of <b>M6</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |
| D5          | <ul style="list-style-type: none"> <li>For <b>D5</b>, <b>three</b> different options should be assessed, but the number of available options may be more than three depending on the genetic disorder.</li> <li>Students are <b>not</b> required to assess more than three available options.</li> <li>Options might focus on a range of factors including patient care, patient well-being, treatments and cures.</li> </ul>                                   |

**Synoptic assessment**

Some of the knowledge, understanding and skills needed to complete this unit will draw on the learning in Units F170 and F171.

This table details these synoptic links.

| Unit F172: Genetics |                          | Unit F170: Fundamentals of human biology |                         |
|---------------------|--------------------------|------------------------------------------|-------------------------|
| Topic Area          |                          | Topic Area                               |                         |
| 1                   | Fundamentals of genetics | 1                                        | Human cells and tissues |

| Unit F172: Genetics |                                      | Unit F171: Health and disease |                                              |
|---------------------|--------------------------------------|-------------------------------|----------------------------------------------|
| Topic Area          |                                      | Topic Area                    |                                              |
| 2                   | Mode of inheritance                  | 1                             | Causes and effects of diseases and disorders |
| 4                   | Gene therapy and genetic engineering | 3                             | The role of immunology                       |

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

### 5.3.2 Unit F173: Biomedical techniques

#### Essential resources required for this unit:

- Science laboratory and relevant equipment (see Teacher/Technician Advice sheet).

#### Unit Aim

Biomedical techniques are practical techniques used in many medical, industrial and quality control laboratories. Biomedical scientists carry out a range of scientific tests to support the diagnosis of ill health in humans. Many health service departments rely on the information from biomedical scientists to complete their diagnoses and select treatment pathways.

In this unit you will learn how to plan and carry out investigations using a variety of quantitative and qualitative laboratory techniques that can be used to assess and analyse biomolecules and biochemicals. The analysis of these biochemicals can reveal different diseases and disorders from samples. Techniques can include chromatography, urinalysis, microscopy and titration. You will also learn about other laboratory techniques that are available in a biomedical laboratory and how these can be used for diagnosis.

| Unit F173: Biomedical techniques                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                 |                         |                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------|
| Topic Area 1: What biomedical science is                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                 |                         |                                           |
| Teaching content                                                                                                                                                                                                                                                                                                                          | Exemplification                                                                                                                                                                                                                                                                                                                                                                 | Opportunities to cover: |                                           |
| <b>1.1 Role of a Biomedical Scientist</b>                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>            | <b>HSW</b>                                |
| <b>1.1.1 The purpose of biomedical science</b> <ul style="list-style-type: none"> <li>□ Contributions to research and medicine</li> <li>□ Specific duties and responsibilities</li> <li>□ Diagnosis and monitoring</li> </ul>                                                                                                             | To include: <ul style="list-style-type: none"> <li>□ Types of diseases and conditions that biomedical scientists can support physicians to diagnose</li> </ul><br>Examples of <b>diseases and conditions</b> may include: <ul style="list-style-type: none"> <li>□ Diabetes</li> <li>□ Kidney and liver diseases</li> <li>□ Allergies</li> </ul>                                |                         | <b>HSW2</b>                               |
| <b>1.1.2 Disciplines associated with biomedical science</b> <ul style="list-style-type: none"> <li>□ Cytopathology</li> <li>□ Cytology</li> <li>□ Clinical Chemistry</li> <li>□ Histopathology</li> <li>□ Haematology</li> <li>□ Immunology</li> <li>□ Medical Microbiology</li> <li>□ Virology</li> <li>□ Transfusion Science</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ The type of analysis conducted by scientists in each discipline</li> <li>□ How each discipline contributes to diagnosis</li> <li>□ The importance of collaboration between disciplines and physicians for diagnosis</li> <li>□ The types of qualitative and quantitative techniques employed by each discipline</li> </ul> |                         | <b>HSW3</b><br><b>HSW6</b>                |
| <b>1.2 Handling Specimens</b>                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>            | <b>HSW</b>                                |
| <b>How specimens in biomedical laboratories are:</b> <ul style="list-style-type: none"> <li>□ Obtained</li> <li>□ Handled</li> <li>□ Transported</li> <li>□ Stored</li> </ul>                                                                                                                                                             | To include: <ul style="list-style-type: none"> <li>□ The importance of effective health and safety protocols when handling biohazardous materials</li> <li>□ The importance of sterility when obtaining and handling samples</li> <li>□ The need for specimen preservatives, storage conditions, and when these are required</li> </ul>                                         |                         | <b>HSW3</b><br><b>HSW4</b><br><b>HSW9</b> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li><input type="checkbox"/> The importance of security in laboratory information systems</li> <li><input type="checkbox"/> The need for effective patient and sample identity protocols</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                         |                                           |
| <b>1.3 Biological variability</b>                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b>                                                            | <b>HSW</b>                                |
| <b>Using reference values and population statistics</b>                                                                                                                                                                                                                                                                                                                                                                               | <p><b>To include:</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> The need for reference values in diagnostics</li> <li><input type="checkbox"/> The limitations of reference values and population statistics, including: <ul style="list-style-type: none"> <li>• Inter- and Intra- individual variation</li> <li>• Effects of age and sex on values</li> <li>• Effects of environment, such as nutrition, time of day, stress on reference values</li> </ul> </li> </ul> <p>Examples of <b>reference values</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Concentration of glucose in urine</li> <li><input type="checkbox"/> Red blood cell count</li> <li><input type="checkbox"/> Ion concentrations</li> </ul>                                                                                                                                                                                | <b>M0.2</b><br><b>M1.2</b>                                              | <b>HSW10</b>                              |
| <b>Topic Area 2: Diagnostic techniques: cells and microscopy</b>                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |                                           |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Opportunities to cover:</b>                                          |                                           |
| <b>2.1 Microscopy</b>                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b>                                                            | <b>HSW</b>                                |
| <p><b>Types of microscopy</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Key features of <ul style="list-style-type: none"> <li>• Light microscopy (LM)</li> <li>• Electron microscopy</li> <li>• Transmission and Scanning</li> <li>• Fluorescence microscopy</li> <li>• Confocal microscopy</li> </ul> </li> <li><input type="checkbox"/> Use of light microscopes to observe cells and tissues</li> </ul> | <p><b>To include:</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How to select the appropriate type of microscopy to use for different biological samples and purposes</li> <li><input type="checkbox"/> The advantages and disadvantages of each type of microscopy in biomedical science, including resolution and magnification</li> <li><input type="checkbox"/> How to measure samples using an eyepiece graticule in eyepiece units and calibrating the units into <math>\mu\text{m}</math> using a stage micrometer</li> <li><input type="checkbox"/> How to determine sizes of biological specimens</li> <li><input type="checkbox"/> The difference between wet and dry mounts of specimens and their appropriateness</li> <li><input type="checkbox"/> How to use a haemocytometer to calculate mean numbers of erythrocytes</li> <li><input type="checkbox"/> The importance of dilution when using a haemocytometer</li> </ul> | <b>M0.1</b><br><b>M0.2</b><br><b>M1.1</b><br><b>M1.4</b><br><b>M2.2</b> | <b>HSW3</b><br><b>HSW4</b><br><b>HSW5</b> |

|                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |                             |
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|                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>□ Common errors, risks and hazards associated with using LM</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Detailed understanding of different types of confocal microscopy</li> <li>□ Detailed understanding of how to prepare biological samples for microscopy not available to schools</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |                             |
| <b>2.2 Cytology and histopathology</b>                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Maths</b>               | <b>HSW</b>                  |
| <b>2.2.1 Cytology</b> <ul style="list-style-type: none"> <li>□ Collecting the cell samples: <ul style="list-style-type: none"> <li>• Exfoliative cytology</li> <li>• Intervention cytology</li> </ul> </li> <li>□ Visualising cell samples: <ul style="list-style-type: none"> <li>• Fixation</li> <li>• Staining</li> <li>• Mounting</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How different cell samples are collected</li> <li>□ The impact of the choice of collection method on the quality of the cell sample</li> <li>□ How to compare healthy specialised cells with abnormal cells</li> <li>□ Potential diseases or disorders indicated by cell abnormalities as seen by LM</li> </ul> <p>Examples of <b>collection techniques</b> may include:</p> <ul style="list-style-type: none"> <li>□ Blood draws</li> <li>□ Skin biopsy</li> <li>□ Fine need aspiration</li> </ul> <ul style="list-style-type: none"> <li>□ Techniques available for visualising cell samples</li> <li>□ How to prepare slides for LM with appropriate stains available to schools</li> <li>□ How to identify normal cell structures and morphology using LM and types of abnormality that could be identified</li> <li>□ How to dispose of cytology samples appropriately</li> </ul> | <b>M0.1</b><br><b>M1.7</b> | <b>HSW6</b><br><b>HSW11</b> |
| <b>2.2.2 Histopathology</b> <ul style="list-style-type: none"> <li>□ Collection of tissue samples</li> <li>□ Visualising tissue samples</li> </ul>                                                                                                                                                                                                           | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How different tissue samples are collected</li> <li>□ The impact of the choice of collection method on the quality of the tissue sample</li> <li>□ How to compare healthy tissues with abnormal tissue</li> <li>□ Techniques available for visualising tissue samples</li> <li>□ Potential diseases or disorders indicated by tissue abnormalities</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>M1.7</b>                | <b>HSW3</b><br><b>HSW6</b>  |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Examples of <b>collection techniques</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Core needle biopsy</li> <li><input type="checkbox"/> Open biopsy</li> <li><input type="checkbox"/> Fine need aspiration</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |                             |
| <b>2.3 Haematology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>               | <b>HSW</b>                  |
| <ul style="list-style-type: none"> <li><input type="checkbox"/> Blood cell counts</li> <li><input type="checkbox"/> Blood film preparation</li> <li><input type="checkbox"/> Staining techniques</li> <li><input type="checkbox"/> Iron levels</li> <li><input type="checkbox"/> Blood typing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How to select the appropriate analysis to carry out for diagnosis</li> <li><input type="checkbox"/> The advantages and disadvantages of each type of analysis</li> <li><input type="checkbox"/> How to carry out research to determine reference values for blood cell counts and iron levels</li> <li><input type="checkbox"/> How to analyse blood films for abnormalities</li> <li><input type="checkbox"/> Common errors, risks and hazards associated with each technique</li> <li><input type="checkbox"/> How to dispose of haematology samples appropriately</li> <li><input type="checkbox"/> Potential diseases or disorders indicated by blood abnormalities</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Haematocrit levels</li> <li><input type="checkbox"/> Detailed knowledge of how iron levels of blood are determined</li> </ul>                  | <b>M0.4</b><br><b>M1.6</b> | <b>HSW4</b><br><b>HSW12</b> |
| <b>2.4 Microbiology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>               | <b>HSW</b>                  |
| <ul style="list-style-type: none"> <li><input type="checkbox"/> Culturing bacteria and fungi effectively and safely <ul style="list-style-type: none"> <li>• Aseptic technique</li> <li>• Preparation of sterile agar plates and nutrient media</li> <li>• Disposal</li> </ul> </li> <li><input type="checkbox"/> The culture of bacteria by the inoculation of agar plates <ul style="list-style-type: none"> <li>• Streak plates</li> <li>• Lawn plates</li> <li>• Pour plates</li> </ul> </li> <li><input type="checkbox"/> The identification of bacteria and fungi through <ul style="list-style-type: none"> <li>• Appropriate staining</li> <li>• Microscopy</li> <li>• Colony morphology</li> <li>• Selective and differential media</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> The techniques required for safe culturing and observation of microorganisms</li> <li><input type="checkbox"/> Sterilisation, disinfection and safe disposal of cultures</li> <li><input type="checkbox"/> How to select different types of growth media in the culturing and identification of microorganisms</li> <li><input type="checkbox"/> How to identify bacteria and fungi by cell and colony morphology</li> <li><input type="checkbox"/> The steps involved in testing for gram-negative and gram-positive bacteria</li> <li><input type="checkbox"/> The role of different types of growth media in the culturing and identification of microorganisms</li> <li><input type="checkbox"/> Advantages and disadvantages of different types of culturing technique</li> <li><input type="checkbox"/> Common errors, risks and hazards associated with microbiological techniques available in schools.</li> </ul> | <b>M4.2</b>                | <b>HSW2</b><br><b>HSW3</b>  |



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|                                                                                                                                                                                                                                                                                                                                                                                                                     | <input type="checkbox"/> Potential diseases or disorders indicated by the presence of bacteria or fungi cultures<br><br>Does not include:<br><input type="checkbox"/> Preparation of specialised growth media<br><input type="checkbox"/> Culturing viruses or parasites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                |                            |
| <b>2.5 Immunological assays</b>                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>                   | <b>HSW</b>                 |
| <input type="checkbox"/> Diagnose infectious diseases<br><input type="checkbox"/> Measure the function of immune cells<br><input type="checkbox"/> Detection of toxins and drugs                                                                                                                                                                                                                                    | To include:<br><input type="checkbox"/> The principles of immunological assays and different types of labelling<br><input type="checkbox"/> The use of assays for qualitative and quantitative assessments<br><input type="checkbox"/> The types of materials detected by immunoassay in biomedical science<br><input type="checkbox"/> The advantages and disadvantages of immunological assays in biomedical sciences, including sensitivity<br><input type="checkbox"/> Potential diseases or disorders monitored and diagnosed by immunological assays<br><br>Examples of <b>uses of immunological assays</b> may include:<br><input type="checkbox"/> Allergy testing<br><input type="checkbox"/> Prostate cancer detection<br><input type="checkbox"/> Pregnancy testing | <b>M0.4</b><br><b>M1.6</b>     | <b>HSW8</b><br><b>HSW9</b> |
| <b>Topic Area 3: Diagnostic techniques: biological molecules</b>                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |                            |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                             | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Opportunities to cover:</b> |                            |
| <b>3.1 Reagent test strips</b>                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>                   | <b>HSW</b>                 |
| <b>Qualitative and quantitative analysis of:</b><br><input type="checkbox"/> Drugs<br><input type="checkbox"/> pH<br><input type="checkbox"/> Glucose<br><input type="checkbox"/> Proteins<br><input type="checkbox"/> Ketones<br><input type="checkbox"/> Hormones<br><input type="checkbox"/> Antibodies<br><input type="checkbox"/> Leukocytes<br><input type="checkbox"/> Other organic and inorganic compounds | To include:<br><input type="checkbox"/> The type of information available from different reagent test strips<br><input type="checkbox"/> How they work and how they are used<br><input type="checkbox"/> The advantages and disadvantages of using reagent test strips, including sensitivity<br><input type="checkbox"/> Hazards associated with their use and associated control measures, including disposal<br><input type="checkbox"/> Potential diseases or disorders indicated by reagent test strips                                                                                                                                                                                                                                                                   | <b>M3.1</b><br><b>M3.2</b>     | <b>HSW6</b>                |

| <b>3.2 Qualitative tests for inorganic substances</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>               | <b>HSW</b>                 |
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| <b>3.2.1 Identification of inorganic substances</b> <ul style="list-style-type: none"> <li>□ Chemical tests for cations               <ul style="list-style-type: none"> <li>• <math>Al^{3+}</math></li> <li>• <math>Ca^{2+}</math></li> <li>• <math>Cu^{2+}</math></li> <li>• <math>Fe^{2+}</math></li> <li>• <math>Fe^{3+}</math></li> <li>• <math>H^+</math></li> <li>• <math>K^+</math></li> <li>• <math>Mg^{2+}</math></li> <li>• <math>Mn^{2+}</math></li> <li>• <math>Na^+</math></li> <li>• <math>NH_4^+</math></li> <li>• <math>Ni^{2+}</math></li> </ul> </li> <li>□ Chemical tests for anions               <ul style="list-style-type: none"> <li>• Carbonate (<math>CO_3^{2-}</math>)</li> <li>• Chloride (<math>Cl^-</math>)</li> <li>• Hydroxide (<math>OH^-</math>)</li> <li>• Iodide (<math>I^-</math>)</li> <li>• Nitrate (<math>NO_3^-</math>)</li> <li>• Nitrite (<math>NO_2^-</math>)</li> <li>• Phosphate (<math>PO_4^{3-}</math>)</li> <li>• Sulfate (<math>SO_4^{2-}</math>)</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li>□ How to perform qualitative analysis for the presence (and absence) of the listed anions and cations</li> <li>□ Common errors, risks and hazards associated with tests available in schools</li> <li>□ The advantages and disadvantages of these tests for diagnosis in biomedical sciences, including sensitivity</li> <li>□ Potential diseases or disorders indicated by abnormal presence or absence of anions and cations in blood and urine</li> </ul> |                            | <b>HSW3</b><br><b>HSW6</b> |
| <b>3.2.2 Alternative techniques using instrumentation</b> <ul style="list-style-type: none"> <li>□ Inductively coupled plasma mass spectrometry (ICP-MS)</li> <li>□ Atomic emission spectroscopy (AES)</li> <li>□ Atomic absorption spectroscopy (AAS)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | To include: <ul style="list-style-type: none"> <li>□ The principles of each instrumental technique and their use to identify ions</li> <li>□ The appropriateness of each technique for different types of material</li> <li>□ The advantages and disadvantages of each technique for diagnosis in biomedical sciences, including sensitivity</li> </ul>                                                                                                                                                         | <b>M1.1</b><br><b>M1.5</b> | <b>HSW8</b>                |
| <b>3.3 Qualitative tests for organic compounds</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>               | <b>HSW</b>                 |
| <b>3.3.1 Chemical tests for organic compounds</b> <ul style="list-style-type: none"> <li>□ Fehling's test for aldehydes</li> <li>□ Benedict's test for sugars</li> <li>□ Emulsion test for lipids</li> <li>□ Sudan III test for lipids</li> <li>□ Biuret test for proteins</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | To include: <ul style="list-style-type: none"> <li>□ How to perform qualitative analysis for the presence of biological organic compounds</li> <li>□ Common errors, risks and hazards associated with tests available in schools</li> <li>□ The advantages and disadvantages of these tests for diagnosis in biomedical sciences, including sensitivity</li> <li>□ Potential diseases or disorders indicated by the abnormal presence or absence of organic compounds in blood or urine</li> </ul>              | <b>M0.4</b><br><b>M3.1</b> | <b>HSW3</b><br><b>HSW5</b> |

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| <b>3.3.2 Alternative techniques and instrumentation</b> <ul style="list-style-type: none"> <li>□ Gas Chromatography (GC)</li> <li>□ Liquid Chromatography (LC)</li> <li>□ Mass Spectrometry (MS)</li> </ul>                                                                                                                                                                                                                                                                                                                                                         | To include: <ul style="list-style-type: none"> <li>□ The principles of each instrumental technique and their use to identify ions</li> <li>□ How these techniques can be combined to produce quantitative information</li> <li>□ The appropriateness of each technique for different types of material</li> <li>□ The advantages and disadvantages of each technique for diagnosis in biomedical sciences, including resolution power and sensitivity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           | <b>M0.2</b><br><b>M0.5</b> | <b>HSW4</b>                 |
| <b>3.4 Separating Techniques for identification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>               | <b>HSW</b>                  |
| <b>Techniques to separate biological materials</b> <ul style="list-style-type: none"> <li>□ Centrifugation</li> <li>□ Flow cytometry</li> <li>□ High Pressure Liquid Chromatography (HPLC)</li> <li>□ Paper Chromatography</li> <li>□ Thin Layer Chromatography (TLC)</li> <li>□ Electrophoresis             <ul style="list-style-type: none"> <li>• DNA</li> <li>• Protein</li> <li>• Cell</li> <li>• Ion</li> </ul> </li> <li>□ Blot             <ul style="list-style-type: none"> <li>• Northern</li> <li>• Southern</li> <li>• Western</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li>□ The principles of each separation technique and how they are performed</li> <li>□ How to carry out paper and thin layer chromatography</li> <li>□ How to use references and read chromatograms to determine the presence or absence of biological materials</li> <li>□ The use of appropriate stains in paper chromatography and TLC</li> <li>□ The role of polymerase chain reaction (PCR) in DNA electrophoresis</li> <li>□ The appropriateness of each separation technique for different types of material</li> <li>□ The advantages and disadvantages of each technique for diagnosis in biomedical sciences, including resolution power</li> </ul><br>Does not include: <ul style="list-style-type: none"> <li>□ Detailed knowledge of PCR and cell lysis procedures</li> </ul> | <b>M2.1</b><br><b>M2.3</b> | <b>HSW9</b><br><b>HSW11</b> |
| <b>3.5 Quantitative analysis of a substance in solution</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>               | <b>HSW</b>                  |
| <b>3.5.1 Titration</b> <ul style="list-style-type: none"> <li>□ Volumetric analysis</li> <li>□ Indicator selection</li> <li>□ Alternative instrumentation for titration             <ul style="list-style-type: none"> <li>• Thermometer</li> <li>• pH meter</li> <li>• Autotitrators</li> </ul> </li> </ul>                                                                                                                                                                                                                                                        | To include: <ul style="list-style-type: none"> <li>□ How to carry out different types of titration to determine concentration</li> <li>□ How to calculate concentrations determined by titration</li> <li>□ How to identify and prepare the appropriate standard solution to use in a titration</li> <li>□ How to select the correct indicator for a titration</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>M0.5</b>                | <b>HSW3</b>                 |

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|                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>□ How to select the correct type of titration to carry out</li> <li>□ The suitability of different types of equipment in a titration to produce accurate results, and their uncertainties</li> <li>□ Common errors, risks and hazards associated with techniques available in schools</li> <li>□ How to use instrumentation in titration:               <ul style="list-style-type: none"> <li>• Thermometer for thermometric titration</li> <li>• pH meter for monitoring pH change</li> <li>• Autotitrators</li> </ul> </li> <li>□ The advantages and disadvantages of each method to determine the concentration of biological molecules, including sensitivity</li> </ul> |                            |             |
| <b>3.5.2 Colorimetry and Spectrophotometry</b> <ul style="list-style-type: none"> <li>□ Blanks and calibration curves</li> <li>□ Wavelength selection</li> <li>□ Serial dilutions</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ How to use a colorimeter and spectrophotometer to determine the concentration of biological molecules</li> <li>□ Types of biological molecules analysed using these methods</li> <li>□ How to select and prepare appropriate blanks to use for calibration and create calibration curves</li> <li>□ How to select the appropriate wavelength for analysing different types of materials</li> <li>□ Common errors, risks and hazards associated with techniques available in schools</li> <li>□ The advantages and disadvantages of each technique to determine the concentration of biological molecules, including sensitivity</li> </ul>                      | <b>M3.2</b><br><b>M3.3</b> | <b>HSW6</b> |
| <b>3.5.3 Biosensors</b>                                                                                                                                                                      | To include: <ul style="list-style-type: none"> <li>□ How biosensors are used to determine the presence and concentration of biological molecules</li> <li>□ Types of biological material analysed using biosensors</li> <li>□ How to select the most appropriate biosensor to use for different biological materials</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      | <b>M1.5</b>                | <b>HSW4</b> |

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|                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>□ The advantages and disadvantages of using biosensors to determine the presence and concentration of biological material, including sensitivity</li> <li>□ Potential diseases or disorders that can be diagnosed using biosensors</li> </ul>                                                                                                                                                                                                                                                                                                                   |                                |              |
| <b>Topic Area 4: Planning a clinical investigation</b>                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                              | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Opportunities to cover:</b> |              |
| <b>4.1 Understanding clinical conditions</b>                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <b>Symptoms and reference values</b>                                                                                                                                                                                                                 | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to carry out research to identify a range of potential diseases and disorders based on a patient's symptoms</li> <li>□ The importance of using reliable sources of information</li> <li>□ How to select the most likely diseases or disorders for a patient by taking into account their medical history</li> <li>□ How to carry out research to find reference values for the tests that are used by biomedical scientists</li> <li>□ How to select appropriate reference values to use that are appropriate for a patient</li> </ul> | <b>M1.3</b>                    | <b>HSW11</b> |
| <b>4.2 Creating a method for an investigation</b>                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                   | <b>HSW</b>   |
| <b>4.2.1 Generating a hypothesis</b>                                                                                                                                                                                                                 | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to write a hypothesis and null hypothesis about a patient's diagnosis based on research</li> <li>□ How to explain the hypothesis using scientific knowledge and details acquired through research</li> <li>□ How to accept or reject a hypothesis</li> </ul>                                                                                                                                                                                                                                                                           |                                | <b>HSW6</b>  |
| <b>4.2.2 Producing a method</b> <ul style="list-style-type: none"> <li>□ A method includes decisions about: <ul style="list-style-type: none"> <li>• Variables</li> <li>• Method</li> <li>• Equipment</li> <li>• Measurements</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to choose appropriate tests and techniques to qualitatively accept or reject a null hypothesis</li> <li>□ Why there are limitations for the types of investigations that can be carried out in schools</li> <li>□ How to justify the choice of tests and techniques appropriate for diagnosis</li> <li>□ The difference between independent, dependent and control variables</li> <li>□ How to identify significant variables to control in an investigation</li> </ul>                                                                | <b>M1.5</b>                    | <b>HSW3</b>  |

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|                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>□ How to decide what values to select for the relevant variables in the investigation</li> <li>□ How data of sufficient quality can be collected through equipment choice</li> <li>□ How to determine the uncertainty associated with different measuring equipment and reduce uncertainty</li> <li>□ How to calibrate equipment to reduce errors</li> </ul>                                                                                                                                                                                                                                                                                                |                                           |             |
| <b>4.2.3 Safe handling of specimens</b>                                                                                                                                                                                                   | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to create and maintain a sterile environment when carrying out diagnostic tests and techniques</li> <li>□ How to plan to carry out diagnostic tests and technique that reduces contamination</li> <li>□ How to handle specimens to reduce the risk of false positive and negatives</li> <li>□ How to maintain the integrity of samples used in investigations</li> <li>□ How to safely dispose of different types of specimen</li> </ul>                                                                                                                                                                                           |                                           | <b>HSW5</b> |
| <b>4.2.4 Risk assessment</b> <ul style="list-style-type: none"> <li>□ Identifying hazardous equipment, chemicals, biological hazards and procedures</li> <li>□ Risks</li> <li>□ Control measures</li> <li>□ Emergency measures</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to complete a risk assessment</li> <li>□ How to differentiate between a hazard and risk</li> <li>□ How to identify appropriate risks and hazards for an investigation</li> <li>□ Hazard symbols and what they represent</li> <li>□ How to select and interpret relevant information from chemical safety data sheets</li> <li>□ How to explain control measures using scientific principles</li> <li>□ Why it is important to be aware of emergency measures before carrying out an investigation</li> <li>□ Why it is important to work safely and with due care and attention in a scientific practical investigation</li> </ul> | <b>M3.1</b>                               | <b>HSW4</b> |
| <b>4.3 Performing a scientific investigation</b>                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Maths</b>                              | <b>HSW</b>  |
| <ul style="list-style-type: none"> <li>□ Types of data available in practical investigations: <ul style="list-style-type: none"> <li>• Qualitative and quantitative data</li> <li>• Continuous and discrete data</li> </ul> </li> </ul>   | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Key features of each type of data</li> <li>□ Appropriate units and conventions for each type of data</li> <li>□ The importance of recording all relevant forms of data</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>M0.1</b><br><b>M0.2</b><br><b>M1.1</b> | <b>HSW3</b> |

|                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                           |                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------|
| <ul style="list-style-type: none"> <li>• Data from observations and measurements (including repeats)</li> </ul> <p>□ Recording data in:</p> <ul style="list-style-type: none"> <li>• Diagrams, images, and video</li> <li>• Results tables</li> <li>• Spreadsheets</li> <li>• Dataloggers</li> </ul> | <p>□ How to select a format for recording data that suits the data being collected</p> <p>□ Use of appropriate column headings and units</p> <p>□ Use of appropriate levels of precision</p>                                                                                                                                                                                                                            |                                           |                            |
| <b>Topic Area 5: Report writing</b>                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                           |                            |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                              | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Opportunities to cover:</b>            |                            |
| <b>5.1 Analysis of data</b>                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Maths</b>                              | <b>HSW</b>                 |
| <b>5.1.1</b> <p>□ Using mathematical skills from Mathematical Skills for Human Biology to analyse data in investigations</p> <ul style="list-style-type: none"> <li>• Processing data</li> <li>• Using graphical techniques to analyse data</li> </ul>                                               | <p>To include:</p> <p>□ How to select which mathematical skills are appropriate to use</p> <p>□ The value of processing raw data for analysis</p> <p>□ How to use appropriate mathematical skills</p> <p>□ How to propagate uncertainties to determine total uncertainty</p> <p>□ How to determine if data is valid</p>                                                                                                 | <b>M1.2</b><br><b>M3.1</b><br><b>M3.2</b> | <b>HSW6</b>                |
| <b>5.1.2</b> <p>□ Types of errors:</p> <ul style="list-style-type: none"> <li>• Measurement</li> <li>• Systematic</li> </ul> <p>□ Anomalous data</p>                                                                                                                                                 | <p>To include:</p> <p>□ Definitions of measurement and systematic error</p> <p>□ How to identify each type of error in an investigation</p> <p>□ How to explain reasons for errors</p> <p>□ How to identify anomalous data in tables and graphs</p> <p>□ Causes and effects of anomalous data</p> <p>□ How to account for anomalous data</p>                                                                            | <b>M1.8</b>                               | <b>HSW2</b>                |
| <b>5.2 Drawing conclusions</b>                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Maths</b>                              | <b>HSW</b>                 |
| <b>Conclusions from data:</b> <p>□ Comparing results to established reference values (secondary data)</p> <p>□ Confidence in conclusions</p> <p>□ Answering the research question</p>                                                                                                                | <p>To include:</p> <p>□ How to write a concise conclusion(s) from primary data and justify the conclusion</p> <p>□ How to select appropriate data from secondary sources to compare results to</p> <p>□ How to make valid comparisons between primary and secondary data</p> <p>□ What is meant by confidence in conclusions for an investigation</p> <p>□ How to explain the impact of limitations on a conclusion</p> |                                           | <b>HSW1</b><br><b>HSW6</b> |

|                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|
|                                                                                                                                                                                                                 | <input type="checkbox"/> How to address the extent to which the hypothesis can be accepted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |             |
| <b>5.3 Evaluating results</b>                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Maths</b> | <b>HSW</b>  |
| <input type="checkbox"/> Evaluating the investigation <ul style="list-style-type: none"> <li>• Equipment</li> <li>• Methods</li> <li>• Outcomes</li> <li>• Sources of information and secondary data</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How to assess the effectiveness of the methods used to collect data</li> <li><input type="checkbox"/> How to explain the limitations and sources of error in collected data</li> <li><input type="checkbox"/> How to determine the reliability of secondary data used in the investigation</li> <li><input type="checkbox"/> How to suggest improvements for an investigation, considering both the techniques used and those that would be available to a biomedical scientist</li> <li><input type="checkbox"/> How to decide if the improvements are appropriate and what impact they will have</li> </ul> | <b>M1.7</b>  | <b>HSW6</b> |

### Assessment criteria

The table below gives the assessment criteria for the tasks in the set assignment for this unit. The assessment criteria indicate what is required in these tasks.

This qualification has a compensatory approach. This means that the unit grade awarded is based on the **total** number of achieved criteria for the unit (see [Section 6.4](#)). Students do **not** have to achieve **all** criteria for a specific grade to achieve that unit grade (e.g. achieve all pass criteria to achieve a pass grade).

[Section 7.4](#) provides full information on how to assess the NEA units and apply the assessment criteria. Students' work must show that all aspects of a criterion have been met in sufficient detail for it to be **successfully achieved** (see [Section 7.4.1](#)). If a student's work does not fully meet a criterion, you must not award that criterion.

The command words used in the assessment criteria are defined in [Appendix B](#).

| Pass                                                                                                                                                                                        | Merit                                                                                                                                 | Distinction                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>P1:</b> Use research to <b>identify</b> a range of potential diseases that the patients might have.                                                                                      | <b>M1:</b> <b>Assess</b> two suspected diseases for <b>each</b> patient in terms of potential likelihood given the symptoms.          |                                                                                                    |
| <b>P2:</b> <b>Create</b> a method for the investigation including the appropriate tests and techniques to investigate the unidentified samples based on suspected diseases of the patients. | <b>M2:</b> <b>Explain</b> the rationale for the tests and techniques chosen based on the suspected diseases identified in <b>M1</b> . | <b>D1:</b> <b>Justify</b> the choice of appropriate equipment for the chosen tests and techniques. |
| <b>P3:</b> <b>Complete</b> an appropriate risk assessment for your investigation.                                                                                                           |                                                                                                                                       |                                                                                                    |



| Pass                                                                                                                           | Merit                                                                                                                      | Distinction                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>P4: Perform</b> the planned investigation safely.                                                                           | <b>M3: Explain</b> how control variables have been managed when undertaking the investigation.                             | <b>D2: Collect</b> sufficient, valid data for all samples with appropriate precision. |
| <b>P5: Explain</b> how the integrity of the samples is maintained.                                                             |                                                                                                                            |                                                                                       |
| <b>P6: Record</b> the data obtained in appropriate ways using correct conventions and units.                                   |                                                                                                                            |                                                                                       |
| <b>P7:</b> Use standard mathematical techniques to <b>process</b> data.                                                        | <b>M4: Calculate</b> percentage uncertainties for the investigation.                                                       | <b>D3: Explain</b> the sources of error and possible reasons for any anomalous data.  |
| <b>P8:</b> Use research to <b>compare</b> your data with established value ranges.                                             | <b>M5: Justify</b> which patient each sample belongs to.                                                                   | <b>D4: Justify</b> which disease each patient has.                                    |
| <b>P9: Analyse</b> the results of the investigation in the context of the suspected diseases for the patients from <b>M1</b> . |                                                                                                                            |                                                                                       |
| <b>P10: Explain</b> the limitations of the data collected.                                                                     | <b>M6: Evaluate</b> the sources of information researched in <b>Task 1</b> and established value ranges in <b>Task 3</b> . | <b>D5: Justify</b> suggestions for any improvements that could be made.               |
| <b>P11: Suggest</b> other tests or techniques that could be undertaken to support the diagnosis suggested for the patients.    | <b>M7: Analyse</b> the strengths of the investigation.                                                                     |                                                                                       |
| <b>P12: Assess</b> the effectiveness of the methods used to collect data.                                                      |                                                                                                                            |                                                                                       |

## Assessment guidance

This assessment guidance gives you information relating to the assessment criteria. There might not be additional assessment guidance for each assessment criterion. It is included only where it is needed.

| Assessment Criteria | Assessment guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1                  | <ul style="list-style-type: none"> <li>Teachers must discuss with students the research they completed independently to inform their research question, giving students the opportunity to say:               <ul style="list-style-type: none"> <li>What research they completed</li> <li>How they completed it</li> <li>Why they used the research methods they did.</li> </ul> </li> <li>Students must use research to identify a range of potential diseases that each patient might have, based on their symptoms.</li> <li>Students must identify <b>at least four</b> potential diseases that the patients might have.</li> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul> |
| P2                  | <ul style="list-style-type: none"> <li>Students must provide a step-by-step method for their investigation. It needs to include all the equipment they wish to use, including size, quantities and PPE, as appropriate.</li> <li>Students should consider the tests and techniques available to them, practical equipment available to them, samples provided and information from <b>P1</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P3                  | <ul style="list-style-type: none"> <li>Students must use the risk assessment template provided to complete a risk assessment for their investigation, considering risks and hazards for each test and technique.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| M1                  | <ul style="list-style-type: none"> <li><b>M1</b> is an extension of <b>P1</b>.</li> <li>Students must give a reasoned judgement for why <b>two</b> diseases are suspected for <b>each</b> patient, in terms of the likelihood given the symptoms.</li> <li>Students must include a hypothesis for the suspected diseases for <b>each</b> patient.</li> <li>The reasoned judgement is informed by relevant facts based on the symptoms given and research completed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |
| M2                  | <ul style="list-style-type: none"> <li><b>M2</b> is an extension of <b>P2</b> and <b>M1</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| D1                  | <ul style="list-style-type: none"> <li><b>D1</b> is an extension of <b>M2</b>.</li> <li>Students might justify the settings of their equipment as part of the choice for the tests and techniques.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P4                  | <ul style="list-style-type: none"> <li>Students must follow their method safely.</li> <li>Students must be able to perform the task safely to achieve this criterion. Staff must intervene if safe working practices are not being followed but where this happens the criteria cannot be awarded.</li> <li>Teachers must complete a 'Teacher Observation Record' for each student to evidence they have met this criterion. Students must also read and sign it.</li> <li>The teacher observation record form should describe how the student performed the planned investigation safely.</li> </ul>                                                                                                                                                                                                                             |
| P6                  | <ul style="list-style-type: none"> <li>A results table may be appropriate for most investigations, but qualitative descriptions are also suitable.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| D2                  | <ul style="list-style-type: none"> <li>The teacher observation record form could comment on the skilful use of apparatus and the accuracy and precision of data collected.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| P7  | <ul style="list-style-type: none"> <li>Students must use mathematical skills identified in Appendix D of the specification to process their data appropriately.</li> <li>Students must show <b>at least one</b> example of their working out in the written evidence.</li> </ul>                                                                                                    |
| P8  | <ul style="list-style-type: none"> <li>Students must use research to determine the correct established value ranges to compare with their data.</li> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul>                                 |
| M4  | <ul style="list-style-type: none"> <li>Students must determine the percentage uncertainty on each piece of equipment used and the combined uncertainty for each repeat.</li> <li>They must show <b>at least one</b> example of their working out in the written evidence.</li> </ul>                                                                                                |
| M5  | <ul style="list-style-type: none"> <li><b>M5</b> is an extension of <b>P9</b>.</li> </ul>                                                                                                                                                                                                                                                                                           |
| D3  | <ul style="list-style-type: none"> <li>This should be done qualitatively only.</li> <li>Students who have no anomalous data to explain should clarify this in their written evidence.</li> </ul>                                                                                                                                                                                    |
| D4  | <ul style="list-style-type: none"> <li><b>D4</b> is an extension of <b>M5</b>.</li> </ul>                                                                                                                                                                                                                                                                                           |
| P12 | <ul style="list-style-type: none"> <li>Students must offer a reasoned judgement of the effectiveness of the methods used to collect data.</li> <li>Students will inform their judgement with relevant information about how well they were able to collect good quality data with the techniques and equipment chosen during the investigation.</li> </ul>                          |
| M6  | <ul style="list-style-type: none"> <li>Students must make reasoned judgements on their confidence in the sources used throughout the investigation, e.g. those used to design the method, create the risk assessment, establish value ranges and the secondary data, with reference to reliability and validity.</li> </ul>                                                         |
| D5  | <ul style="list-style-type: none"> <li>Give valid reasons for improvements to the investigation that would improve the conclusion(s) or help answer the research question.</li> <li>Processed data should be used to support any recommendations.</li> <li>If no improvements can be recommended, then this needs to be justified using evidence from the investigation.</li> </ul> |

### Synoptic assessment

Some of the knowledge, understanding and skills needed to complete this unit will draw on the learning in Unit F170.

This table details these synoptic links.

| Unit F173: Biomedical techniques |                                             | Unit F170: Fundamentals of human biology |                         |
|----------------------------------|---------------------------------------------|------------------------------------------|-------------------------|
| Topic Area                       |                                             | Topic Area                               |                         |
| 2                                | Diagnostic techniques: cells and microscopy | 1                                        | Human cells and tissues |
|                                  |                                             | 4                                        | Basics of microbiology  |

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

### 5.3.3 Unit F174: Nutrition and metabolism

#### Unit Aim

Good nutrition is vital for the healthy functioning of the human body. The wrong balance of nutrients in the body's cells can lead to different disorders and long-term effects. When considering 'good nutrition', it's important to understand that different groups of people have different dietary requirements. This unit considers the dietary requirements for specific groups and includes the processes of digestion, absorption and assimilation, the long-term effects of poor diet; and the consequences of being unable to incorporate nutrients into body cells. The unit also explores different metabolic pathways involving nutrients vital to maintaining body functions. The unit is completed by considering the control mechanisms that regulate certain nutrients in the body and how disorders can be diagnosed, monitored, and treated.

In this unit, you will learn to identify biomolecules required for the maintenance of a healthy body and learn how food labels provide a guide for recommended daily intake. You will learn about the dietary needs of different individuals and the health issues associated with poor diet. You will also study the challenging topics of metabolic pathways and how hormones control not only the levels of certain nutrients in the body but also how they are involved with hunger. Finally, you will learn how to be able to research some of the techniques used to diagnose, monitor and treat some of the conditions associated with nutritional disorders.

| Unit F174: Nutrition and metabolism                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------|
| Topic Area 1: Nutrients required for a healthy body                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |              |
| Teaching content                                                                                                                                                                       | Exemplification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Opportunities to cover:    |              |
| 1.1 Macronutrients- major food groups                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Maths                      | HSW          |
| <b>Carbohydrates, proteins and lipids</b> <ul style="list-style-type: none"> <li>□ Sources of different macromolecules</li> <li>□ Roles of macromolecules in the human body</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ Recall that proteins are formed from amino acids</li> <li>□ Draw the general structure of an amino acid</li> <li>□ Recall that carbohydrates are formed from carbon, hydrogen and oxygen</li> <li>□ Know that starch is formed from long chains of glucose units</li> <li>□ Draw the general structure of glucose</li> <li>□ Know that a triglyceride is formed from fatty acids and a molecule of glycerol</li> <li>□ Which foods are rich in proteins, carbohydrates and lipids</li> <li>□ Why some molecules are considered essential and others non-essential</li> <li>□ Why macromolecules are required in different quantities</li> <li>□ Role of proteins, carbohydrates and lipids in maintaining healthy body</li> </ul> | <b>M1.5</b><br><b>M4.1</b> | <b>HSW12</b> |

|                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------|
|                                                                                                                                                                | <input type="checkbox"/> How macromolecule amounts may be affected by food processing and storage, including: <ul style="list-style-type: none"> <li>• Preparation (such as peeling)</li> <li>• Cooking</li> <li>• Freezing and defrosting</li> <li>• Canning</li> </ul> Does not include: <input type="checkbox"/> The detailed process of how food processing and storage affects macromolecule amounts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                           |              |
| <b>1.2 Micronutrients</b>                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Maths</b>                              | <b>HSW</b>   |
| <b>Mineral and vitamin requirements</b> <input type="checkbox"/> Main minerals and vitamins and their sources <input type="checkbox"/> Roles in the human body | To include: <input type="checkbox"/> Which foods provide different minerals and vitamins <input type="checkbox"/> Know vitamins are organic molecules with complex chemical structures <input type="checkbox"/> Know minerals are chemical elements that are required as essential nutrients by organisms <input type="checkbox"/> Why we need to obtain vitamins and minerals from food <input type="checkbox"/> Roles of vitamins and minerals in maintaining a healthy body <input type="checkbox"/> How and why foods may need to be fortified with vitamins and minerals <input type="checkbox"/> How vitamins and minerals amounts may be affected by food processing and storage, including: <ul style="list-style-type: none"> <li>• Preparation (such as peeling)</li> <li>• Cooking</li> <li>• Freezing and defrosting</li> <li>• Canning</li> </ul> Does not include: <input type="checkbox"/> The detailed process of how food processing and storage affects vitamin and mineral amounts | <b>M0.3</b><br><b>M3.1</b>                | <b>HSW10</b> |
| <b>1.3 From food to body cells</b>                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Maths</b>                              | <b>HSW</b>   |
| <b>1.3.1 Importance of digestion</b> <input type="checkbox"/> Mechanical digestion <input type="checkbox"/> Chemical digestion                                 | To include: <input type="checkbox"/> How and why we break down large food pieces to smaller pieces <input type="checkbox"/> How and why we breakdown large food molecules into smaller molecules, including: <ul style="list-style-type: none"> <li>• Hydrolysis reaction</li> </ul> <input type="checkbox"/> How problems with digestion of food can lead to malfunctions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>M0.1</b><br><b>M2.1</b><br><b>M4.1</b> | <b>HSW9</b>  |

|                                                                                                                                                                                                                                                           | <input type="checkbox"/> How surface area is calculated and impact of change in surface area<br><br>Does not include<br><input type="checkbox"/> Details of digestive system                                                                                                                                                                                                                                                                                                                                                                    |                               |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|
| <b>1.3.2 Importance of absorption and assimilation</b><br><input type="checkbox"/> How the body gets nutrients from digestive system into the blood stream<br><input type="checkbox"/> How the body incorporates nutrients into cells, tissues and organs | To include:<br><input type="checkbox"/> Adaptations of small intestine<br><input type="checkbox"/> Role of structures in the small intestine, including villi<br><input type="checkbox"/> How nutrients become parts of cells such as amino acids being made into new proteins in the cell<br><br>Does not include:<br><input type="checkbox"/> Mechanism of absorption<br><input type="checkbox"/> Details of digestive system other than intestinal wall<br><input type="checkbox"/> Details of the reactions involved in metabolism in liver |                               | <b>HSW1</b>  |
| <b>Topic Area 2: Diets and disorders</b>                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                               |              |
| Teaching content                                                                                                                                                                                                                                          | Exemplification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Opportunities to cover:       |              |
| <b>2.1 Dietary requirements</b>                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                  | <b>HSW</b>   |
| <b>2.1.1 Dietary reference values (DRVs)</b><br><input type="checkbox"/> Balanced diet<br><input type="checkbox"/> Recommended daily intake<br><input type="checkbox"/> Safe intakes of minerals and vitamins                                             | To include:<br><input type="checkbox"/> Why a balanced diet is needed for an adequate intake of nutrients for maintaining health<br><input type="checkbox"/> How DRVs may change dependent on age, gender, activity, pregnancy and lactation<br><input type="checkbox"/> Calculations to include percentage increases/decreases in nutrients and differences in DRVs                                                                                                                                                                            | <b>M0.4<br/>M1.8<br/>M3.1</b> | <b>HSW11</b> |
| <b>2.1.2 Food labels</b><br><input type="checkbox"/> Guidance offered by food labels                                                                                                                                                                      | To include:<br><input type="checkbox"/> What guidance is offered by food labels with regards to nutritional values<br><input type="checkbox"/> How the red, amber, green system is used<br>Calculations to include converting actual mass of nutrients into percentages, for example in a 150 g can                                                                                                                                                                                                                                             | <b>M0.3<br/>M0.4</b>          | <b>HSW5</b>  |
| <b>2.2 Malnutrition</b>                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                  | <b>HSW</b>   |
| <b>2.2.1 Diet-related nutrient deficiencies</b><br><input type="checkbox"/> Problems caused by lack of macronutrients<br><input type="checkbox"/> Problems caused by mineral and vitamin deficiencies                                                     | To include:<br><input type="checkbox"/> How deficiencies and unbalanced diets may lead to malfunction and disease including symptoms associated with: <ul style="list-style-type: none"> <li>• Starvation</li> <li>• Kwashiorkor</li> <li>• Rickets</li> <li>• Gum disease (and scurvy)</li> </ul>                                                                                                                                                                                                                                              |                               | <b>HSW6</b>  |

|                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------|
|                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Night blindness</li> <li>• Spina bifida</li> <li>• Anaemia</li> </ul>                                                                                                                                                                                                                                                                                                                                               |                                |              |
| <b>2.2.2 Malabsorption and allergies</b> <ul style="list-style-type: none"> <li>□ Inability to incorporate nutrients into the body</li> <li>□ Food allergies</li> </ul>                | To include: <ul style="list-style-type: none"> <li>□ How inability to digest or absorb nutrients may lead to disorders</li> <li>□ Causes and symptoms of disorders associated with malabsorption or food allergy including lactose intolerance, coeliac disease and anaphylactic shock</li> </ul>                                                                                                                                                            | <b>M1.6</b>                    | <b>HSW11</b> |
| <b>2.2.3 Nutrients in excess</b> <ul style="list-style-type: none"> <li>□ Metabolic disorders</li> <li>□ Excess intake</li> </ul>                                                      | To include: <ul style="list-style-type: none"> <li>□ Why excess nutrients may result in metabolic disorders</li> <li>□ Causes and symptoms of disorders</li> </ul> <p>Examples of <b>causes and symptoms of disorders</b> may include:</p> <ul style="list-style-type: none"> <li>□ Phenylketonuria</li> <li>□ Diabetes</li> <li>□ Obesity</li> <li>□ Non-alcoholic fatty liver disease</li> <li>□ Coronary heart disease</li> <li>□ Hypertension</li> </ul> | <b>M0.1</b>                    | <b>HSW5</b>  |
| <b>Topic Area 3: Metabolic pathways and control mechanisms</b>                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                |              |
| <b>Teaching content</b>                                                                                                                                                                | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Opportunities to cover:</b> |              |
| <b>3.1 Metabolic pathways</b>                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Maths</b>                   | <b>HSW</b>   |
| <b>3.1.1 Macromolecules in metabolism</b> <ul style="list-style-type: none"> <li>□ Use of macromolecules in metabolism</li> </ul>                                                      | To include: <ul style="list-style-type: none"> <li>□ How different macromolecules release different amounts of energy</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Detail of reactions or ATP breakdown</li> </ul>                                                                                                                                                                                                           | <b>M0.2</b><br><b>M0.5</b>     |              |
| <b>3.1.2 Metabolic rates</b> <ul style="list-style-type: none"> <li>□ Metabolic requirements for energy</li> <li>□ Comparison of metabolic rates</li> </ul>                            | To include: <ul style="list-style-type: none"> <li>□ Why individuals may have different metabolic rates</li> <li>□ How metabolic rates can be calculated</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Detail of pathways or reactions</li> </ul>                                                                                                                                                                             | <b>M2.3</b><br><b>M2.4</b>     |              |
| <b>3.2 Role of the liver in metabolism</b>                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Maths</b>                   | <b>HSW</b>   |
| <b>3.2.1 Metabolism of macromolecules in liver</b> <ul style="list-style-type: none"> <li>□ Carbohydrate metabolism</li> <li>□ Protein metabolism</li> <li>□ Fat metabolism</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ Role of the liver in maintaining blood glucose levels</li> <li>□ Deamination and transamination of amino acids</li> <li>□ Fatty acid synthesis</li> <li>□ Cholesterol</li> </ul>                                                                                                                                                                                                                        |                                | <b>HSW10</b> |

|                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                           |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------|
|                                                                                                                                                                                   | Does not include:<br><input type="checkbox"/> Detail of reactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                           |              |
| <b>3.2.2 Storage of nutrients</b><br><input type="checkbox"/> Carbohydrate store<br><input type="checkbox"/> Vitamin and mineral store                                            | To include:<br><input type="checkbox"/> Glycogen store<br><input type="checkbox"/> Stores fat-soluble vitamins and minerals<br><br>Examples of <b>fat-soluble vitamins and minerals</b> may include:<br><input type="checkbox"/> Vitamin A<br><input type="checkbox"/> Iron<br><br>Does not include:<br><input type="checkbox"/> Details of reactions or metabolic pathways                                                                                                                                                                                                                                   |                                           | <b>HSW10</b> |
| <b>3.2.3 Detoxification</b><br><input type="checkbox"/> Ammonia<br><input type="checkbox"/> Drugs<br><input type="checkbox"/> Alcohol<br><input type="checkbox"/> Bile production | To include:<br><input type="checkbox"/> How the liver deals with toxins in the diet and waste products of metabolism<br><br>Examples of <b>how the liver deals with toxins</b> include:<br><input type="checkbox"/> Removal of toxins, for example, alcohol<br><input type="checkbox"/> Removal of ammonia<br><input type="checkbox"/> Conversion of medicinal drugs into non-toxic products<br><input type="checkbox"/> Removal of worn out and damaged red blood cells<br><br>Does not include:<br><input type="checkbox"/> Details of reactions<br><input type="checkbox"/> Details of excretion by kidney |                                           | <b>HSW11</b> |
| <b>3.3 Control mechanisms for nutrient metabolism</b>                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>                              | <b>HSW</b>   |
| <b>3.3.1 Regulation of food intake</b><br><input type="checkbox"/> Role of hormones in control of hunger                                                                          | To include:<br><input type="checkbox"/> How hormones leptin and the 'hunger' hormone ghrelin control appetite<br><input type="checkbox"/> Why changes to normal levels of these hormones may affect health<br><input type="checkbox"/> How hormone levels are determined including an evaluation as to accuracy of results<br><br>Does not include:<br><input type="checkbox"/> No details of homeostatic mechanism required                                                                                                                                                                                  | <b>M3.1</b><br><b>M3.2</b><br><b>M3.3</b> | <b>HSW8</b>  |
| <b>3.3.2 Regulation of blood glucose</b><br><input type="checkbox"/> Role of hormones in control of blood glucose                                                                 | To include:<br><input type="checkbox"/> How a negative feedback mechanism results in normal blood glucose levels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>M1.5</b><br><b>M1.6</b>                | <b>HSW6</b>  |



|                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------|
|                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>□ Why changes to normal levels of these hormones may affect health</li> <li>□ How hormone levels are determined including an evaluation as to accuracy of results</li> </ul>                                                                                                                                                                                                                                                                                                                                                                |                                |              |
| <b>3.3.3 Osmoregulation</b> <ul style="list-style-type: none"> <li>□ Regulation of salt intake</li> <li>□ Importance of the kidneys in maintaining water potential of the blood</li> </ul>                                       | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Why sodium chloride (salt) and water potential needs to be controlled</li> <li>□ How changes in salt intake can affect health</li> <li>□ Why salt intake and water potential differs depending on activity and lifestyle</li> <li>□ How to use calculations involving secondary data to compare salt levels of individuals to normal levels</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Kidney structure</li> <li>□ Other kidney functions</li> <li>□ Mechanism of osmosis</li> </ul> | <b>M3.4</b><br><b>M3.6</b>     | <b>HSW3</b>  |
| <b>Topic Area 4: Diagnosis, monitoring and treatment for nutritional/metabolic disorders</b>                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                          | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Opportunities to cover:</b> |              |
| <b>4.1 Diagnostic techniques</b>                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Maths</b>                   | <b>HSW</b>   |
| <b>4.1.1 Clinical assessments</b> <ul style="list-style-type: none"> <li>□ Data collection</li> </ul>                                                                                                                            | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Roles of health care staff in obtaining patient information</li> <li>□ How different professionals have different roles to play in gathering information and monitoring individuals</li> </ul> <p>For example the <b>roles to play in gathering information</b> with regards to:</p> <ul style="list-style-type: none"> <li>□ Lifestyle</li> <li>□ Family history</li> <li>□ Symptoms</li> <li>□ Dietary information</li> <li>□ Use of surveys</li> </ul>                                                              | <b>M1.2</b><br><b>M1.6</b>     | <b>HSW8</b>  |
| <b>4.1.2 Use of scanning techniques</b> <ul style="list-style-type: none"> <li>□ Endoscopy</li> <li>□ Ultrasound</li> <li>□ Magnetic resonance imaging (MRI)</li> <li>□ X-ray</li> <li>□ Computerised tomography (CT)</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Advantages and disadvantages of scanning techniques in diagnosing and monitoring gastrointestinal disorders associated with nutritional problems</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |                                | <b>HSW11</b> |

| <b>4.2 Monitoring</b>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>               | <b>HSW</b>  |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------|
| <b>4.2.1 Use of body mass index (BMI) and growth charts</b>    | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Why individuals need to be monitored</li> <li><input type="checkbox"/> How BMI is calculated</li> <li><input type="checkbox"/> Why average BMI charts are used for comparison</li> <li><input type="checkbox"/> How growth charts and percentiles for children are used for comparison</li> <li><input type="checkbox"/> Advantages and disadvantages of using BMI and growth charts</li> </ul>                                                                                                                                                                         | <b>M2.1</b><br><b>M2.4</b> | <b>HSW3</b> |
| <b>4.2.2 Biomarkers</b>                                        | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Overview of the techniques used to monitor these biomarkers, for example, <ul style="list-style-type: none"> <li>• Blood tests</li> <li>• Urine tests</li> <li>• Saliva tests</li> <li>• Tissue biopsies</li> </ul> </li> <li><input type="checkbox"/> Advantages and disadvantages of techniques to monitor biomarkers</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Details of chemical reactions involved</li> <li><input type="checkbox"/> Details of exactly how monitoring tests are carried out</li> </ul> |                            | <b>HSW9</b> |
| <b>4.2.3 Biosensors and monitors</b>                           | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How these allow self-monitoring and targeted measurement of nutrients, for example, glucose</li> <li><input type="checkbox"/> Overview of how a biosensor is used to measure blood glucose</li> <li><input type="checkbox"/> Advantages and disadvantages of biosensors and monitors</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Details of chemical reactions involved</li> </ul>                                                                                                                              | <b>M1.5</b>                | <b>HSW4</b> |
| <b>4.3 Treatments and health care</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>               | <b>HSW</b>  |
| <b>4.3.1 Types of treatment and medical interventions for:</b> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How having a healthier, more balanced diet prevents malnutrition</li> <li><input type="checkbox"/> Why different types of diabetes have different methods for treatment and monitoring</li> <li><input type="checkbox"/> How lifestyle changes can be part of treatment and diet plans for obesity and non-alcoholic fatty liver disease</li> </ul>                                                                                                                                                                                                                     | <b>M0.2</b><br><b>M1.8</b> | <b>HSW8</b> |

|                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |              |
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|                                                                                                                                                                                                                                                       | <input type="checkbox"/> How medication is used to reduce cholesterol and bariatric surgery are used for treating certain individuals                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |              |
| <b>4.3.2 Role of governments and health/social care providers</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Clinics</li> <li><input type="checkbox"/> Support groups</li> <li><input type="checkbox"/> Food agencies</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Why specialist clinics and nurses specific to each disorder are important</li> <li><input type="checkbox"/> How support groups such as weight loss groups can help individuals with treatment and diet plans</li> <li><input type="checkbox"/> The role of health care professionals in providing education, advice and offering routine check ups</li> <li><input type="checkbox"/> The importance of communication between professionals when developing food strategies and diet plans for individuals</li> </ul> |  | <b>HSW11</b> |
| <b>4.3.3 Complementary therapies</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Alternative practices to support health and healing</li> </ul>                                                                                   | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Advantages and disadvantages of:               <ul style="list-style-type: none"> <li>• Therapies to promote well-being</li> <li>• Alternative methods</li> </ul> </li> </ul> Examples of <b>alternative methods</b> include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Hypnotherapy</li> <li><input type="checkbox"/> Meditation</li> <li><input type="checkbox"/> Counselling</li> </ul>                                                                                                     |  | <b>HSW10</b> |

### Assessment criteria

The table below gives the assessment criteria for the tasks in the set assignment for this unit. The assessment criteria indicate what is required in these tasks.

This qualification has a compensatory approach. This means that the unit grade awarded is based on the **total** number of achieved criteria for the unit (see [Section 6.4](#)). Students do **not** have to achieve **all** criteria for a specific grade to achieve that unit grade (e.g. achieve all pass criteria to achieve a pass grade).

[Section 7.4](#) provides full information on how to assess the NEA units and apply the assessment criteria. Students' work must show that all aspects of a criterion have been met in sufficient detail for it to be **successfully achieved** (see [Section 7.4.1](#)). If a student's work does not fully meet a criterion, you must not award that criterion.

The command words used in the assessment criteria are defined in [Appendix B](#).

| Pass                                                                                                                                       | Merit                                                                                                                              | Distinction                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>P1: Explain</b> why the individual requires a specialised diet.                                                                         | <b>M1:</b> Use research to <b>describe</b> the details of medical guidance given in a similar situation to that in the case study. | <b>D1: Analyse</b> the benefits of having a specialised diet for the individual's physical and mental well-being. |
| <b>P2:</b> Use research to <b>describe</b> how the macronutrient requirements for the individual varies from an average person.            | <b>M2: Explain</b> how the role of metabolism influences the creation of the specialised diet.                                     |                                                                                                                   |
| <b>P3:</b> Use research to <b>describe</b> how the micronutrient requirements for the individual varies from an average person.            |                                                                                                                                    |                                                                                                                   |
| <b>P4: Create</b> an appropriate specialised diet.                                                                                         | <b>M3: Explain</b> the potential risks and side-effects of the specialised diet for the individual.                                | <b>D2: Discuss</b> the advantages and disadvantages of the specialised diet for the individual.                   |
| <b>P5: Create</b> appropriate and customisable meal plan(s).                                                                               | <b>M4:</b> Use appropriate calculations to <b>process</b> data when creating your meal plan(s).                                    | <b>D3: Justify</b> your choice of meal plan(s) for inclusion in the specialised diet.                             |
| <b>P6: Explain</b> how the meals in the meal plan(s) need to be prepared and stored.                                                       |                                                                                                                                    |                                                                                                                   |
| <b>P7: Analyse</b> how the physiological health of the individual could be affected by the specialised diet.                               | <b>M5: Discuss</b> the use of external providers to support the individual with the specialised diet.                              |                                                                                                                   |
| <b>P8: Analyse</b> the impact of the specialised diet on the social, emotional and mental well-being needs of the individual.              |                                                                                                                                    |                                                                                                                   |
| <b>P9: Identify</b> appropriate techniques for monitoring the individual on the specialised diet.                                          | <b>M6: Justify</b> the monitoring techniques chosen for the individual.                                                            |                                                                                                                   |
| <b>P10: Describe</b> appropriate interventions that may be required based on the monitoring results.                                       |                                                                                                                                    |                                                                                                                   |
| <b>P11: Summarise</b> additional information that could increase confidence in the suitability of the specialised diet for the individual. | <b>M7: Analyse</b> how the additional information from <b>P11</b> would have been useful when creating the specialised diet.       | <b>D5: Evaluate</b> the limitations of your meal plan(s) for the individual following the specialised diet.       |
| <b>P12: Suggest</b> why the meal plan(s) may need to be adapted for another individual following the same specialised diet.                |                                                                                                                                    |                                                                                                                   |

## Assessment guidance

This assessment guidance gives you information relating to the assessment criteria. There might not be additional assessment guidance for each assessment criterion. It is included only where it is needed.

| Assessment Criteria | Assessment guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task 1              | <ul style="list-style-type: none"> <li>The research element of the criteria in this Task does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criteria.</li> </ul>                                                                                                                                                                                                                         |
| P1                  | <ul style="list-style-type: none"> <li>Students need to review information about nutritional requirements that are specific to the needs of the individual in the case study.</li> <li>They must recognise the needs of the individual in the case study and explain why a specialised diet is required.</li> </ul>                                                                                                                                                 |
| P2                  | <ul style="list-style-type: none"> <li>Students must use research to describe the macronutrient requirements of the individual in the case study.</li> <li>Students must describe how the macronutrient requirements for the individual varies from the average person in terms of the average nutritional requirements and recommended values for daily intake.</li> </ul>                                                                                         |
| P3                  | <ul style="list-style-type: none"> <li>Students must use research to describe the micronutrient requirements of the individual in the case study.</li> <li>Students must describe how the micronutrient requirements for the individual varies from the average person in terms of the average nutritional requirements and recommended values for daily intake.</li> </ul>                                                                                         |
| M1                  | <ul style="list-style-type: none"> <li>Students must research and describe medical guidance that would be given to an individual in a similar situation. This should include details of monitoring and treatment of any disorders.</li> <li>The guidance should be from appropriate medical professionals relevant to the case study context.</li> </ul>                                                                                                            |
| P5                  | <ul style="list-style-type: none"> <li>Students must create an appropriate meal plan(s) relevant to the context of the case study. Meal plan(s) should be created to last the timeframe specified in the case study.</li> <li>The meal plan(s) should be customisable to show relevant substitutions that could be made for at least one meal each day for the duration of the timeframe specified in the case study.</li> </ul>                                    |
| M4                  | <ul style="list-style-type: none"> <li>Students must show evidence of processing data using appropriate calculations for creating the meal plan(s) in <b>P5</b>.</li> <li>The calculation(s) used will depend on the context of the case study but should be relevant and provide information to support the student in creating the meal plan(s).</li> <li>Students must show <b>at least one</b> example of their working out in the written evidence.</li> </ul> |
| D3                  | <ul style="list-style-type: none"> <li>Students must give valid reasons for their choice of meals in the meal plan(s) for the specialised diet, the customisable elements of the meal plan(s), and the preparation and storage requirements.</li> </ul>                                                                                                                                                                                                             |
| P9                  | <ul style="list-style-type: none"> <li>The monitoring techniques might focus on how any of the physiological, social, emotional, and/or mental well-being of the individual can be monitored.</li> </ul>                                                                                                                                                                                                                                                            |
| M6                  | <ul style="list-style-type: none"> <li><b>M6</b> is an extension of <b>P9</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           |
| P11                 | <ul style="list-style-type: none"> <li>Students must consider what additional information would have been useful in order to increase the confidence in the suitability of the specialised diet. Students will summarise what additional information they would have wanted.</li> </ul>                                                                                                                                                                             |
| M7                  | <ul style="list-style-type: none"> <li><b>M7</b> is an extension of <b>P11</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                          |

**Synoptic assessment**

Some of the knowledge, understanding and skills needed to complete this unit will draw on the learning in Units F170 and F171.

This table details these synoptic links.

| Unit F174: Nutrition and metabolism |                                                                         | Unit F170: Fundamentals of human biology |                                                              |
|-------------------------------------|-------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------|
| Topic Area                          |                                                                         | Topic Area                               |                                                              |
| 1                                   | Nutrients required for a healthy body                                   | 2                                        | Human physiology, organs and systems                         |
| 3                                   | Metabolic pathways and control mechanisms                               | 3                                        | Key concepts in endocrinology, neurobiology and reproduction |
| 4                                   | Diagnosis, monitoring and treatment for nutritional/metabolic disorders | 2                                        | Human physiology, organs and systems                         |

| Unit F174: Nutrition and metabolism |                                                                         | Unit F171: Health and disease |                                                 |
|-------------------------------------|-------------------------------------------------------------------------|-------------------------------|-------------------------------------------------|
| Topic Area                          |                                                                         | Topic Area                    |                                                 |
| 2                                   | Diets and disorders                                                     | 1                             | Causes and effects of diseases and disorders    |
|                                     |                                                                         | 2                             | Curative, management and preventative therapies |
| 4                                   | Diagnosis, monitoring and treatment for nutritional/metabolic disorders | 4                             | Techniques for diagnosis and monitoring         |

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

### 5.3.4 Unit F175: Human reproduction

#### Unit Aim

Reproduction creates new life. This unit explores the role of the human reproductive system in creating new life and the way in which science can be used to help control this process. Science plays a part in monitoring pregnancy and helping those finding it difficult to conceive.

In this unit you will study how life is created through reproduction. You will explore the development of the zygote, embryo and foetus and the process of pregnancy and antenatal care. You will learn about contraception and how some individuals find it difficult to conceive. You will explore how modern medicine can assist these individuals to have children by identifying the causes of infertility and enabling individuals to receive treatment for their infertility.

| Unit F175: Human reproduction                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------|
| Topic Area 1: Conception and pregnancy                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |              |
| Teaching content                                                                                                                                                           | Exemplification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Opportunities to cover:    |              |
| <b>1.1 Menstrual cycle</b>                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>               | <b>HSW</b>   |
| <input type="checkbox"/> Menstrual phase<br><input type="checkbox"/> Follicular phase<br><input type="checkbox"/> Ovulation phase<br><input type="checkbox"/> Luteal phase | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How hormones regulate the menstrual cycle</li> <li><input type="checkbox"/> How to determine the 'fertility window'</li> <li><input type="checkbox"/> How to use results from blood tests to determine whether ovulation is occurring</li> <li><input type="checkbox"/> How irregular or abnormal ovulation can impact fertility</li> <li><input type="checkbox"/> How anovulation can be treated with fertility drugs</li> </ul>                 | <b>M3.4</b><br><b>M3.5</b> | <b>HSW11</b> |
| <b>1.2 Fertilisation and implantation</b>                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>               | <b>HSW</b>   |
| <input type="checkbox"/> Fertilisation<br><input type="checkbox"/> Zygote formation                                                                                        | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How the acrosome reaction forms a zygote</li> <li><input type="checkbox"/> How the cortical reaction prevents the zygote from having an abnormal number of chromosomes</li> <li><input type="checkbox"/> Comparison between <i>in vitro</i> fertilisation (IVF), artificial insemination (IUI) and intracytoplasmic sperm injection (ICSI)</li> <li><input type="checkbox"/> Use of images to explain medically assisted fertilisation</li> </ul> |                            | <b>HSW9</b>  |
| <b>1.3 Development from zygote to foetus</b>                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>               | <b>HSW</b>   |
| <input type="checkbox"/> Development of the zygote into an embryo<br><input type="checkbox"/> Development of the foetus                                                    | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Key stages of development</li> <li><input type="checkbox"/> Comparison between IVF, IUI and ICSI treatments</li> </ul>                                                                                                                                                                                                                                                                                                                            |                            | <b>HSW11</b> |

| <b>1.4 Contraception</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>                   | <b>HSW</b><br><b>HSW5</b> |
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| <input type="checkbox"/> Main methods of contraception <ul style="list-style-type: none"> <li>• Barrier methods               <ul style="list-style-type: none"> <li>○ Internal and external condoms</li> <li>○ Cap</li> <li>○ Diaphragm</li> </ul> </li> <li>• Chemical methods               <ul style="list-style-type: none"> <li>○ Combined pill</li> <li>○ Progesterone only pill</li> <li>○ Contraceptive injection and patch</li> <li>○ Intrauterine system (IUS)</li> <li>○ Intrauterine device (IUD)</li> </ul> </li> <li>• Emergency contraception</li> <li>• Natural methods</li> <li>• Surgical procedures – sterilisation</li> <li>• Use of spermicides</li> </ul>                                                                                | To include: <ul style="list-style-type: none"> <li>□ Key features of each method</li> <li>□ Impact of contraception methods on fertility and ability to conceive</li> <li>□ Impact of hormonal treatments and other medications on efficacy of contraception methods</li> <li>□ Tubal sterilisation and vasectomy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |                           |
| <b>Topic Area 2: Pregnancy (antenatal) care</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                           |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Opportunities to cover:</b> |                           |
| <b>2.1 First antenatal appointment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b>                   | <b>HSW</b><br><b>HSW3</b> |
| <input type="checkbox"/> Information that may be collected during the appointment: <ul style="list-style-type: none"> <li>• About the foetus' biological parents</li> <li>• Domestic abuse</li> <li>• Female genital mutilation (FGM)</li> <li>• Health issues</li> <li>• Lifestyle</li> <li>• Other pregnancies or children</li> <li>• Physical and mental health</li> <li>• Smoking, alcohol and drug use</li> <li>• Support network</li> </ul> <input type="checkbox"/> Tests carried out during the appointment: <ul style="list-style-type: none"> <li>• Blood pressure</li> <li>• Blood tests for general health, blood group, HIV, syphilis and hepatitis B</li> <li>• Body mass index (BMI)</li> <li>• Urine test for signs of pre-eclampsia</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ How to use the information collected to identify the physical, psychological and personal needs of the patient</li> <li>□ How to use the information collected to assess the health and well-being of the patient and foetus</li> <li>□ How the information collected may have an impact on the physical and psychological health of the patient and foetus</li> <li>□ How to use the results from the tests to assess the physical health and well-being of the patient and foetus</li> <li>□ How to use the information collected to assess and support the personal needs of the patient</li> <li>□ How to use the information collected to provide healthcare advice on promoting and supporting the health and wellbeing needs of the patient and foetus</li> </ul> | <b>M1.1</b><br><b>M1.5</b>     |                           |



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| <ul style="list-style-type: none"> <li>□ Advice and information that may be given about antenatal clinical investigations (tests and scans) and antenatal activities:             <ul style="list-style-type: none"> <li>• Antenatal care</li> <li>• Antenatal classes</li> <li>• A healthy pregnancy diet</li> <li>• Pregnancy exercise</li> <li>• Tests and scans offered during pregnancy</li> </ul> </li> <li>□ Role of health professionals involved in antenatal care</li> </ul>                                   | <ul style="list-style-type: none"> <li>□ How to use the information collected to suggest appropriate antenatal clinical investigations and activities to promote and support the health and well-being needs of the patient and foetus</li> <li>□ Key features and advantages of:             <ul style="list-style-type: none"> <li>• Antenatal care</li> <li>• Antenatal classes</li> <li>• A healthy pregnancy diet</li> <li>• Pregnancy exercise</li> <li>• Tests and scans offered during pregnancy</li> </ul> </li> <li>□ How the tests and scans offered during pregnancy can be used to monitor the physical health of the patient and foetus</li> <li>□ How information is shared between healthcare professionals</li> <li>□ How to use the information collected to write an antenatal care plan</li> </ul> |                            |              |
| <b>2.2 Antenatal care plan</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>               | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ Key components             <ul style="list-style-type: none"> <li>• Medical history</li> <li>• Care professionals involved</li> <li>• Care professional roles</li> <li>• Information about further antenatal clinical investigations that may be needed</li> <li>• Information about antenatal activities that may be needed or advised</li> <li>• Any further advice given to the patient to promote the health and wellbeing of the patient and foetus</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to write an antenatal care plan</li> <li>□ Importance of including the key components of the care plan</li> <li>□ Advantages and disadvantages of following an antenatal care plan</li> <li>□ Advantages and disadvantages of undertaking the antenatal clinical investigations and activities suggested in the antenatal care plan</li> <li>□ The possible physical, psychological and personal effects of undertaking the antenatal clinical investigations and activities suggested in the antenatal care plan on the patient and foetus</li> <li>□ Communication skills for different audiences</li> </ul>                                                                                                                                         |                            | <b>HSW11</b> |
| <b>2.3 Monitoring foetal development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>               | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ Use of techniques to monitor pregnancy and development:             <ul style="list-style-type: none"> <li>• Amniocentesis</li> <li>• Blood tests</li> <li>• Chorionic villus sampling</li> <li>• 3D and colour scan</li> <li>• Nuchal translucency (NT) scan</li> <li>• Ultrasound</li> </ul> </li> </ul>                                                                                                                                                                      | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How techniques can determine:             <ul style="list-style-type: none"> <li>• Age of foetus</li> <li>• Chromosomal abnormalities</li> <li>• Developmental problems</li> <li>• Due date</li> <li>• Pregnancy complications</li> <li>• Size of foetus</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>M1.2</b><br><b>M1.3</b> | <b>HSW9</b>  |

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| <ul style="list-style-type: none"> <li>□ Role of health professionals during pregnancy</li> </ul>                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>□ How to use medical data to assess and explain risks associated with tests and procedures</li> <li>□ The possible physical and psychological effects of the tests and procedures on the mother and foetus</li> <li>□ Advantages and disadvantages of the tests and procedures</li> <li>□ How pregnancy is monitored by health professionals</li> </ul>                                                                                  |                                |              |
| <b>2.4 Complications during pregnancy</b>                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                   | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ Ectopic pregnancies</li> <li>□ Gestational diabetes</li> <li>□ Multiple pregnancies</li> <li>□ Preeclampsia</li> <li>□ Premature birth</li> </ul>                                                                                                                                                                                                                  | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Key features of complications</li> <li>□ How to use medical information to assess and diagnose pregnancy complications</li> <li>□ How to use medical information to suggest possible clinical interventions and/or further tests</li> <li>□ The physical and psychological effects of pregnancy complications on the patient and foetus</li> <li>□ How IVF and ICSI may increase pregnancy complications</li> </ul> | <b>M1.3</b>                    | <b>HSW9</b>  |
| <b>2.5 Legislation and regulatory boards</b>                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                   | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ National Institute for Health and Care Excellence (NICE)</li> <li>□ Integrated care board (ICB)</li> </ul>                                                                                                                                                                                                                                                         | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How legislation and regulatory boards impact antenatal care and maternity services</li> <li>□ How regulatory boards ensure safe and effective antenatal care</li> </ul>                                                                                                                                                                                                                                             |                                | <b>HSW7</b>  |
| <b>Topic Area 3: Infertility</b>                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                     | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Opportunities to cover:</b> |              |
| <b>3.1 Diagnosing infertility</b>                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                   | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ Information collected during an initial GP assessment:             <ul style="list-style-type: none"> <li>• Age</li> <li>• How long they have been trying to conceive</li> <li>• Lifestyle</li> <li>• Medicines being taken</li> <li>• Previous miscarriages or previous children</li> <li>• Sexual history</li> <li>• Use of contraception</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Initial physical pelvic examination results and impact on fertility</li> <li>□ How to use initial consultation information to determine possible causes of infertility, including variations in sex traits</li> <li>□ How to use the consultation and medical information collected to identify the physical, psychological and personal needs of the patient(s)</li> </ul>                                         | <b>M3.1</b>                    | <b>HSW11</b> |

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| <ul style="list-style-type: none"> <li>□ Risk factors that affect fertility:               <ul style="list-style-type: none"> <li>• Age</li> <li>• Alcohol</li> <li>• Body mass index (BMI)</li> <li>• Drug use</li> <li>• Environmental and occupational exposures</li> <li>• Medications</li> <li>• Sexually Transmitted Infections (STI)</li> <li>• Smoking</li> <li>• Stress</li> </ul> </li> <li>□ Referral process to a fertility clinic</li> <li>□ Role of different health professionals in diagnosing infertility</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>□ How to use the information collected to assess the health and well-being of the patient(s)</li> <li>□ How to use consultation information to write a reproductive health plan</li> <li>□ Impact of a fertility diagnosis on the health and well-being of the patient(s)</li> <li>□ How information is shared between different health professionals</li> <li>□ Communication skills for different audiences</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                |                                           |                            |
| <b>3.2 Causes of infertility</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>                              | <b>HSW</b>                 |
| <ul style="list-style-type: none"> <li>□ The common causes of infertility :               <ul style="list-style-type: none"> <li>• Abnormal sperm</li> <li>• Autoimmune conditions</li> <li>• Blocked or damaged fallopian tubes</li> <li>• Damaged testicles</li> <li>• Endometriosis and fibroids</li> <li>• Failure to ovulate as a result of polycystic ovary syndrome (PCOS), thyroid problems and premature ovulation failure</li> <li>• Hypogonadism</li> <li>• Low sperm count</li> <li>• Medicines being taken</li> <li>• Pelvic inflammatory disease (PID)</li> <li>• Previous miscarriages or previous children</li> <li>• Sperm immobility</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to use blood test results to determine hormone levels and to see if they are within the 'normal' range</li> <li>□ How to use ultrasound images to view the uterus and ovaries to look for:               <ul style="list-style-type: none"> <li>• Scarring</li> <li>• Endometriosis</li> <li>• Ovarian tumours or cysts</li> <li>• Fibroids</li> </ul> </li> <li>□ How to use laparoscopy images to examine the uterus, oviducts and ovaries to determine if there are any blockages in the oviducts</li> <li>□ How to use semen analysis to determine sperm count, motility and malformation and see if they are within the 'normal' parameters</li> <li>□ How to use hormone data to determine possible cause of male infertility</li> <li>□ How certain types of medicines can cause infertility</li> </ul> | <b>M0.2</b><br><b>M1.6</b><br><b>M3.1</b> | <b>HSW6</b><br><b>HSW9</b> |
| <b>3.3 Treatment options</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>                              | <b>HSW</b>                 |
| <ul style="list-style-type: none"> <li>□ Preconception care and advice               <ul style="list-style-type: none"> <li>• Assess any complications from previous pregnancies</li> <li>• BMI</li> <li>• Diet</li> <li>• Exposure to environmental toxins</li> <li>• Folic Acid</li> <li>• Medical conditions</li> <li>• Rubella</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                     | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How preconception care and use of complementary therapies can improve fertility</li> <li>□ Key features of fertility investigations and tests</li> <li>□ How to use consultation and medical information to suggest appropriate healthcare advice and treatment options</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>M1.6</b>                               | <b>HSW5</b>                |

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| <ul style="list-style-type: none"> <li>□ Complementary and alternative therapies               <ul style="list-style-type: none"> <li>• Acupuncture</li> <li>• Nutritional therapy</li> </ul> </li> <li>□ Fertility investigation and tests               <ul style="list-style-type: none"> <li>• Hysterosalpingography</li> <li>• Laparoscopy</li> <li>• Hormone profiles</li> <li>• Semen analysis</li> </ul> </li> <li>□ Assisted reproduction</li> <li>□ Role of healthcare professionals involved in fertility treatment</li> </ul> | <ul style="list-style-type: none"> <li>□ Advantages and disadvantages of the treatment options</li> <li>□ Comparison of the success rates of the treatment options to improve fertility</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                |              |
| <b>3.4 Reproductive health plan</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>                   | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ Key components               <ul style="list-style-type: none"> <li>• Medical history</li> <li>• Care professionals involved</li> <li>• Care professional roles</li> <li>• Information and advice given about further fertility clinical investigations and tests</li> <li>• Information and advice given about fertility treatments</li> <li>• Additional advice given to the patient</li> </ul> </li> </ul>                                                                                    | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to write a reproductive health plan</li> <li>□ Importance of including the key components of the health plan</li> <li>□ Advantages and disadvantages of following a reproductive health plan</li> <li>□ Advantages and disadvantages of undertaking the fertility clinical investigations, treatments and advice suggested in the reproductive health plan</li> <li>□ The possible physical, psychological and personal effects of undertaking the fertility clinical investigations, treatments and advice suggested in the reproductive health plan on the patient(s)</li> <li>□ Communication skills for different audiences</li> </ul> |                                | <b>HSW11</b> |
| <b>Topic Area 4: Assisted reproduction (AR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Opportunities to cover:</b> |              |
| <b>4.1 Assisted reproduction options</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>                   | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li>□ Range of options available:               <ul style="list-style-type: none"> <li>• Medical treatment such as clomiphene, tamoxifen, metformin and gonadotrophins for infrequent or lack of ovulation</li> <li>• Surgical procedures to treat endometritis, fibroids and blocked oviducts</li> <li>• IUI</li> <li>• IVF to include the protocols frequently used and the hormones administered</li> <li>• ICSI</li> <li>• Egg and sperm donation</li> </ul> </li> </ul>                           | <p>To include:</p> <ul style="list-style-type: none"> <li>□ Key features of each option</li> <li>□ Advantages and disadvantages of the options available</li> <li>□ How the treatment may overcome infertility</li> <li>□ How the options available depend on the cause of infertility</li> <li>□ How to use consultation and medical information to suggest a suitable AR option</li> <li>□ How medical information is shared and communicated between different audiences</li> </ul>                                                                                                                                                                                                                     | <b>M1.3</b><br><b>M3.1</b>     | <b>HSW9</b>  |

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| <input type="checkbox"/> Role of different health professionals working in the assisted reproduction field                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                               |              |             |
| <b>4.2 Undergoing AR tests and treatment</b>                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b> | <b>HSW</b>  |
| <input type="checkbox"/> Determining eligibility for fertility tests and treatments<br><input type="checkbox"/> Success rates of AR techniques                                                                                                                 | To include:<br><input type="checkbox"/> How to use patient information and current regulations to determine eligibility for fertility tests and treatment<br><input type="checkbox"/> Comparison of success rates                                                                                                                                                             | <b>M1.6</b>  |             |
| <b>4.3 Legislation and regulatory boards</b>                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                               | <b>Maths</b> | <b>HSW</b>  |
| <input type="checkbox"/> Regulatory boards: <ul style="list-style-type: none"> <li>• Human Fertilisation and Embryology Authority (HFEA)</li> <li>• Integrated care board (ICB)</li> <li>• National Institute for Health and Care Excellence (NICE)</li> </ul> | To include:<br><input type="checkbox"/> How legislation impacts on assisted reproduction techniques<br><input type="checkbox"/> How AR is regulated in the UK<br><input type="checkbox"/> Role of regulatory boards for patients and clinics<br><input type="checkbox"/> How to use data provided by HFEA<br><input type="checkbox"/> Ethical considerations of AR techniques | <b>M1.6</b>  | <b>HSW6</b> |

### Assessment criteria

The table below gives the assessment criteria for the tasks in the set assignment for this unit. The assessment criteria indicate what is required in these tasks.

This qualification has a compensatory approach. This means that the unit grade awarded is based on the **total** number of achieved criteria for the unit (see [Section 6.4](#)). Students do **not** have to achieve **all** criteria for a specific grade to achieve that unit grade (e.g. achieve all pass criteria to achieve a pass grade).

[Section 7.4](#) provides full information on how to assess the NEA units and apply the assessment criteria. Students' work must show that all aspects of a criterion have been met in sufficient detail for it to be **successfully achieved** (see [Section 7.4.1](#)). If a student's work does not fully meet a criterion, you must not award that criterion.

The command words used in the assessment criteria are defined in [Appendix B](#).

| Pass                                                                                                                                                                        | Merit                                                                                                                                            | Distinction                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1: Create</b> a reproductive health plan containing all key components to meet the needs of the patient(s) in Case Study A.                                             | <b>M1:</b> Use research to <b>explain</b> the appropriateness of the reproductive health plan for the patient(s) in Case Study A.                | <b>D1: Analyse</b> the specific roles of the healthcare professionals, legislation, and regulatory boards in relation to their involvement in the reproductive health plan created in <b>P1</b> . |
| <b>P2: Explain</b> possible causes of infertility for the patient(s) in Case Study A.                                                                                       |                                                                                                                                                  |                                                                                                                                                                                                   |
| <b>P3: Explain</b> the advantages and disadvantages of different treatment options in relation to the context of the patient(s) in Case Study A.                            |                                                                                                                                                  |                                                                                                                                                                                                   |
| <b>P4: Explain</b> the rationale of the treatment options and further tests chosen for the patient(s) in the reproductive health plan, including the likelihood of success. |                                                                                                                                                  |                                                                                                                                                                                                   |
| <b>P5: Create</b> an antenatal care plan containing all key components to meet the needs of the patient in Case Study B.                                                    | <b>M3:</b> Use research to <b>explain</b> the appropriateness of the antenatal care plan for the patient in Case Study B.                        | <b>D2: Analyse</b> the specific roles of the healthcare professionals, legislation, and regulatory boards in relation to their involvement in the antenatal care plan created in <b>P5</b> .      |
| <b>P6: Explain</b> possible effects on the mother and the foetus of undertaking the antenatal care plan in Case Study B.                                                    |                                                                                                                                                  |                                                                                                                                                                                                   |
| <b>P7: Explain</b> the advantages and disadvantages of the antenatal care plan for the patient.                                                                             |                                                                                                                                                  |                                                                                                                                                                                                   |
| <b>P8: Explain</b> the rationale of the interventions and further tests identified chosen for the patient in the antenatal care plan.                                       |                                                                                                                                                  |                                                                                                                                                                                                   |
| <b>P9: Create</b> an appropriate presentation for the <b>chosen</b> Case Study, including the fundamentals of the plan.                                                     | <b>M5: Deliver</b> the presentation effectively, with clear explanations of rationale beyond what is included in the presentation documentation. | <b>D3: Justify</b> the content of the <b>chosen</b> presentation by detailing the scientific reasoning behind its inclusion.                                                                      |
| <b>P10: Explain</b> how the presentation has been focused with the patient(s) as the intended audience.                                                                     | <b>M6: Explain</b> appropriate adaptations to the presentation so that it can be used to communicate to other members of the healthcare team.    |                                                                                                                                                                                                   |

| Pass                                                                                 | Merit                                                                       | Distinction                                                                                                          |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>P11: Summarise</b> the feedback received for your <b>chosen</b> plan.             | <b>M7: Analyse</b> the strengths and weaknesses of your <b>chosen</b> plan. | <b>D4: Justify</b> the content of the <b>chosen</b> plan by detailing the scientific reasoning behind its inclusion. |
| <b>P12: Suggest</b> how the presentation created in <b>Task 3</b> could be improved. |                                                                             | <b>D5: Assess</b> the impact on the mental well-being of the patient(s) involved in your <b>chosen</b> plan.         |

## Assessment guidance

This assessment guidance gives you information relating to the assessment criteria. There might not be additional assessment guidance for each assessment criterion. It is included only where it is needed.

| Assessment Criteria | Assessment guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| P1                  | <ul style="list-style-type: none"> <li>Students must create a logical reproductive health plan which is presented in a clear order and within an appropriate timescale.</li> <li>Students must include all key components as listed in subtopic area 3.5 Reproductive health plan.</li> </ul>                                                                                                                                                                                                                                                                                                                                           |
| P2                  | <ul style="list-style-type: none"> <li>Students must use the information and background provided in Case Study A to explain possible causes of infertility for the patient(s).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P3                  | <ul style="list-style-type: none"> <li>Students must explain the advantages and disadvantages of different options that <b>could</b> be used for the patient.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P4                  | <ul style="list-style-type: none"> <li>Students must explain the rationale of the treatment options and further tests that they have <b>chosen</b> for the patient(s) in the reproductive health plan.</li> <li>Students must include an explanation of the likelihood of the success of each treatment option and test included.</li> </ul>                                                                                                                                                                                                                                                                                            |
| M1                  | <ul style="list-style-type: none"> <li><b>M1</b> is an extension of <b>P1</b>.</li> <li>Students must use research to provide rationale for the appropriateness of the reproductive health plan they have produced for the patient(s) in Case Study A.</li> <li>Students must apply their research to the information and background provided in Case Study A and the different treatment options available.</li> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul>                        |
| D1                  | <ul style="list-style-type: none"> <li>Students must analyse the role of the most appropriate healthcare professionals needed to treat and support the patient(s) (for example, doctor, fertility nurse, embryologist, etc) as appropriate to the reproductive health plan.</li> <li>Students must analyse the legislation and regulatory boards that uphold the safety and quality of the treatment options identified in the reproductive health plan.</li> <li>The specific healthcare professionals, legislation and regulatory boards will depend on the case study context. All relevant information must be included.</li> </ul> |

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| P5     | <ul style="list-style-type: none"> <li>Students must create a logical antenatal care plan which is presented in a clear order and within an appropriate timescale.</li> <li>Students must include all key components as listed in subtopic area 2.2 Antenatal care plan.</li> </ul>                                                                                                                                                                                                                                                                                                                                                |
| P6     | <ul style="list-style-type: none"> <li>Possible effects might include physical, psychological and personal effects, and might have a positive or negative impact.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| M3     | <ul style="list-style-type: none"> <li><b>M3</b> is an extension of <b>P5</b>.</li> <li>Students must use research to provide rationale for the appropriateness of the antenatal care plan they have produced for the patient in Case Study B.</li> <li>Students must apply their research to the information and background provided in Case Study B and the different treatment options available.</li> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul>                           |
| D2     | <ul style="list-style-type: none"> <li>Students must analyse the role of the most appropriate healthcare professionals needed to treat and support the patient(s) (for example, doctor, midwife, etc) as appropriate to the antenatal care plan.</li> <li>Students must analyse the legislation and regulatory boards that uphold the safety and quality of the interventions and/or further tests identified in the antenatal care plan.</li> <li>The specific healthcare professionals, legislation and regulatory boards discussed will depend on the case study context. All relevant information must be included.</li> </ul> |
| Task 3 | <ul style="list-style-type: none"> <li>Students can either deliver the presentation to the teacher, peers or a combination of both. If the presentation is delivered to peers only, this must be recorded, so that the teacher can use the recording to complete the Teacher Observation Record for <b>M5</b> (you do <b>not</b> need to submit this for moderation).</li> </ul>                                                                                                                                                                                                                                                   |
| P9     | <ul style="list-style-type: none"> <li>Students must create a presentation for the patient(s) identified in the chosen case study. The presentation should be in the format they feel is most appropriate, which could include a poster, a PowerPoint presentation, a flow diagram, etc.</li> <li>There must be sufficient detail in the presentation to demonstrate the key components of the plan appropriate for the patient(s).</li> </ul>                                                                                                                                                                                     |
| P10    | <ul style="list-style-type: none"> <li>Students must explain how the presentation created for <b>P9</b> was focused for the patient(s) as the intended audience.</li> <li>Students must explain how the presentation was written so that it was relevant and accessible for the patient(s).</li> <li>Students might choose to consider ways that scientific terminology might be re-phrased, amended or why they would need to use a particular scientific term.</li> </ul>                                                                                                                                                        |
| M5     | <ul style="list-style-type: none"> <li><b>M5</b> is an extension of <b>P9</b>.</li> <li>Teachers must complete a 'Teacher Observation Record' for each student to evidence they have met the criteria. Students must also read and sign it.</li> <li>The Teacher Observation Record form should describe in detail how the student delivered the presentation effectively, with clear explanations of rationale beyond what is included in the presentation documentation.</li> </ul>                                                                                                                                              |



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| M6  | <ul style="list-style-type: none"> <li>Having created the presentation for the patient(s) in <b>P9</b>, students must now consider how it could be adapted for other members of the healthcare team.</li> <li>Students might choose to create a further presentation to highlight the adaptations needed or they might choose to explain the adaptations in a different format, e.g. a table.</li> <li>Students must explain the adaptations suggested so that the members of the healthcare team would be able to understand their contribution to the plan.</li> <li>Students could consider how the scientific terminology used in the presentation might be modified to be communicated to a specialist audience.</li> </ul> |
| D3  | <ul style="list-style-type: none"> <li>Students must justify the content of the chosen presentation by detailing the scientific reasoning.</li> <li>Students will use their understanding of the unit content to provide valid reasons for the content's inclusion.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P11 | <ul style="list-style-type: none"> <li>Students must clearly express the most important points stemming from the feedback received for their treatment plan in a short and clear form.</li> <li>The feedback for the treatment plan can be provided by the teacher and/or other students.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| D4  | <ul style="list-style-type: none"> <li>Students must justify the content of the chosen plan for the patient(s) by detailing the scientific reasoning.</li> <li>Students will use their understanding of the unit content to provide valid reasons for the content's inclusion.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Synoptic assessment

Some of the knowledge, understanding and skills needed to complete this unit will draw on the learning in Units F170 and F171.

This table details these synoptic links.

|                               |             |                                          |                                                               |
|-------------------------------|-------------|------------------------------------------|---------------------------------------------------------------|
| Unit F175: Human reproduction |             | Unit F170: Fundamentals of human biology |                                                               |
| Topic Area                    |             | Topic Area                               |                                                               |
| 3                             | Infertility | 3                                        | Key concepts in endocrinology, neurobiology, and reproduction |

|                               |                            |                               |                                              |
|-------------------------------|----------------------------|-------------------------------|----------------------------------------------|
| Unit F175: Human reproduction |                            | Unit F171: Health and disease |                                              |
| Topic Area                    |                            | Topic Area                    |                                              |
| 2                             | Pregnancy (antenatal) care | 4                             | Techniques for diagnosis and monitoring      |
| 3                             | Infertility                | 1                             | Causes and effects of diseases and disorders |

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

### 5.3.5 Unit F176: The brain

#### Unit Aim

The brain is a fascinating organ. The study of the brain involves a number of clinical and laboratory investigations and the analysis of data collected by neuroscientists. We still do not have a complete understanding about the functions of the brain but many advances have been made in the diagnosis and treatment of various disorders.

In this unit you will gain a greater insight into the structure and function of the nervous system, including the spinal cord, brain and nerves. This will involve the study of photomicrographs using online research to produce annotated, biological drawings. You will also explore the complex world of neuron communication and the control of the body. The challenging topics of nociception (sensing nociceptor pain), neurotransmitters and drug control will form the basis of case study analyses. This unit will also enable you to obtain an insight into the interpretation of nerve impulses as shown by electroencephalogram (EEG) recordings. Finally, the diagnosis and treatment of brain disorders and traumatic brain injury (TBI) will be outlined for you to gain an enhanced understanding of the scientific method. You will also learn how to evaluate the communication of science to different audiences.

| <b>Unit F176: The brain</b>                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |                            |
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| <b>Topic Area 1: Structure and function of the nervous system</b>                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |                            |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                            | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Opportunities to cover:</b> |                            |
| <b>1.1 The brain</b>                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>                   | <b>HSW</b>                 |
| <b>Brain anatomy/structure and function</b> <ul style="list-style-type: none"> <li>□ Skull and meninges</li> <li>□ Cerebrum</li> <li>□ Cerebellum</li> <li>□ Hypothalamic-pituitary-adrenal axis (HPA)</li> <li>□ Brain stem               <ul style="list-style-type: none"> <li>• Pons</li> <li>• Medulla (oblongata)</li> <li>• Midbrain</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li>□ How the location of different parts of the brain as revealed by photographic images (generated by different scanning techniques) and shown in biological drawings for vertical and transverse sections</li> <li>□ The key function(s) of the structures listed</li> <li>□ How to draw, annotated low power plans of the brain from computed tomography (CT)/Magnetic resonance imaging (MRI) images</li> <li>□ How different types of drawings are used to share information about brain anatomy and function to different audiences</li> <li>□ How the brain carries out both central nervous system (CNS) and autonomic nervous system (ANS) functions</li> <li>□ Why the skull and meninges present challenges during brain surgery</li> </ul><br>Does not include: <ul style="list-style-type: none"> <li>□ Histology of brain tissues</li> </ul> |                                | <b>HSW1</b><br><b>HSW2</b> |

| <b>1.2 The spinal cord</b>                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Maths</b>               | <b>HSW</b>                 |
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| <b>Spinal cord anatomy (transverse section - TS) and function</b> <ul style="list-style-type: none"> <li>□ Vertebrae</li> <li>□ Meninges</li> <li>□ Grey matter</li> <li>□ White matter</li> <li>□ Central canal</li> <li>□ Dorsal and ventral roots</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ The key function(s) of the structures listed</li> <li>□ How to draw, annotated low power plans of the spinal cord from CT/MRI images</li> <li>□ How different types of drawings are used to share information about spinal cord anatomy and function to different audiences</li> <li>□ How a lumbar puncture can be performed to add drugs/anaesthetics to the CNS and to take samples of cerebrospinal fluid</li> <li>□ Why cervical breaks of the vertebral column/spine are more damaging than lumbar breaks</li> <li>□ Limitations of surgical interventions to regenerate damaged regions of spinal cord</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Histology of spinal cord tissues</li> </ul> | <b>M0.1</b><br><b>M1.1</b> | <b>HSW1</b><br><b>HSW9</b> |
| <b>1.3 Nerves</b>                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Maths</b>               | <b>HSW</b>                 |
| <b>Nerve anatomy (TS) and function</b> <ul style="list-style-type: none"> <li>□ Cranial and spinal nerves</li> <li>□ Endoneurium, perineurium and epineurium</li> <li>□ Fascicles</li> <li>□ Myelin sheath</li> </ul>                                           | <p>To include:</p> <ul style="list-style-type: none"> <li>□ The key function(s) of the structures listed</li> <li>□ How to draw, annotated low power plans of a nerve from light microscopy (LM) or CT/MRI images</li> <li>□ How different types of drawings may be needed to share information about nerve anatomy and function to different audiences</li> <li>□ Benefits and limitations of using scan images to identify damaged nerves</li> <li>□ How repetitive sports injuries can cause damage to nerves</li> <li>□ How traumatic injury of nerves can lead to loss of motor and sensory functions</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Histology of nerve tissues</li> </ul>                                                                      | <b>M0.1</b><br><b>M1.1</b> | <b>HSW1</b><br><b>HSW9</b> |

| <b>Topic Area 2: Neuron communication and control</b>                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                |                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------|
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Exemplification</b>         |                                            |
|                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Opportunities to cover:</b> |                                            |
| <b>2.1 Neuron communication</b>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b>                   | <b>HSW</b>                                 |
| <b>2.1.1 Action potentials</b> <ul style="list-style-type: none"> <li>□ Resting and action potentials</li> <li>□ Depolarisation, polarisation and repolarisation</li> <li>□ Absolute and relative refractory periods</li> </ul>                                                                                                                                      | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How sodium and potassium ions are exchanged across the axon membrane to generate an action potential</li> <li>□ How to interpret the different phases of nerve impulse transmission</li> <li>□ Why myelinated neurons are capable of increasing the speed of neuronal transmission</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Cytology of neurons</li> </ul>                                                                                                                                                                                                                                          | <b>M3.3</b><br><b>M3.4</b>     |                                            |
| <b>2.1.2 Structure and function of the synapse</b> <ul style="list-style-type: none"> <li>□ Different types of synaptic connections</li> <li>□ Detailed components of the synapse</li> <li>□ Stages of neuron impulse transmission across the synapse</li> <li>□ Route of neurotransmitter synthesis, release, recognition, reabsorption and re-synthesis</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How synapses provide a junction between one neuron and the next but also link the nervous system to the effectors, including muscle cells/fibres</li> <li>□ How the nerve impulse is transmitted across the synapse</li> <li>□ What is the relevance of mitochondria in the pre-synaptic knob</li> <li>□ Why pyramidal neurons in the brain have many dendrites</li> <li>□ The advantages and disadvantages of drugs acting as agonists, antagonists, activators and inhibitors</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Postsynaptic ionic exchange along the axon of the second neuron</li> </ul> |                                | <b>HSW1</b>                                |
| <b>2.2 Nervous control</b>                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Maths</b>                   | <b>HSW</b>                                 |
| <b>2.2.1 Control of movement and balance</b> <ul style="list-style-type: none"> <li>□ Shared functions of motor cortex and cerebellum in brain</li> <li>□ Significance of proprioceptors</li> </ul>                                                                                                                                                                  | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How the motor cortex in the cerebrum is involved in conscious control of movement but the cerebellum provides fine control of muscle contraction and balance/posture</li> <li>□ How proprioceptors act as pressure receptors to detect the changes in muscle contraction/relaxation and convey impulses to the motor cortex and cerebellum</li> </ul>                                                                                                                                                                                                                                                                                   |                                | <b>HSW1</b><br><b>HSW3</b><br><b>HSW11</b> |

|                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                    |                     |
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| <ul style="list-style-type: none"> <li>□ Link between visual stimuli and voluntary muscle contraction</li> </ul>                                                                                             | <ul style="list-style-type: none"> <li>□ What are the reasons for poor balance, including brain injuries caused by repetitive sports trauma</li> <li>□ How simple experiments can demonstrate the link between visual stimuli and voluntary muscle contraction</li> <li>□ How different forms of communication may be needed to share information about brain injuries/disorders and their impact on movement and balance to different audiences</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ Calcium influx and sliding-filament theory</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                    |                     |
| <p><b>2.2.2 Control of heartbeat</b></p> <ul style="list-style-type: none"> <li>□ Role of midbrain</li> <li>□ Nervous connections with the heart</li> <li>□ Receptors in carotid and aortic nodes</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How the midbrain, in particular the medulla (oblongata), acts as both the cardiovascular and respiratory centre</li> <li>□ Benefits and limitations of autonomic nervous system (ANS) control of heartbeat</li> <li>□ Why the control of heartbeat and pulmonary ventilation rate is linked</li> <li>□ How the sinoatrial node (SAN) in the wall of the heart is connected to the brain via sympathetic and parasympathetic/vagus nerves to accelerate and decelerate heartbeat rate, respectively</li> <li>□ How the atrioventricular node (AVN) transmits impulses across the wall of the heart via the AVN, bundle of His and Purkyne tissue</li> <li>□ Why impulse transmission from the medulla (oblongata) is affected by sensory impulses received from receptors in the carotid and aortic nodes</li> <li>□ How electrical activity in the heart can be monitored via electrocardiogram (ECG) readings, to show tachycardia, atrial fibrillation and bradycardia</li> <li>□ What is the impact of heart surgery on the bundle of His and Purkyne tissues</li> </ul> | <p><b>M3.3</b><br/><b>M3.4</b></p> | <p><b>HSW11</b></p> |

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|                                                                                                                                                                                                                                     | <input type="checkbox"/> How different forms of communication may be needed to share information about brain injuries/disorders and their impact on the control of heartbeat to different audiences<br><br>Does not include:<br><input type="checkbox"/> Heart and blood vessel defects                                                                                                                                                                                                                                                                                                                                                                                                          |                                |             |
| <b>Topic Area 3: Nociception, neurotransmitters and drugs</b>                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |             |
| <b>Teaching content</b>                                                                                                                                                                                                             | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Opportunities to cover:</b> |             |
| <b>3.1 Nociception</b>                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Maths</b>                   | <b>HSW</b>  |
| <b>3.1.1 Nervous receptors</b><br><input type="checkbox"/> Different types of receptors<br><br><br><br><br><br><br><br><br><br><input type="checkbox"/> Generator and action potentials at the receptor                             | To include:<br><input type="checkbox"/> The types of receptors include: <ul style="list-style-type: none"> <li>• Proprioceptors</li> <li>• Photoreceptors</li> <li>• Chemoreceptors</li> <li>• Touch receptors</li> <li>• Nociceptors/pain receptors</li> </ul> <input type="checkbox"/> How that receptors are connected to sensory neurons within the spinal reflex arc<br><input type="checkbox"/> Why there are different types of receptors at different locations in the body<br><br><input type="checkbox"/> How the all-or-nothing law is linked to stimulus threshold when a receptor is stimulated<br><br>Does not include:<br><input type="checkbox"/> Ionic exchange at the receptor | <b>M2.1</b>                    | <b>HSW3</b> |
| <b>3.1.2 Sensing pain (nociception)</b><br><input type="checkbox"/> Definition of a nociceptor<br><br><br><input type="checkbox"/> Sensing nociceptor pain<br><br><br><br><br><br><input type="checkbox"/> Locations of nociceptors | To include:<br><input type="checkbox"/> What is the structure and function of a nociceptor<br><br><input type="checkbox"/> How the sense of pain is closely linked to the activity of nociceptors at the cellular level, including the link between nociceptors, sensory and motor neurons<br><br><input type="checkbox"/> How nociceptors can detect different levels of pain due to their location in the dermis of the skin, mucosa and cornea of the eye, but also deeper in the body, including at the skeletal muscles/joints, bladder, visceral organs and digestive tract                                                                                                                | <b>M2.1</b>                    | <b>HSW5</b> |

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| <input type="checkbox"/> Pain gate control theory                                                                                                                                                                                                  | <input type="checkbox"/> How pain is detected when a stimulus reaches a threshold to break through the 'gates' controlling entry to the brain<br><input type="checkbox"/> The advantages and disadvantages of using nerve blocks, massage, exercise, transcutaneous electrical nerve stimulation (TENS) and cognitive behavioural therapy (CBT) to control pain<br><input type="checkbox"/> How different forms of communication may be needed to share information about nociceptor pain to different audiences<br><br>Does not include:<br><input type="checkbox"/> Detailed analysis of nociceptor models                                                                                                                     |              |              |
| <b>3.2 Neurotransmitters</b>                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Maths</b> | <b>HSW</b>   |
| <b>3.2.1 Different types of neurotransmitter</b><br><input type="checkbox"/> Function of different types of neurotransmitter including: <ul style="list-style-type: none"> <li>• Excitatory</li> <li>• Inhibitory</li> <li>• Modulatory</li> </ul> | To include:<br><input type="checkbox"/> How the antagonistic action of excitatory and inhibitory neurotransmitters functions<br><br>Does not include:<br><input type="checkbox"/> The chemistry of neurotransmitters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              | <b>HSW11</b> |
| <b>3.2.2 Problems with neurotransmitters</b><br><input type="checkbox"/> Insufficient or excess quantities released by neurons<br><input type="checkbox"/> Reabsorbed too quickly<br><input type="checkbox"/> Readily deactivated by enzymes       | To include:<br><input type="checkbox"/> How reduction in the function of neurotransmitters has a direct effect on neuron activity<br><input type="checkbox"/> How the loss of neurons in the brain in Parkinson's disease leads to a significant reduction in neurotransmitter activity<br><input type="checkbox"/> How epilepsy causes seizures resulting from an interruption in neuron activity in the brain<br><input type="checkbox"/> How different forms of communication may be needed to share information about brain disorders, including Parkinson's disease and epilepsy, to different audiences<br><br>Does not include:<br><input type="checkbox"/> Histology of tissues affected by neurotransmitter malfunction |              | <b>HSW12</b> |

| <b>3.3 Drugs</b>                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>                   | <b>HSW</b>                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------------|
| <b>Drugs used to modify function of the brain and nervous system</b> <ul style="list-style-type: none"> <li>□ Medicinal/therapeutic drugs</li> <li>□ Recreational drugs</li> <li>□ Fitness-enhancing drugs</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How drugs can be used for many purposes including medicine/therapies, for recreation and fitness training</li> <li>□ What are the key features of a prescription drug schedule when used for treatment and as a therapy</li> <li>□ Why dopamine injections are used under clinical conditions</li> <li>□ The advantages and disadvantages of using serotonin as an anti-depressant</li> <li>□ Benefits and harms of recreational drug use</li> <li>□ How fitness-enhancing drugs are detected before and after sporting events</li> <li>□ How different forms of communication may be needed to share information about the use of drugs to different audiences</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ The chemistry of drugs</li> </ul> | <b>M1.6</b><br><b>M3.1</b>     | <b>HSW11</b>                 |
| <b>Topic Area 4: The diagnosis and treatment of brain disorders/injuries</b>                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |                              |
| <b>Teaching content</b>                                                                                                                                                                                               | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Opportunities to cover:</b> |                              |
| <b>4.1 Diagnosis of brain disorders/injuries</b>                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b>                   | <b>HSW</b>                   |
| <b>Clinical assessment</b> <ul style="list-style-type: none"> <li>□ Causes of brain disorder and injury</li> <li>□ Clinical assessments carried out by a general practitioner (GP) or physician</li> </ul>            | <p>To include:</p> <ul style="list-style-type: none"> <li>□ The difference between the cause of brain disorders (inherited or age-related development) and traumatic brain injuries (TBI) (physical damage to the head/skull)</li> <li>□ How brain disorders and injuries have a differential impact on the health and wellbeing of patients</li> <li>□ How brain disorders and injuries can be identified by the analysis of scans (CT, MRI and ultrasound) and external symptoms (site of bleeding)</li> <li>□ How disorders can be monitored over time, including Parkinson's disease and epilepsy</li> </ul>                                                                                                                                                                                                                           | <b>M0.3</b>                    | <b>HSW11</b><br><b>HSW12</b> |



|                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                            |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------|
| <ul style="list-style-type: none"> <li>□ Use of tissue samples/biopsy</li> <li>□ Causes and diagnosis of mental health issues</li> </ul>                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>□ How brain disorders and injuries can present a range of symptoms including necrosis and haematoma</li> <li>□ Why the results of clinical assessments may be referred to neurologists</li> <li>□ How brain tissues can be sampled and observed via biopsy/pathology procedures to detect diseased and necrotic tissue</li> <li>□ How different forms of communication may be needed to share information about brain disorders/injuries to different audiences</li> <li>□ How mental health issues can be linked to a variety of causes including: <ul style="list-style-type: none"> <li>• Traumatic/physical brain injury</li> <li>• Post-traumatic stress disorder (PTSD)</li> <li>• Childhood abuse</li> <li>• Bereavement</li> <li>• Long term chronic condition</li> <li>• Drug/alcohol misuse</li> <li>• Social disadvantage</li> </ul> </li> <li>□ How healthcare professionals can diagnose mental health issues</li> <li>□ Why some patients with mental health issues are signposted to other professionals</li> <li>□ Benefits of promoting mental health awareness in the context of wellbeing</li> </ul> <p>Does not include:</p> <ul style="list-style-type: none"> <li>□ fMRI technology when used by neurologists</li> </ul> |                            |              |
| <b>4.2 Treatment and care of brain disorders/injuries</b>                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Maths</b>               | <b>HSW</b>   |
| <b>4.2.1 Types of treatment</b> <ul style="list-style-type: none"> <li>□ The key components of a treatment plan including: <ul style="list-style-type: none"> <li>• Recent medical history of patient</li> <li>• Cause of brain injury/disorder</li> <li>• Emergency treatment given</li> <li>• Medications/drugs given</li> <li>• Surgical procedures carried out</li> </ul> </li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ How to create a treatment plan</li> <li>□ How a treatment plan consists of a series of components, within a given timescale, designed to meet the physical and psychological needs of a patient and identifying the contributions of healthcare professionals and non-specialists, for example patient and their family/friends</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>M1.6</b><br><b>M1.7</b> | <b>HSW11</b> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <ul style="list-style-type: none"> <li>• Post-operative drug schedule required</li> <li>• Treatments (physical and psychological) required to aid rehabilitation</li> <li>• Contributions to be made by the healthcare professionals and non-specialists</li> <li>• Potential lifestyle changes needed to aid recovery</li> <li>• Personal support available at home or in a care setting</li> <li>• Other factors influencing recovery</li> </ul> <p>□ Brain surgery</p> <p>□ Use of therapeutic drugs</p> <p>□ Lifestyle modifications</p> | <ul style="list-style-type: none"> <li>□ Why treatment plans are likely to enable the rehabilitation of the patient</li> <li>□ How different forms of communication may be needed to share information about personalised treatment plans for brain injuries/disorders to different audiences</li> <li>□ Why brain surgery is highly-specialised in response to the physical basis of a brain injury or long term disorder</li> <li>□ How brain surgery is generally invasive, requiring the temporary removal of part of the skull and meninges</li> <li>□ How robotic surgery is carried out to enable fine control of techniques</li> <li>□ How ethical decisions must be considered when brain surgery is undertaken, including quality of life</li> <li>□ How therapeutic drugs can reduce symptom expression and further complications following a brain injury and/or the progress of a brain disorder</li> <li>□ The key features of an effective schedule or regime for the use of therapeutic drugs</li> <li>□ How a variety of lifestyle modifications can be used to treat brain disorders/injuries or reduce the impact of symptoms</li> <li>□ Benefits and limitations of managed aerobic exercise, rest periods, awareness of mental and physical wellbeing and the use of medical aids to carry out daily tasks</li> </ul> |  |  |
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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------|
| <input type="checkbox"/> Therapeutics for neurodegenerative diseases and brain injuries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <input type="checkbox"/> How different therapeutics are applied to slow the progress of neurodegenerative diseases, including the use of L-dopa for Parkinson's Disease<br><br>Does not include:<br><input type="checkbox"/> Exercise routines<br><input type="checkbox"/> Details of wellbeing programmes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |              |
| <b>4.2.2 Support via teams of healthcare professionals</b><br><input type="checkbox"/> Teams of healthcare professionals, including: <ul style="list-style-type: none"> <li>• Doctor/neurologist</li> <li>• Physiotherapist</li> <li>• Nurse</li> <li>• Occupational therapist</li> <li>• Health care support worker</li> <li>• Clinical psychologist</li> </ul><br><input type="checkbox"/> Roles of healthcare professionals within personalised treatment plans for patients with brain disorders/injuries<br><br><input type="checkbox"/> Different social care settings | To include:<br><input type="checkbox"/> Benefits and limitations of the support available via a team of healthcare professionals to support patients with brain disorders or injuries<br><input type="checkbox"/> Why not all types of healthcare professionals are involved in the treatment and support of patients with brain disorders or injuries (affected by the form of treatment/support required)<br><input type="checkbox"/> How does a team of healthcare professionals work together to provide appropriate support for patients with brain disorders/injuries<br><input type="checkbox"/> How does a team of healthcare professional share plans and outcomes with the patient and their family<br><br><input type="checkbox"/> How different healthcare professionals treat and support patients with brain disorders/injuries: <ul style="list-style-type: none"> <li>• Doctors and neurologists</li> <li>• Physiotherapists</li> <li>• Nurses</li> <li>• Occupational therapists</li> <li>• Healthcare support workers</li> <li>• Clinical psychologists</li> </ul><br><input type="checkbox"/> Why choose care at home for patients with brain disorders/injuries rather than care in a nursing home<br><br>Does not include:<br><input type="checkbox"/> Legal aspects of care |  | <b>HSW11</b> |

| <b>Topic Area 5: Monitoring and scanning the brain</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------|
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Exemplification</b>                    |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Opportunities to cover:</b>            |             |
| <b>5.1 Monitoring via electroencephalogram (EEG) readings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                              | <b>HSW</b>  |
| <b>Use of EEG readings</b> <ul style="list-style-type: none"> <li>□ Location of EEG sensors when placed on the patient</li> <li>□ Appearance of EEG readings</li> <li>□ Clinical application of EEG readings to analyse sleep patterns</li> </ul>                                                                                                                                                                                                                                                                              | To include: <ul style="list-style-type: none"> <li>□ Know why EEG sensors are placed on different parts of the body</li> <li>□ How EEG readings are used to detect electrical activity (transmission of nerve impulses) within the brain</li> <li>□ Benefits and limitations of using EEG readings to monitor brain disorders/injuries</li> <li>□ How an EEG can be used to analyse sleep patterns including the local brain clock and post-operative recovery rates</li> </ul>                                                                                                                                                                                                                                                                                                                                                        | <b>M3.1</b><br><b>M3.6</b>                | <b>HSW1</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Does not include: <ul style="list-style-type: none"> <li>□ The physics of EEG equipment</li> <li>□ The detailed interpretation of EEG readings</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                           |             |
| <b>5.2 Scanning techniques</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Maths</b>                              | <b>HSW</b>  |
| <b>Use of scanning techniques</b> <ul style="list-style-type: none"> <li>□ Features of CT, MRI, positron emission tomography (PET), X-ray and ultrasound scans</li> <li>□ Specialised scanning techniques for brain study, including:               <ul style="list-style-type: none"> <li>• Functional MRI (fMRI)</li> <li>• Iron beam scanning electron microscopy (FIB-SEM)</li> <li>• Serial section transmission electron microscopy (TEM)</li> <li>• Analysing scanned images for sports injuries</li> </ul> </li> </ul> | To include: <ul style="list-style-type: none"> <li>□ The advantages and disadvantages of CT, MRI, PET, X-ray and ultrasound scans when diagnosing/treating various brain disorders or injuries</li> <li>□ How to interpret scanned images</li> <li>□ When is it more effective to choose CT, MRI, PET, X-ray or ultrasound scanning techniques to diagnose a particular form of brain disorder/injury</li> <li>□ Why some forms of brain injury and conditions require the use of highly-specialised scanning techniques</li> <li>□ How fMRI is used in brain research and in the support of clinical interventions</li> <li>□ How FIB-SEM and serial section TEM techniques are used to observe neuronal connections/circuits in the brain</li> <li>□ How scanned images are used to identify sports injuries to the brain</li> </ul> | <b>M0.1</b><br><b>M0.2</b><br><b>M0.3</b> | <b>HSW9</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Does not include: <ul style="list-style-type: none"> <li>□ Physics of scanning equipment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |             |

## Assessment criteria

The table below gives the assessment criteria for the tasks in the set assignment for this unit. The assessment criteria indicate what is required in these tasks.

This qualification has a compensatory approach. This means that the unit grade awarded is based on the **total** number of achieved criteria for the unit (see [Section 6.4](#)). Students do **not** have to achieve **all** criteria for a specific grade to achieve that unit grade (e.g. achieve all pass criteria to achieve a pass grade).

[Section 7.4](#) provides full information on how to assess the NEA units and apply the assessment criteria. Students' work must show that all aspects of a criterion have been met in sufficient detail for it to be **successfully achieved** (see [Section 7.4.1](#)). If a student's work does not fully meet a criterion, you must not award that criterion.

The command words used in the assessment criteria are defined in [Appendix B](#).

| Pass                                                                                                                                                                                                                             | Merit                                                                                                                                       | Distinction                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1: Interpret</b> the scan image to identify those regions of the brain likely to be affected by the TBI.<br><b>P2: Draw</b> a fully annotated low-power plan diagram to show parts of the brain anatomy affected by the TBI. | <b>M1: Evaluate</b> the advantages and disadvantages of using different scanning techniques for the diagnosis of the TBI in the case study. | <b>D1: Justify</b> why an EEG should be used to confirm the impact of the TBI on nerve impulse transmission in the patient's brain. |
| <b>P3:</b> Use research to <b>describe</b> how the patient's symptoms relate to the TBI in the case study.                                                                                                                       | <b>M2: Describe</b> the wider impact of the patient's injuries on their physical and mental wellbeing.                                      | <b>D2: Explain</b> whether the spinal cord and nerves are affected by the TBI in the case study.                                    |
| <b>P4:</b> Use research to <b>describe</b> how a range of relevant potential treatments could be appropriate for the TBI patient.                                                                                                | <b>M3: Evaluate two</b> physical treatments and <b>two</b> psychological treatments which are needed to aid recovery of the patient.        | <b>D3: Analyse</b> how the options chosen for pain management affect the patient on a cellular level.                               |
| <b>P5: Create</b> a logical treatment plan, containing all key components to meet the physical, psychological and personal needs of the patient.                                                                                 |                                                                                                                                             |                                                                                                                                     |
| <b>P6: Design</b> a relevant schedule for drug prescription for the TBI patient.                                                                                                                                                 |                                                                                                                                             |                                                                                                                                     |
| <b>P7: Describe</b> what contributions are required to be made by the specialists and non-specialists involved in the treatment plan.                                                                                            | <b>M4: Discuss</b> the use of different teams of healthcare professionals to support the patient.                                           |                                                                                                                                     |

| Pass                                                                                                                                                                   | Merit                                                                                                                                                                                | Distinction                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>P8: Create</b> an appropriate presentation of the treatment plan for the specialists identified in <b>Task 2</b> .                                                  | <b>M5: Explain</b> the most appropriate way for scientific terminology used in the presentation for the specialists to be communicated with the non-specialists.                     | <b>D4: Justify</b> the content of the presentation by detailing the scientific reasoning behind its inclusion.               |
| <b>P9: Suggest four</b> adaptations to the presentation so that it can be used to communicate the treatment plan to the non-specialists in the case study effectively. | <b>M6: Explain</b> the adaptations suggested to the presentation in <b>P9</b> so that the non-specialists in the case study can understand their contribution to the treatment plan. |                                                                                                                              |
| <b>P10: Draw</b> a simplified low power plan diagram to show parts of the brain anatomy affected by the TBI for the non-specialists in the case study.                 |                                                                                                                                                                                      |                                                                                                                              |
| <b>P11: Summarise</b> the feedback received for your treatment plan.                                                                                                   | <b>M7: Assess</b> the strengths and weaknesses of the information used in the creation of treatment plan for the TBI patient.                                                        | <b>D5: Justify</b> any potential improvements to the information used in the creation of treatment plan for the TBI patient. |
| <b>P12: Analyse</b> the strengths and weaknesses of the materials created to present information to the specialists and suggested adaptations for the non-specialists. |                                                                                                                                                                                      |                                                                                                                              |

### Assessment guidance

This assessment guidance gives you information relating to the assessment criteria. There might not be additional assessment guidance for each assessment criterion. It is included only where it is needed.

| Assessment Criteria | Assessment guidance                                                                                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1                  | <ul style="list-style-type: none"> <li>Students need to interpret the scan image shown in the case study for the TBI patient.</li> <li>Students must recognise the prominent part(s) of the brain damaged at the site of the injury and the part(s) showing signs of damage, as relevant to the scan from the case study.</li> </ul>           |
| P2                  | <ul style="list-style-type: none"> <li>The interpretation of the scan image could be written only but to achieve <b>P2</b> a diagrammatic model must be included to demonstrate the parts of the brain affected by the TBI. This could be presented via either a vertical section (VS) or transverse section (TS) of brain anatomy.</li> </ul> |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P3 | <ul style="list-style-type: none"> <li>The symptoms shown by the TBI patient are outlined in the case study. Symptoms may have been recorded before and/or following surgery.</li> <li>Students must research how symptoms of TBIs link to brain structure and function.</li> <li>Students must apply their research to the information from the case study.</li> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul>             |
| M1 | <ul style="list-style-type: none"> <li>The case study confirms that the image is the product of a scanning technique. The image reveals the site of injury and of damaged tissue.</li> <li>Students must evaluate the advantages and disadvantages of the scanning technique from the case study.</li> <li>Students must also evaluate the advantages and disadvantages of using <b>two other</b> scanning techniques for the diagnosis of the TBI in the case study.</li> </ul>                                                                                             |
| M2 | <ul style="list-style-type: none"> <li>Students must describe the wider impact of the patient's injuries on their physical and mental well-being.</li> <li>The patient's injuries could be considered to be any from the range of symptoms and behaviours shown by the patient in the case study.</li> </ul>                                                                                                                                                                                                                                                                 |
| D1 | <ul style="list-style-type: none"> <li>Students need to give valid reasons why some of the symptoms shown by the patient in the case study are the product of a change to nerve impulse transmission. This forms the justification that the change can be confirmed via an EEG.</li> </ul>                                                                                                                                                                                                                                                                                   |
| D2 | <ul style="list-style-type: none"> <li>Students must explain whether the spinal cord and nerves are affected by the TBI for the patient in the case study. This might involve the link between the spinal cord and the brain, as well as the role of cranial versus spinal nerves.</li> </ul>                                                                                                                                                                                                                                                                                |
| P4 | <ul style="list-style-type: none"> <li>Students must identify a range of <b>at least three</b> potential physical treatments and <b>at least three</b> psychological treatments that could be appropriate for the TBI patient.</li> <li>For each treatment students must describe how each treatment is appropriate for the TBI patient in the case study.</li> <li>The research element of this criterion does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criterion.</li> </ul>               |
| P6 | <ul style="list-style-type: none"> <li>Students must design a relevant drug prescription for the TBI patient based on the information in the case study. The drugs prescribed could be to either treat or reduce the symptoms shown by the patient.</li> <li>An explanation of how drugs affect nerve impulse transmission is <b>not</b> expected for this assessment criterion.</li> </ul>                                                                                                                                                                                  |
| P7 | <ul style="list-style-type: none"> <li><b>P7</b> is an extension of the treatment plan created in <b>P4</b>.</li> <li>Students must describe the contributions of the most appropriate specialists needed to treat and support the patient (for example, doctor, physiotherapist, clinical psychologist, etc) as appropriate to the case study.</li> <li>Students must describe the contributions of the most appropriate non-specialists needed to support the patient (for example, the patient, family members, carers, etc) as appropriate to the case study.</li> </ul> |

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| M3  | <ul style="list-style-type: none"> <li>• <b>M3</b> is an extension of <b>P4</b>.</li> <li>• Students must evaluate <b>two</b> physical treatments and <b>two</b> psychological treatments in the context of the patient. The treatments evaluated need to come from those described in <b>P4</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| M4  | <ul style="list-style-type: none"> <li>• <b>M4</b> is an extension of <b>P7</b>.</li> <li>• Students must discuss how different teams of healthcare professionals will be used to support the patient.</li> <li>• The specific healthcare teams discussed will depend on the case study context. All relevant healthcare teams should be discussed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                          |
| D3  | <ul style="list-style-type: none"> <li>• Students must analyse how the options chosen for pain management, as part of the treatment plan and/or drug prescription schedule, affect the patient on a cellular level.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P8  | <ul style="list-style-type: none"> <li>• Students must create a presentation for the specialists identified in the treatment plan in <b>Task 2</b>. The presentation should be in the format they feel is most appropriate, which could include a poster, a PowerPoint presentation, a flow diagram, etc.</li> <li>• There must be sufficient detail in the presentation to demonstrate the key components of the treatment plan appropriate for the specialists.</li> </ul>                                                                                                                                                                                                                                                                            |
| P9  | <ul style="list-style-type: none"> <li>• Having created the presentation for the specialists, students must consider how it could be adapted to be relevant and accessible for the non-specialists from the case study.</li> <li>• Students might choose to create a further presentation to highlight the adaptations needed or they might choose to suggest adaptations in a different format, for example a table.</li> <li>• Adaptations suggested should focus on the changes to the presentation required, for example different parts of the plan which should be concentrated on, information which could be removed or added, etc. Amendments should not focus on changes to scientific terminology which will be considered in M5.</li> </ul> |
| M5  | <ul style="list-style-type: none"> <li>• Students must explain the most appropriate way for <b>at least three</b> examples of scientific terminology used in the presentation for the specialists to be modified to be communicated with the non-specialists.</li> <li>• Students could choose to consider ways the terminology might be scaffolded, re-phrased, amended or why they would need to use a particular scientific term as it is.</li> </ul>                                                                                                                                                                                                                                                                                                |
| M6  | <ul style="list-style-type: none"> <li>• <b>M6</b> is an extension of <b>P9</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| D4  | <ul style="list-style-type: none"> <li>• Students must justify the content of the presentation for the specialists by detailing the scientific reasoning.</li> <li>• Students will use their understanding of the unit content to provide valid reasons for the content's inclusion.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| P11 | <ul style="list-style-type: none"> <li>• Students must clearly express the most important points stemming from the feedback received for their treatment plan in a short and clear form.</li> <li>• The feedback for their treatment plan might be provided by the teacher and/or other students.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| M7  | <ul style="list-style-type: none"> <li>• The information used in the creation of the treatment plan might include the <b>case study</b>, <b>Task 1</b> and/or <b>Task 2</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



**Synoptic assessment**

Some of the knowledge, understanding and skills needed to complete this unit will draw on the learning in Units F170 and F171.

This table details these synoptic links.

| Unit F176: The brain |                                          | Unit F170: Fundamentals of human biology |                                                              |
|----------------------|------------------------------------------|------------------------------------------|--------------------------------------------------------------|
| Topic Area           |                                          | Topic Area                               |                                                              |
| 1                    | Structure and function of nervous system | 3                                        | Key concepts in endocrinology, neurobiology and reproduction |
| 2                    | Neuron communication and control         | 2                                        | Human physiology, organs and systems                         |

| Unit F176: The brain |                                                         | Unit F171: Health and disease |                                                 |
|----------------------|---------------------------------------------------------|-------------------------------|-------------------------------------------------|
| Topic Area           |                                                         | Topic Area                    |                                                 |
| 4                    | The diagnosis and treatment of brain disorders/injuries | 2                             | Curative, management and preventative therapies |
|                      |                                                         | 4                             | Techniques for diagnosis and monitoring         |

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

### 5.3.6 Unit F177: Drug development

#### Unit Aim

There are many different types of diseases and medical conditions and thousands of medicines and drugs that have been produced to help people who need treatment. Medicines are used to treat or prevent disease and have been used for thousands of years. Many different herbs and plants have been used, not only in the past but also now, to provide natural materials from which modern medicines are extracted and developed. Drug manufacture is changing and now drugs are usually made synthetically or semi-synthetically. The process of drug development is long and expensive, and so scientists must carefully consider a variety of factors before moving through each stage.

In this unit you will look at the different properties of pharmaceutical drugs and how these properties influence the development of future drugs. You will learn how the stages in the development of a drug, including pre-clinical and clinical trials are completed. You will consider the importance of clinical trials to determining the efficacy and safety of the potential drug. You will also learn the stages in the development and the discovery of a commercial drug/medicine and how pre-clinical and clinical trials, associated with the safety of the drug, are completed. Finally, you learn how to prepare a presentation for a panel that represents stakeholders who will approve funding for a new drug that is being developed.

| Unit F177: Drug development                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |                            |
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| Topic Area 1: Pharmaceutical drugs                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |                            |
| Teaching content                                                                                                                                                                                                                                                                                                                               | Exemplification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Opportunities to cover:    |                            |
| 1.1 Classification of drugs                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Maths                      | HSW                        |
| <b>The classification of pharmaceutical drugs</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Stimulants</li> <li><input type="checkbox"/> Depressants</li> <li><input type="checkbox"/> Hallucinogens</li> <li><input type="checkbox"/> Cannabinoids</li> <li><input type="checkbox"/> Opioids</li> </ul>                 | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Purpose of each type of pharmaceutical drug</li> <li><input type="checkbox"/> How pharmaceutical drugs are classified</li> </ul><br><b>Examples of how pharmaceutical drugs are classified</b> may include: <ul style="list-style-type: none"> <li><input type="checkbox"/> General structure</li> <li><input type="checkbox"/> Mechanism of action</li> <li><input type="checkbox"/> Intended therapeutic use</li> <li><input type="checkbox"/> Potential for abuse</li> </ul><br>Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Detailed chemical mechanisms of actions</li> <li><input type="checkbox"/> Detailed structure</li> </ul> |                            | <b>HSW8</b><br><b>HSW9</b> |
| 1.2 Properties of drugs                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Maths                      | HSW                        |
| <b>General properties of drugs</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Pharmacodynamics</li> <li><input type="checkbox"/> Pharmacokinetics</li> <li><input type="checkbox"/> Toxicity</li> <li><input type="checkbox"/> Adverse drug reactions</li> <li><input type="checkbox"/> Drug-drug interactions</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How each property needs to be considered when developing a new drug</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>M1.6</b><br><b>M2.1</b> | <b>HSW6</b>                |

| <b>1.3 Actions of drugs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                   | <b>HSW</b>   |
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| <b>Mechanism of action of drugs</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Receptor activation</li> <li><input type="checkbox"/> Agonists and antagonists</li> <li><input type="checkbox"/> Enzyme inhibition</li> <li><input type="checkbox"/> Transporter inhibition</li> <li><input type="checkbox"/> Non-specific drug action</li> <li><input type="checkbox"/> Gene expression modulation</li> </ul>                                                      | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> The general steps of mechanism of action:               <ul style="list-style-type: none"> <li>• Binding</li> <li>• Activation</li> <li>• Signal transduction</li> <li>• Effect</li> </ul> </li> <li><input type="checkbox"/> Advantages and disadvantages of each drug action</li> </ul> Does not include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Detailed process of each mechanism of action</li> </ul>      |                                | <b>HSW11</b> |
| <b>1.4 Drug delivery</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                   | <b>HSW</b>   |
| <b>Routes of drug delivery:</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Oral</li> <li><input type="checkbox"/> Rectal</li> <li><input type="checkbox"/> Injectable</li> <li><input type="checkbox"/> Transdermal</li> <li><input type="checkbox"/> Inhalational</li> <li><input type="checkbox"/> Topical</li> <li><input type="checkbox"/> Transnasal</li> <li><input type="checkbox"/> Vaginal</li> <li><input type="checkbox"/> Intraosseous</li> </ul>      | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> How the drug travels through the body from each delivery method</li> <li><input type="checkbox"/> How the method of delivery affects the amount of drug reaching the site of action</li> <li><input type="checkbox"/> How the chemical properties of the drug affect the permissible drug delivery</li> <li><input type="checkbox"/> Advantages and disadvantages of each route of drug delivery into the body</li> </ul>                |                                | <b>HSW4</b>  |
| <b>Topic Area 2: Process of drug development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |              |
| <b>Teaching content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Opportunities to cover:</b> |              |
| <b>2.1 The process of drug development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Maths</b>                   | <b>HSW</b>   |
| <b>2.1.1 The phases of drug development</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Discovery</li> <li><input type="checkbox"/> Preclinical Research</li> <li><input type="checkbox"/> Clinical Research</li> <li><input type="checkbox"/> Regulatory Approval</li> <li><input type="checkbox"/> Post market surveillance</li> </ul>                                                                                                                            | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> The purpose of each phase of drug development</li> <li><input type="checkbox"/> The challenges of drug development</li> </ul> Examples of <b>the challenges of drug development</b> may include: <ul style="list-style-type: none"> <li><input type="checkbox"/> Cost</li> <li><input type="checkbox"/> Development time</li> <li><input type="checkbox"/> Failure rate</li> <li><input type="checkbox"/> Regulatory approval</li> </ul> | <b>M0.3</b>                    | <b>HSW4</b>  |
| <b>2.1.2 The researchers involved in drug development</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Research Scientist</li> <li><input type="checkbox"/> Computational Biologist</li> <li><input type="checkbox"/> Pharmacologist</li> <li><input type="checkbox"/> Toxicologist</li> <li><input type="checkbox"/> Clinical Scientist</li> <li><input type="checkbox"/> Regulatory Affairs specialist</li> <li><input type="checkbox"/> Medical writer</li> </ul> | To include: <ul style="list-style-type: none"> <li><input type="checkbox"/> The role of each researcher</li> <li><input type="checkbox"/> Which phase(s) each researcher is most likely to be involved in</li> </ul>                                                                                                                                                                                                                                                                                                 |                                | <b>HSW11</b> |

| <b>2.2 Discovery</b>                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>               | <b>HSW</b>   |
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| <b>Discovery of new drugs</b> <ul style="list-style-type: none"> <li>□ New insights into a disease process and identifying new targets</li> <li>□ Designing new compounds</li> <li>□ Screening natural products</li> <li>□ Existing treatments with unanticipated effects</li> <li>□ New technologies</li> </ul> | To include: <ul style="list-style-type: none"> <li>□ The importance of discovering new drugs</li> <li>□ The use of computer modelling to determine viable potential drug candidates to go onto preclinical research</li> <li>□ The use of cell lines to determine viable potential drug candidates to go onto preclinical research</li> </ul>                                                                                                                                                                                                                                                                                                                                                  | <b>M1.1</b><br><b>M1.2</b> | <b>HSW1</b>  |
| <b>2.3 Preclinical research</b>                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>               | <b>HSW</b>   |
| <b>The purpose of preclinical research in animals</b>                                                                                                                                                                                                                                                            | To include: <ul style="list-style-type: none"> <li>□ The need for testing drug candidates in animals before humans</li> </ul> <p>Examples of <b>the need for testing drug candidates in animals</b> may include:</p> <ul style="list-style-type: none"> <li>□ Best dosage</li> <li>□ Best method of delivery</li> <li>□ Side effects and toxicity</li> <li>□ Potential benefits</li> <li>□ How it is absorbed, distributed, metabolised and excreted</li> </ul>                                                                                                                                                                                                                                | <b>M1.6</b>                | <b>HSW12</b> |
| <b>2.4 Clinical research</b>                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Maths</b>               | <b>HSW</b>   |
| <b>2.4.1 The process of testing drug candidates in humans</b> <ul style="list-style-type: none"> <li>□ Phase 1: A small number of healthy volunteers</li> <li>□ Phase 2: A larger group of volunteers with the condition</li> <li>□ Phase 3: Several thousand patients with the condition</li> </ul>             | To include: <ul style="list-style-type: none"> <li>□ What factors researchers need to consider when designing each phase</li> <li>□ Why it's important to consider these factors when designing clinical research phases</li> <li>□ What researchers need to consider when selecting participants for clinical research</li> </ul> <p>Examples of <b>factors to be considered when designing each phase</b> may include:</p> <ul style="list-style-type: none"> <li>□ How long the study will last</li> <li>□ What assessments will be conducted</li> <li>□ What data will be collected and when</li> <li>□ How many participants are needed</li> <li>□ Efficacy and dosage results</li> </ul> | <b>M1.2</b><br><b>M1.6</b> | <b>HSW3</b>  |

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|                                                                                                                                                                                                                                                                                                                                                    | <p>Examples of <b>what needs to be considered when selecting participants</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Age</li> <li><input type="checkbox"/> Sex</li> <li><input type="checkbox"/> Ethnicity</li> <li><input type="checkbox"/> Severity of condition</li> </ul>                                                                                                                                                                                    |              |              |
| <b>2.4.2 Limiting research bias</b>                                                                                                                                                                                                                                                                                                                | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Why it is important to limit research bias</li> <li><input type="checkbox"/> Methods to limit research bias in clinical research</li> </ul>                                                                                                                                                                                                                                                                             |              | <b>HSW3</b>  |
| <b>2.4.3 Importance of clinical research</b>                                                                                                                                                                                                                                                                                                       | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How researchers determine safe and effective dosages</li> <li><input type="checkbox"/> The role of clinical research in determining side-effects</li> <li><input type="checkbox"/> Advantages and disadvantages of each phase of clinical research</li> </ul>                                                                                                                                                           |              | <b>HSW10</b> |
| <b>2.5 Regulatory approval</b>                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b> | <b>HSW</b>   |
| <b>2.5.1 Regulatory approval</b><br>Purpose of regulatory approval                                                                                                                                                                                                                                                                                 | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> What must be submitted to regulators for a license: <ul style="list-style-type: none"> <li>• Preclinical data and analyses</li> <li>• All clinical trial data and analyses</li> <li>• Proposed labelling</li> <li>• Safety updates</li> <li>• Drug abuse information</li> <li>• Directions for use</li> </ul> </li> </ul>                                                                                               |              | <b>HSW8</b>  |
| <b>2.5.2 Legislation</b> <ul style="list-style-type: none"> <li><input type="checkbox"/> Medicines Act 1968</li> <li><input type="checkbox"/> Human Medicines Regulations 2012</li> <li><input type="checkbox"/> Medicines for Human Use (Clinical Trials) Regulations 2004</li> <li><input type="checkbox"/> Drug Trafficking Act 1994</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How each piece of legislation influences drug development</li> <li><input type="checkbox"/> The role of the Medicines and Healthcare Products Regulatory Agency (MHRA) in relation to legislation</li> <li><input type="checkbox"/> Key principles that underpin the legislation: <ul style="list-style-type: none"> <li>• Safety</li> <li>• Efficacy</li> <li>• Quality</li> <li>• Transparency</li> </ul> </li> </ul> |              | <b>HSW7</b>  |
| <b>2.6 Post market surveillance</b>                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Maths</b> | <b>HSW</b>   |
| <b>Post market surveillance</b>                                                                                                                                                                                                                                                                                                                    | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Importance of post-market surveillance</li> <li><input type="checkbox"/> Benefits and challenges of post-market surveillance</li> </ul>                                                                                                                                                                                                                                                                                 |              | <b>HSW11</b> |

| Topic Area 3: Factors influencing drug development                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |       |
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| Teaching content                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Exemplification         |       |
|                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Opportunities to cover: |       |
| 3.1 Stakeholders                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Maths                   | HSW   |
| <b>Stakeholder groups involved in drug development</b> <ul style="list-style-type: none"> <li>□ Researchers</li> <li>□ Pharmaceutical companies</li> <li>□ Academic institutions</li> <li>□ Regulatory agencies</li> <li>□ Patient advocacy groups</li> <li>□ Healthcare providers</li> <li>□ Funding providers</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li>□ The role of each stakeholder group in drug development</li> <li>□ How these stakeholder groups collaborate to develop drugs</li> <li>□ How to communicate effectively to these different stakeholder groups</li> <li>□ What constitutes success for different stakeholder groups involved in drug development</li> </ul> <p>Examples of <b>stakeholders</b> may include:</p> <ul style="list-style-type: none"> <li>□ <b>Researchers</b> – pharmacologist, clinical researcher, medical writer</li> <li>□ <b>Pharmaceutical companies</b> – Pharmacologists, quality assurance professionals, regulatory affairs professionals</li> <li>□ <b>Academic institutions</b> – Research technicians, toxicologists, clinicians</li> <li>□ <b>Regulatory agencies</b> – Clinical reviewers, regulatory affairs professionals,</li> <li>□ <b>Patient advocacy groups</b> – Policymakers, advocates, patients, legal experts</li> <li>□ <b>Healthcare providers</b> – Bioethicists, legal experts, nurses</li> <li>□ <b>Funding providers</b> – government agencies, philanthropic organisations, venture capitalists</li> </ul> | M1.7                    | HSW11 |
| 3.2 Ethical considerations                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Maths                   | HSW   |
| <b>Ethical considerations in drug development</b>                                                                                                                                                                                                                                                                          | <p>To include:</p> <ul style="list-style-type: none"> <li>□ What the ethical considerations are when developing drugs</li> <li>□ How each ethical consideration may affect the process of drug development</li> <li>□ How each ethical consideration can be addressed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         | HSW9  |

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|                                                                                                                                                                                   | <p>Examples of <b>ethical considerations</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Safety of patients</li> <li><input type="checkbox"/> Efficacy of drugs</li> <li><input type="checkbox"/> Informed consent of patients</li> <li><input type="checkbox"/> Fair distribution of drugs</li> <li><input type="checkbox"/> Use of animals in research</li> <li><input type="checkbox"/> Payment of research participants</li> <li><input type="checkbox"/> Marketing of drugs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |              |
| <b>3.3 Market considerations</b>                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                   | <b>HSW</b>   |
| <b>Market considerations affecting decisions around drug development</b>                                                                                                          | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How each market consideration can impact the process of drug development</li> <li><input type="checkbox"/> The importance of considering market factors when deciding which drugs to develop</li> <li><input type="checkbox"/> How market factors may affect decisions through the drug development process</li> </ul> <p>Examples of <b>market considerations</b> may include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Size of target market</li> <li><input type="checkbox"/> Unmet medical need</li> <li><input type="checkbox"/> Cost of drug development</li> <li><input type="checkbox"/> Regulatory environment</li> <li><input type="checkbox"/> Competition</li> <li><input type="checkbox"/> Reimbursement landscape</li> <li><input type="checkbox"/> Patient advocacy</li> <li><input type="checkbox"/> Public Perception</li> </ul> | <b>M1.6</b>                    | <b>HSW9</b>  |
| <b>Topic Area 4: Producing a clinical research proposal</b>                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |              |
| <b>Teaching content</b>                                                                                                                                                           | <b>Exemplification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Opportunities to cover:</b> |              |
| <b>4.1 Clinical Research Proposal</b>                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maths</b>                   | <b>HSW</b>   |
| <ul style="list-style-type: none"> <li><input type="checkbox"/> Producing a pitch</li> <li><input type="checkbox"/> Communicating the pitch to a range of stakeholders</li> </ul> | <p>To include:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> How to write a clinical research proposal</li> <li><input type="checkbox"/> How to design a presentation of the clinical research proposal that is appropriate for stakeholders involved in drug development</li> <li><input type="checkbox"/> How to communicate an appropriate clinical research proposal to a variety of drug development stakeholders</li> <li><input type="checkbox"/> How to assess the quality of a clinical research proposal pitch</li> <li><input type="checkbox"/> How to obtain appropriate feedback on a research proposal pitch and then summarise the feedback</li> </ul>                                                                                                                                                                                                                                                                     |                                | <b>HSW11</b> |

## Assessment criteria

The table below gives the assessment criteria for the tasks in the set assignment for this unit. The assessment criteria indicate what is required in these tasks.

This qualification has a compensatory approach. This means that the unit grade awarded is based on the **total** number of achieved criteria for the unit (see [Section 6.4](#)). Students do **not** have to achieve **all** criteria for a specific grade to achieve that unit grade (e.g. achieve all pass criteria to achieve a pass grade).

[Section 7.4](#) provides full information on how to assess the NEA units and apply the assessment criteria. Students' work must show that all aspects of a criterion have been met in sufficient detail for it to be **successfully achieved** (see [Section 7.4.1](#)). If a student's work does not fully meet a criterion, you must not award that criterion.

The command words used in the assessment criteria are defined in [Appendix B](#).

| Pass                                                                                                                                               | Merit                                                                                                                               | Distinction                                                                                                                        |
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| <b>P1:</b> Use research to <b>compare</b> the properties of other drugs with a similar aim to the new drug being developed.                        | <b>M1: Explain</b> how the properties of the new drug will affect the development process.                                          |                                                                                                                                    |
| <b>P2:</b> Use research to <b>describe</b> the effects of other drugs with a similar aim as the new drug being developed.                          | <b>M2:</b> Use research to <b>summarise</b> the different market factors which may impact on the development of the new drug.       |                                                                                                                                    |
| <b>P3:</b> Use research to <b>explain</b> <b>three</b> ways that specific legislation will affect the development of the new drug being developed. |                                                                                                                                     |                                                                                                                                    |
| <b>P4: Create</b> a written proposal describing the clinical trial phases of the development of the new drug.                                      | <b>M3: Explain</b> the chosen participation groups in each phase of the clinical trials in terms of their validity and reliability. | <b>D1: Justify</b> the decisions made in the written proposal with scientific rationale.                                           |
| <b>P5: Explain</b> how it can be determined whether the suggested dosage is safe and effective during the development of the new drug.             |                                                                                                                                     | <b>D2: Evaluate</b> the risk of side effects beyond those identified in pre-clinical trials for the new drug.                      |
| <b>P6: Explain</b> how the properties of the new drug influence the purpose of each phase of the clinical trial.                                   |                                                                                                                                     |                                                                                                                                    |
| <b>P7: Explain</b> the roles of the various stakeholders involved in the development of the new drug.                                              | <b>M4: Discuss</b> potential success criteria for the various stakeholders of the new drug.                                         | <b>D3: Assess</b> the ethical considerations of the development of the new drug.                                                   |
| <b>P8: Create</b> an appropriate presentation which summarises the drug development proposal.                                                      | <b>M5: Explain</b> how the presentation has been tailored to all of the different members of the panel.                             | <b>D4: Justify</b> the inclusion and omission of content from the written proposal in the presentation using scientific reasoning. |



| Pass                                                                                                                                                    | Merit                                                                                             | Distinction                                                                                                                            |
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| <b>P9: Deliver</b> the presentation to the intended audience, with explanations of rationale beyond what is included in the presentation documentation. |                                                                                                   |                                                                                                                                        |
| <b>P10: Summarise</b> the feedback received for your presentation.                                                                                      | <b>M6: Discuss</b> the strengths and weaknesses of your drug development proposal.                | <b>D5: Assess</b> how your drug development proposal could be improved to provide the greatest chance of success of receiving funding. |
| <b>P11: Analyse</b> how the presentation of your pitch could be improved.                                                                               |                                                                                                   |                                                                                                                                        |
| <b>P12: Explain</b> how <b>three</b> other pieces of information would have been useful when creating the drug development proposal.                    | <b>M7: Evaluate</b> how the information suggested in <b>P12</b> might have affected the proposal. |                                                                                                                                        |

### Assessment guidance

This assessment guidance gives you information relating to the assessment criteria. There might not be additional assessment guidance for each assessment criterion. It is included only where it is needed.

| Assessment Criteria | Assessment guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task 1              | <ul style="list-style-type: none"> <li>The research element of the criteria in this Task does <b>not</b> need to be completed under teacher supervised conditions but is necessary in order for students to access the criteria.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| P1                  | <ul style="list-style-type: none"> <li>Students must research the properties of other drugs with a similar aim to the new drug being developed.</li> <li>'Other drugs with a similar aim' might be, for example, other drugs to treat infections (could be to treat a different area of the body than given in the scenario) or the type of drug (e.g. antimicrobial drugs, antibacterial, antifungal, anti-inflammatory, antiviral).</li> <li>'Properties' means different features such as dosage, resistance, routes of administration, strength.</li> <li>Students must use their research to compare the properties of other drugs with the new drug being developed.</li> </ul> |
| P2                  | <ul style="list-style-type: none"> <li>The competitor drugs focused on in <b>P2</b> must be the drugs compared to the new drug in <b>P1</b>.</li> <li>Students must describe the effects of similar drugs on the market - including side-effects.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P3                  | <ul style="list-style-type: none"> <li>Students must use research to explain <b>three</b> ways that specific legislation will affect the development of the new drug being developed.</li> <li>The three different ways could come from one or multiple pieces of legislation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                             |
| P4                  | <ul style="list-style-type: none"> <li>The written proposal must cover the clinical trial phases of clinical research, regulatory approval and post market surveillance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P5                  | <ul style="list-style-type: none"> <li>Students must focus on the specific features of the new drug in the case study to explain how to determine that the suggested dosage given is safe and would fulfil the aim whilst limiting the side-effects given.</li> <li>Students can use their research from <b>Task 1</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M4     | <ul style="list-style-type: none"> <li>• <b>M4</b> is an extension of <b>P7</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| D1     | <ul style="list-style-type: none"> <li>• Students must justify the decisions made in the written proposal using scientific rationale.</li> <li>• Students will use their understanding of the unit content to provide valid reasons for the decisions made.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Task 3 | <ul style="list-style-type: none"> <li>• Presentations will need to be aimed at a length of 5 minutes, but flexibility should be allowed.</li> <li>• Students can either deliver the presentation to the teacher, peers or a combination of both. If the presentation is delivered to peers only, this must be recorded, so that the teacher can use the recording to complete the Teacher Observation Record for <b>P9</b> (you do <b>not</b> need to submit this for moderation).</li> <li>• The focus of other members of the drug development team is from the scenario. There is no requirement for the presentation to take place in front of a certain number of other students.</li> <li>• Students can create their presentation in the format they feel is most appropriate. This could include a poster, a PowerPoint presentation, a flow diagram, etc.</li> </ul> |
| P9     | <ul style="list-style-type: none"> <li>• Teachers must complete a Teacher Observation Record for each student to evidence they have met the criteria. Students must also read and sign it.</li> <li>• The Teacher Observation Record form should describe in detail how the student delivered the presentation to the intended audience, with explanations of rationale beyond what is included in the presentation documentation.</li> <li>• The intended audience is the panel members given in the scenario.</li> </ul>                                                                                                                                                                                                                                                                                                                                                     |
| D4     | <ul style="list-style-type: none"> <li>• Students must apply knowledge and understanding from the unit content learnt to give valid reasons for the inclusion or omission of content from the written proposal in their presentation. This will form their justification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| P10    | <ul style="list-style-type: none"> <li>• Students must clearly express the most important points stemming from the feedback received for their presentation in a short and clear form.</li> <li>• The feedback for the presentation might be provided by the teacher and/or other students.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| M7     | <ul style="list-style-type: none"> <li>• <b>M7</b> is an extension of <b>P12</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Synoptic assessment

Some of the knowledge, understanding and skills needed to complete this unit will draw on the learning in Unit F171.

This table details these synoptic links.

| Unit F177: Drug development |                             | Unit F171: Health and disease |                                                |
|-----------------------------|-----------------------------|-------------------------------|------------------------------------------------|
| Topic Area                  |                             | Topic Area                    |                                                |
| 1                           | Pharmaceutical drugs        | 2                             | Curative management and preventative therapies |
| 2                           | Process of drug development | 5                             | Reporting, research and confidentiality        |

More information about synoptic assessment in these qualifications can be found in [Section 6.2 Synoptic Assessment](#).

## 6 Assessment and grading

### 6.1 Overview of the assessment

|                     |                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Entry code          | H049                                                                                                                   |
| Qualification title | Cambridge OCR Level 3 Alternative Academic Qualification<br>Cambridge Advanced National in Human Biology (Certificate) |
| GLH                 | 180*                                                                                                                   |
| Reference           | 610/3945/7                                                                                                             |
| Total units         | Has three units: <ul style="list-style-type: none"> <li>• Mandatory units F170, F172, F173</li> </ul>                  |

|                     |                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Entry code          | H149                                                                                                                                                                 |
| Qualification title | Cambridge OCR Level 3 Alternative Academic Qualification<br>Cambridge Advanced National in Human Biology (Extended Certificate)                                      |
| GLH                 | 360*                                                                                                                                                                 |
| Reference           | 610/3946/9                                                                                                                                                           |
| Total units         | Has six units: <ul style="list-style-type: none"> <li>• Mandatory units F170, F171, F172, F173</li> <li>• and two other units from F174, F175, F176, F177</li> </ul> |

\*the GLH includes assessment time for each unit

**Unit F170: Fundamentals of human biology**

80 GLH

1 hour 15 minute written exam

60 marks (60 UMS)

Set and marked by us

Calculators are required in this exam.

The exam has one part and a range of item types will be used in this assessment including:

- Forced choice/controlled response questions - typically 1 mark but a maximum of four marks for a single MCQ.
- Short answer, closed response questions (with or without diagrams) – typically 1 to 4 marks.
- Short answer with calculation/working – typically 1 to 4 marks.
- Extended constructed response with points-based mark scheme – 1 mark per factor or feature to a stated maximum, typically 1 to 4 marks.

**Unit F171: Health and disease**

80 GLH

1 hour 15 minute written exam

60 marks (60 UMS)

Set and marked by us

Calculators are required in this exam.

The exam has one part and a range of item types will be used in this assessment including:

- Forced choice/controlled response questions - typically 1 mark but a maximum of four marks for a single MCQ.
- Short answer, closed response questions (with or without diagrams) – typically 1 to 4 marks.
- Short answer with calculation/working – typically 1 to 4 marks.
- Extended constructed response with points-based mark scheme – typically 1 to 4 marks, 1 mark per factor or feature to a stated maximum.
- Extended constructed response with levels of response mark scheme – one 6 mark question and one 9 mark question.

### **Unit F172: Genetics**

50 GLH

Our set assignments

Centre-assessed and moderated by us

This set assignment has 3 practical tasks.

We have estimated that this assignment will take about 15 hours of supervised time and 12 hours of unsupervised time to complete.

### **Unit F173: Biomedical techniques**

50 GLH

Our set assignments

Centre-assessed and moderated by us

This set assignment has 4 practical tasks.

It should take about 20 hours of supervised time and 5 hours of unsupervised time to complete.

### **Unit F174: Nutrition and metabolism**

50 GLH

Our set assignments

Centre-assessed and moderated by us

This set assignment has 4 practical tasks.

It should take about 15 hours of supervised time and 10 hours of unsupervised time to complete.

### **Unit F175: Human reproduction**

50 GLH

Our set assignments

Centre-assessed and moderated by us

This set assignment has 4 practical tasks.

We have estimated that this assignment will take about 20 hours of supervised time and 4 hours of unsupervised time to complete.

**Unit F176: The brain**

50 GLH

Our set assignments

Centre-assessed and moderated by us

This set assignment has 4 practical tasks.

It should take about 20 hours of supervised time and 4 hours of unsupervised time to complete.

**Unit F177: Drug development**

50 GLH

Our set assignments

Centre-assessed and moderated by us

This set assignment has 4 practical tasks.

It should take about 15 hours of supervised time and 10 hours of unsupervised time to complete.

The set assignments for NEA units are on our secure website, [Teach Cambridge](#). Each NEA assignment is live for two years. The intended cohort is shown on the front cover. It is important you use the correct NEA set assignment for each cohort, as starting a new cohort of Year 12 students on an NEA set assignment that has already been live for one year will mean that these students will only have one year to work on the assignment.

## 6.2 Synoptic assessment

Synoptic assessment is a built-in feature of these qualifications. It means that students need to use an appropriate selection of their knowledge, understanding and skills developed across each qualification in an integrated way and apply them to a key task or tasks.

This helps students to build a holistic understanding of the subject and the connections between different elements of learning, so they can go on to apply what they learn from these qualifications to new and different situations and contexts.

The externally assessed units allow students to gain underpinning knowledge and understanding relevant to human biology. The NEA units draw on and strengthen this learning by assessing it in an applied and practical way.

It is important to be aware of the synoptic links between the units so that teaching, learning and assessment can be planned accordingly. Then students can apply their learning in ways which show they are able to make connections across the qualification. [Section 5.3](#) shows the synoptic links for each unit.

## 6.3 Transferable skills

These qualifications give students the opportunity to gain broad, transferable skills and experiences that they can apply in future study, employment and life.

Higher Education Institutions (HEIs) have told us that developing some of these skills helps students to transition into higher education.

These skills include:

- Communication
- Creativity
- Critical thinking
- Independent learning
- Presentation skills
- Problem solving
- Referencing
- Reflection
- Research skills
- Self-directed study
- Time management
- Writing for different purposes

## 6.4 Grading and awarding grades

### Externally assessed units

We mark all the externally assessed units.

Each external assessment is marked according to a mark scheme, and the mark achieved will determine the unit grade awarded (pass, merit or distinction). We determine grade boundaries for each of the external assessments in each assessment series.

If a student doesn't achieve the mark required for a pass grade, we issue an unclassified result for that unit. The marks achieved in the external assessment will contribute towards the student's overall qualification grade, even if a pass is not achieved in the unit assessment.

### NEA units

You will assess the NEA units and we will moderate them.

Each unit has specified pass, merit and distinction assessment criteria. The assessment criteria for each unit are provided with the unit content in [Section 5.3](#) of this specification. You must judge whether students have met the criteria or not.

A unit grade can be awarded at pass, merit or distinction. The number of assessment criteria needed to achieve each grade has been built into each assignment. These are referred to as design thresholds. The table below shows the design thresholds for each grade outcome for the NEA assessments in these qualifications. The unit grade awarded is based on the **total** number of achieved criteria for the unit. The total number of achieved criteria for each unit can come from achievement of any of the criteria (pass, merit or distinction). This is **not** a 'hurdles-based' approach, so students do **not** have to achieve **all** criteria for a specific grade to achieve that grade (e.g. all pass criteria to achieve a pass).

The number of assessment criteria achieved for an NEA unit will be classed as the raw mark. You will assess students' work and identify the number of criteria (raw marks) achieved for each NEA unit. Our moderators will moderate samples of work from your centre. This moderation process may result in the number of assessment criteria (raw marks) achieved being changed. The final

raw mark achieved after moderation has taken place will be converted into a mark on the Uniform Mark Scale (UMS) and will contribute towards the student's overall qualification grade. (More information about UMS is in the section [Calculating the qualification grades](#).)

To make sure we can keep outcomes fair and comparable over time, we will review the performance of the qualifications through their lifetime. The review process might lead to changes in these design thresholds if any unexpected outcomes or significant changes are identified.

|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>Unit size (GLH)</b>                                 | <b>50</b> |
| Number of pass criteria                                | 12        |
| Number of merit criteria                               | 7         |
| Number of distinction criteria                         | 5         |
| Total number of criteria needed for a unit pass        | 10        |
| Total number of criteria needed for a unit merit       | 15        |
| Total number of criteria needed for a unit distinction | 20        |
| Total number of criteria available for the unit        | 24        |

If a student doesn't achieve enough criteria to achieve a unit pass, we will issue an unclassified result for that unit. The number of criteria achieved will be converted into a mark on the Uniform Mark Scale (UMS) and will contribute towards the student's overall qualification grade, even if a pass is not achieved in the unit assessment. More information about this is in the section below ([Calculating the qualification grades](#)).

## Qualifications

The overall qualification grades are:

### Certificate and Extended Certificate

- Distinction\* (D\*)
- Distinction (D)
- Merit (M)
- Pass (P)
- Unclassified (U)

## Calculating the qualification grades

When we work out students' overall grades, we need to be able to compare performance on the same unit in different assessments over time and between different units. We use a Uniform Mark Scale (UMS) to do this.

A student's uniform mark for each externally assessed unit is calculated from the student's raw mark on that unit. A student's uniform mark for each NEA unit is calculated from the number of criteria the student achieves for that unit. The raw mark or number of criteria achieved are converted to the equivalent mark on the uniform mark scale. Marks between grade boundaries are converted on a pro rata basis.

When unit results are issued, the student's unit grade and uniform mark are given. The uniform mark is shown out of the maximum uniform mark for the unit (for example, 48/60).

The student's uniform marks for each unit will be aggregated to give a total uniform mark for the qualification. The student's overall grade will be determined by the total uniform mark.



The tables below show:

- the maximum raw marks or number of criteria, and uniform marks for each unit in the qualifications
- the uniform mark boundaries for each of the assessments in each qualification
- the minimum total mark for each overall grade in the qualifications.

#### Certificate Qualification:

| Unit                 | Maximum raw mark/number of criteria | Maximum uniform mark (UMS) | Distinction* (UMS) | Distinction (UMS) | Merit (UMS) | Pass (UMS) |
|----------------------|-------------------------------------|----------------------------|--------------------|-------------------|-------------|------------|
| F170                 | 60                                  | 60                         | -                  | 48                | 36          | 24         |
| F172                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| F173                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| Qualification Totals | 108                                 | 150                        | 135                | 120               | 90          | 60         |

#### Extended Certificate Qualification:

| Unit                 | Maximum raw mark/number of criteria | Maximum uniform mark (UMS) | Distinction* (UMS) | Distinction (UMS) | Merit (UMS) | Pass (UMS) |
|----------------------|-------------------------------------|----------------------------|--------------------|-------------------|-------------|------------|
| F170                 | 60                                  | 60                         | -                  | 48                | 36          | 24         |
| F171                 | 60                                  | 60                         | -                  | 48                | 36          | 24         |
| F172                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| F173                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| F174                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| F175                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| F176                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| F177                 | 24                                  | 45                         | -                  | 36                | 27          | 18         |
| Qualification Totals | 216                                 | 300                        | 270                | 240               | 180         | 120        |

You can find a marks calculator on the qualification page of our [website](#) to help you convert raw marks/number of achieved criteria into uniform marks.

## 6.5 Performance descriptors

Performance descriptors indicate likely levels of attainment by representative students performing at the pass, merit and distinction grade boundaries at Level 3.

The descriptors must be interpreted in relation to the content in the units and the qualification as a whole. They are not designed to define that content. The grade achieved will depend on how far the student has met the assessment criteria overall. Shortcomings in some parts of the assessment might be balanced by better performance in others.

### **Level 3 Pass**

At pass, students show adequate knowledge and understanding of the basic elements of much of the content being assessed. They can develop and apply their knowledge and understanding to some basic and familiar contexts, situations and problems.

Responses to higher order tasks involving detailed discussion, evaluation and analysis are often limited.

Many of the most fundamental skills and processes relevant to the subject are executed effectively but lack refinement, producing functional outcomes. Demonstration and application of more advanced skills and processes might be attempted but not always executed successfully.

### **Level 3 Merit**

At merit, students show good knowledge and understanding of many elements of the content being assessed. They can sometimes develop and apply their understanding to different contexts, situations and problems, including some which are more complex or less familiar.

Responses to higher order tasks involving detailed discussion, evaluation and analysis are likely to be mixed, with some good examples at times and others which are less accomplished.

Skills and processes relevant to the subject, including more advanced ones, are developed in terms of range and quality. They generally lead to outcomes which are of good quality, as well as being functional.

### **Level 3 Distinction**

At distinction, students show thorough knowledge and understanding of most elements of the content being assessed. They can consistently develop and apply their understanding to different contexts, situations and problems, including those which are more complex or less familiar.

Responses to higher order tasks involving detailed discussion, evaluation and analysis are successful in most cases.

Most skills and processes relevant to the subject, including more advanced ones, are well developed and consistently executed, leading to high quality outcomes.

## 7 Non examined assessment (NEA) units

This section gives guidance on completing the NEA units. In the NEA units, students build a portfolio of evidence to meet the assessment criteria for the unit.

Assessment for these qualifications **must** adhere to JCQ's [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#).

Do **not** use JCQ's Instructions for Conducting non-examination assessments (GCE & GCSE Specifications) – these are only relevant to GCE and GCSE specifications.

The NEA units are centre-assessed and externally moderated by us.

You **must** read and understand all the rules and guidance in this section **before** your students start the set assignments.

If you have any questions, please contact us for help and support.

### 7.1 Preparing for NEA unit delivery and assessment

#### 7.1.1 Centre and teacher/assessor responsibilities

We assume the teacher is the assessor for the NEA units.

**Before** you apply to us for [approval](#) to offer these qualifications you must be confident your centre can fulfil all the responsibilities described below. Once you're approved, you can offer any of our general qualifications, Cambridge Nationals or Cambridge Advanced Nationals **without** having to seek approval for individual qualifications.

Here's a summary of the responsibilities that your centre and teachers must be able to fulfil. It is the responsibility of the head of centre<sup>1</sup> to make sure our requirements are met. The head of centre must ensure that:

- there are enough trained or qualified people to teach and assess the expected number of students you have in your cohorts.
- teaching staff have the relevant level of subject knowledge and skills to deliver and assess these qualifications.
- teaching staff will fully cover the knowledge, understanding and skills requirements in teaching and learning activities.
- allowed combinations of units are considered at the start of the course to be confident that all students can access a valid route through the qualifications.
- all necessary resources are available for teaching staff and students during teaching and assessment activities. This gives students every opportunity to meet the requirements of the qualification and reach the highest grade possible.
- there is a system of internal standardisation in place so that all assessment decisions for centre-assessed assignments are consistent, fair, valid and reliable (see [Section 7.4.3](#)).
- there is enough time for effective teaching and learning, assessment and internal standardisation.

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<sup>1</sup> This is the most senior officer in the organisation, directly responsible for the delivery of Cambridge OCR qualifications. For example, the headteacher or principal of a school/college. The head of centre accepts full responsibility for the correct administration and conduct of Cambridge OCR exams.

- robust processes are in place to make sure that students' work is individual and confirmed as authentic (see [Section 7.2.1](#)).
- our set assignments are used for students' summative assessments. You must make sure that students use the assignment that is live for the period during which they are taking their summative assessment.
- our set assignments are **not** used for practice. This includes both assignments that are currently live or live assignments that have expired. Sample assessment material for each of the NEA units is available on our [website](#). This sample assessment material can be used for practice purposes.
- students understand what they need to do to achieve the criteria.
- students understand what it means when we say work must be authentic and individual and they (and you) follow our requirements to make sure their work is their own.
- students know they must not reference another individual's personal details in any evidence produced for summative assessment, in accordance with the Data Protection Act 2018 and the UK General Data Protection Regulations (UK GDPR). It is the student's responsibility to make sure evidence that includes another individual's personal details is anonymised.
- outcomes submitted to us are correct and are accurately recorded and adhere to the published deadlines.
- assessment of set assignments adheres to the JCQ [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#) and JCQ [AI Use in Assessments: Protecting the Integrity of Qualifications](#).
- a declaration is made at the point you're submitting any work to us for assessment that confirms:
  - all assessment is conducted according to the specified regulations.
  - students' work is authentic.
  - marks have been transcribed accurately.

[Section 7.3.2](#) has more information about student and centre declarations.

(Failing to meet the assessment requirements might be considered as malpractice.)

- centre records and students' work are kept according to these requirements:
  - students' work **must** be kept until **after** the unit has been awarded and any review of results or appeals processed. We cannot consider any review if the work has not been kept.
  - internal standardisation and assessment records must be kept securely for a minimum of three years after the date we've issued a certificate for a qualification.
- all cases of suspected malpractice involving teachers or students are reported (see [Section 7.3.1](#)).

### 7.1.2 Health and safety

In UK law, health and safety is primarily the responsibility of the employer. In a school or college, the employer could be a local education authority, the governing body or board of trustees. Employees (teachers/lecturers, technicians etc.), have a legal duty to cooperate with their employer on health and safety matters. Various regulations, but especially the COSHH Regulations 2002 (as amended) and the Management of Health and Safety at Work Regulations 1999, require that before any activity involving a hazardous procedure or harmful microorganisms is carried out, or hazardous chemicals are used or made, the employer must carry out a risk assessment. A useful summary of the requirements for risk assessment in school or college science can be found at: [www.ase.org.uk](http://www.ase.org.uk).

For members, the CLEAPSS® guide, PS90, *Making and recording risk assessments in school science*<sup>2</sup> offers appropriate advice.

Most education employers have adopted nationally available publications as the basis for their model risk assessments.

Where an employer has adopted model risk assessments, an individual school or college then has to review them, to see if there is a need to modify or adapt them in some way to suit the particular conditions of the establishment.

Such adaptations might include a reduced scale of working, deciding that the fume cupboard provision was inadequate or the skills of the learners were insufficient to attempt particular activities safely. The significant findings of such risk assessment should then be recorded in a '*point of use text*', for example on schemes of work, published teachers guides, work sheets, etc. There is no specific legal requirement that detailed risk assessment forms should be completed for each practical activity, although a minority of employers may require this.

Where project work or investigations, sometimes linked to work-related activities, are included in specifications this may well lead to the use of novel procedures, chemicals or microorganisms, which are not covered by the employer's model risk assessments. The employer should have given guidance on how to proceed in such cases. Often, for members, it will involve contacting CLEAPSS®.

## 7.2 Requirements and guidance for delivering and marking the set assignments

The assignments are:

- set by us.
- taken under supervised conditions (unless we specify otherwise in the assessment guidance).
- assessed by the teacher.
- moderated by us.

You can find the set assignments on our secure website, [Teach Cambridge](#).

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<sup>2</sup> These, and other CLEAPSS® publications, are on the CLEAPSS® Science Publications website [www.cleapss.org.uk](http://www.cleapss.org.uk). Note that CLEAPSS® publications are only available to members. For more information about CLEAPSS® go to [www.cleapss.org.uk](http://www.cleapss.org.uk)

The set assignments give an approximate time that it will take to complete all the tasks. These timings are for guidance only, but should be used by you, the teacher, to give students an indication of how long to spend on each task. You can decide how the time should be allocated between each task or part task. Students can complete the tasks and produce the evidence across several sessions. Students' evidence (either hard copy or digital) must be kept securely by the teacher and access to assessment responses must be controlled. Students aren't permitted to access their work in between the assessment sessions.

We will publish a new set assignment each year and they will be live for two years. Each new set assignment will be released on 1 June for teacher planning. You must not start delivery of live assignments with students until the live assessment dates, which are shown on the front cover. We strongly recommend you use the set assignment released in the same calendar year as the new cohort starts to ensure they have two years for that assignment. You may potentially disadvantage students if you allow new cohorts to use assignments which have already been live for a year. This is because the assignments for each unit are designed for students to access throughout their two years of study. This enables resubmission opportunities across academic years if needed. Students are allowed one resubmission of work based on the same live assignment. [Section 7.4.5](#) provides more information about resubmissions.

You must:

- only download set assignments from our secure website, [Teach Cambridge](#), and use a set assignment that is live for assessment for all summative assessment of students.
- have registered students for the subject and made unit entries before submitting NEA work for moderation.
- not share the set assignments with anyone from outside of your centre. These must only be shared with appropriate centre staff and students taking the assessments.

(More information about maintaining the integrity of assessment materials is in the JCQ document [General Regulations for Approved Centres General and Vocational qualifications](#).)

- make sure students know that they must not share assessment material or their own work with others, including posting or sharing on social media.

(More information is in the JCQ guidance [Information for candidates Using social media and examinations/assessments](#).)

The rest of this section is about how to manage the delivery and marking of the set assignments so that assessment is valid and reliable. Please note that failing to meet these requirements might be considered as malpractice.

Here is a summary of what you need to do.

You **must**:

- be familiar with the requirements set out in the JCQ document [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#).
- have covered the knowledge, understanding and skills with your students and be sure they are ready for assessment **before** you start the summative assessment. This may include students practising applying their learning and receiving feedback from teachers in preparing to take the assessment.
- use the correct live set assignment for summative assessment of the students. The dates for which set assignments are live for summative assessment and the intended cohort are shown on the front cover. These assignments are available on [Teach Cambridge](#).
- give students the [Student Guide](#) before they start the assessment.

- familiarise yourself with the assessment guidance relating to the tasks. The assessment guidance for each unit is in [Section 5](#) after the assessment criteria grids and with the student tasks in the assignments.
- make sure students are clear about the tasks they must complete and the assessment criteria they are attempting to meet.
- students need to be supervised in all 'practical' work to ensure that they are following health and safety protocols.
- in a number of units there are specific criteria which require safe working; where this is the case, the criteria cannot be achieved if staff have to intervene during the assessment to ensure the students' safety. In such instances, staff should assist the student to ensure their safety and so that they can continue with the subsequent assessment tasks, but they cannot be credited for the criteria directly addressing the practical skills where they have had to be helped unless the assessment guidance states otherwise.
- give students a reasonable amount of time to complete the assignments and be fair and consistent to all students. The estimated time we think each assignment should take is stated in the set assignments. In that time students can work on the tasks under the specified conditions until the date that you collect the work for centre assessment.
- tell the students the resources they can use in the assignment before they start the assessment tasks.
- only give students templates provided by us. Where a template is needed for a task, we have provided it in the assignment. The Assessment Guidance will tell students if they **must** use the template(s) we provide. Where this is not stated as a mandatory requirement ('must use'), we **strongly** recommend that students use the template(s) we provide. You must **not** give students any other templates to use when completing their live assignments. Students may create their own templates to record their evidence. If they use templates accessed from other sources, they **must** make sure that:
  - the source is referenced.
  - the template is not pre-populated with responses for which the students may gain marks.
- monitor students' progress to make sure work is capable of being assessed against the assessment criteria, on track for being completed in good time and is the student's own work:
  - NEA work must be completed in the centre under teacher supervision. Supervision is not invigilation. A supervised classroom does not require exam conditions in that classroom. We do not require the internet to be disabled in the supervised classroom. NEA assessments would typically be completed in normal curriculum time:
    - work **must** be completed in the supervised sessions with enough supervision to make sure that it can be authenticated as the student's own work. This means that any source (such as the internet) that may be used by the student when they complete their work must be used appropriately to support the students' **own** work. Students must not claim work sourced through the internet (or any other source) as their own (see [Section 7.2.3](#)). The supervising teacher must be the teacher who will authenticate the students' work. (See [Section 7.2.1](#) for more information on ways to authenticate work.) You must be familiar with the requirements of the JCQ document [AI Use in Assessments: Protecting the Integrity of Qualifications](#) before assessment starts.
    - there may be exceptions to the requirement for supervised conditions if there is work to complete to support the assignment tasks (e.g. research). The assignment and assessment guidance will specify if there are exceptions.



- Where students are allowed to complete work outside of supervised conditions (e.g. research that may be allowed between supervised sessions) you **must** make sure that they only bring notes relating to the work they are allowed to complete unsupervised into the supervised sessions (e.g. notes relating to the research they have done) and to make sure any work they have done is independent. They must not use unsupervised time as an opportunity to:
    - Create drafts of work for their tasks.
    - Gather information to use in other aspects of their tasks.
  - if you provide any material to prepare students for the set assignment, you must adhere to the rules on using referencing and on acceptable levels of guidance to students. This is in [Section 7.2.3](#) and [7.3](#).
  - students must produce their work independently (see [Sections 7.2.1](#) and [7.3](#)).
  - you must make sure students know to keep their work and passwords secure and know that they must not share completed work with other students, use any aspect of another student's work or share their passwords.
- complete the **Teacher Observation Record** that is with the assignments for tasks that state it is needed. This must be submitted with the students' evidence. You **must** follow the guidance given with the form when completing it. Electronic signatures may be used on the forms.
  - use the assessment criteria to assess students' work.
  - before submitting a final outcome to us, you can mark students' completed work and allow them to repeat any part of the assignment, reworking their original evidence. We call this a reattempt. Students must have completed the whole assignment before you mark their work. Any feedback you give to students on the marked work, must:
    - be factual: telling the student what you have observed, not what to do to improve their work.
    - be recorded.
    - be available to the moderator.

(See [Section 7.3 on Feedback](#) and [Section 7.4.4 on reattempting work](#).)

You **must not**:

- create your own assignments for students to use for practice or live assessment.
- change any part of the set assignments (scenarios or tasks).
- mark students' work in stages, providing feedback at each stage. This would be iterative assessment which is not allowed.
- accept multiple reattempts of work where small changes have been made in response to feedback. Marking and feedback must not be an iterative process.
- allow teachers or students to add, amend or remove any work **after** submission for moderation.
- give detailed advice and suggestions to individuals or the whole class on how work may be improved to meet the assessment criteria. This includes giving access to student work as an exemplar.
- allow students access to their assignment work between teacher supervised sessions. (There may be exceptions where students are allowed to complete work independently (e.g. research). Any exceptions will be stated in the assignments.)



- practise the live set assignment tasks with the students. We provide Sample Assignments for you to use for practice purposes.

### 7.2.1 Ways to authenticate work

All NEA work must be completed under teacher supervision (unless the assessment guidance for a specific task or sub-task advises otherwise). In addition, you must complete enough checks to be confident that the work you mark is the student's own and was produced independently.

You should discuss work in progress with students, including asking them questions such as what they are planning/doing and why. This will make sure that work is being completed in a planned and timely way and will give you opportunities to check the authenticity of the work. This is not an opportunity to offer additional guidance to students.

You **must**:

- have read and understood the JCQ documents [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#) and [AI Use in Assessments: Protecting the Integrity of Qualifications](#).
- make sure students and other teachers understand what constitutes plagiarism and other forms of malpractice (e.g. collusion and copying).
- not accept plagiarised work as evidence.
- use questioning as appropriate to confirm authenticity.
- make sure students and teachers fill in authentication declarations (see [Section 7.3.2](#)).

### 7.2.2 Group work

Group work is not allowed for the NEA assignments in these qualifications.

### 7.2.3 Plagiarism

Students must use their own words when they produce final written pieces of work to show they have genuinely applied their knowledge and understanding. When students use their own words, ideas and opinions, it reduces the possibility of their work being identified as plagiarised.

Plagiarism is:

- the submission of someone else's work as your own
- failure to acknowledge a source correctly, including any use of written material, the internet or Artificial Intelligence (AI).

You might find the following JCQ documents helpful:

- [Plagiarism in Assessments](#)
- [AI Use in Assessments: Protecting the Integrity of Qualifications](#)

Due to increasing advancements in AI technology, we strongly recommend that you are familiar with the likely outputs from AI tools. This could include using AI tools to produce responses to some of the assignment tasks, so that you can identify typical formats and wording that these may produce. This may help you identify any cases of potential plagiarism from students using AI tools to generate written responses.

Plagiarism makes up a large percentage of cases of suspected malpractice reported to us by our moderators. You must **not** accept plagiarised work as evidence.

Plagiarism often happens innocently when students do not know that they must reference or acknowledge their sources or aren't sure how to do this. It's important to make sure your students understand:

- the meaning of plagiarism and what penalties may be applied.
- that they can refer to research, quotations or evidence produced by somebody else, but they must list and reference their sources and clearly mark quotations.
- quoting someone else's work, even when it's properly sourced and referenced, doesn't evidence understanding. The student must 'do' something with that information to show they understand it. For example, if a student has to analyse data from an experiment, quoting data doesn't show that they understand what it means. The student must interpret the data and, by relating it to their assignment, say what they think it means. The work must clearly show how the student is using the material they have referenced to inform their thoughts, ideas or conclusions.

We have [The OCR Guide to Referencing](#) on our website. We have also produced a [poster](#) about referencing and plagiarism which may be useful to share with your students.

Teach your students how to reference and explain why it's important to do it. At Key Stage 5 they must:

- use quote marks to show the beginning and end of the copied work.
- list the html address for website text and the date they downloaded information from the website.
- show the name of the AI source used and the date the content was generated for computer-generated content (such as an AI Chatbot).
- for other publications, list:
  - the name of the author.
  - the name of the resource/book/printed article.
  - the year in which it was published.
  - the page number.

Teach your students to:

- always reference material copied from the internet or other sources. This also applies to infographics (graphical information providing data or knowledge).
- always identify information they have copied from teaching handouts and presentations for the unit, using quote marks and stating the text is from class handouts.

### **Identifying copied/plagiarised work**

Inconsistencies throughout a student's work are often indicators of plagiarism. For example:

- different tones of voice, sentence structure and formality across pieces of work.
- use of American expressions, spellings and contexts (such as American laws and guidelines).
- dated expressions and references to past events as being current.
- sections of text in a document where the font or format is inconsistent with other sections.

## What to do if you think a student has plagiarised

If you identify plagiarised work during assessment or internal standardisation, you must:

- consider the plagiarism when judging the number of assessment criteria achieved. (You must not award assessment criteria where the work is plagiarised.)
- record that there is plagiarism in the work on the unit recording sheet (URS) or your recording document, and that you have adjusted the number of assessment criteria achieved to take account of the plagiarism. You will also need to include this information on the Moderation Service if the student's work is requested as part of the moderation sample. This helps our moderators to understand why such criterion may look to be achieved but have not been awarded. (More information about recording documents is in [Section 7.5.2](#).)
  - if the work is requested as part of the moderation sample, it must be provided to our moderator with the other work requested.

If plagiarism is identified during ongoing monitoring of students' work, you can address this in your centre (for example, by instructing the student(s) involved to re-do the affected tasks).

If plagiarism is identified when the work has been submitted to you as final for marking, you must:

- report the student(s) for plagiarism in line with the JCQ document [Suspected Malpractice Policies and Procedures](#)
  - fill in the **JCQ form M1**.

In line with JCQ's policies and procedures on suspected malpractice, the penalties applied for plagiarism will usually result in the work not being allowed (disqualification) or the mark being significantly reduced.

## 7.3 Feedback

### Feedback to students on work in progress towards summative assessment

You can discuss work in progress towards summative assessment with students to make sure it's being done in a planned and timely way. It also provides an opportunity to check the authenticity of the work. You must intervene if there's a health and safety risk (and reflect this in your assessment if the student's ability to operate safely and independently is part of the criteria).

Generic guidance to the whole class is also allowed. This could include reminding students to check they have provided evidence to cover all key aspects of the task. Individual students can be prompted to double check for gaps in evidence providing that specific gaps are not pointed out to them.

You can give general feedback and support if one or more students are struggling to get started on an aspect of the assignment or following a break between sessions working on the assignment. For example, if a student is seeking more guidance that suggests they are not able to apply knowledge, skills and understanding to complete their evidence, you can remind them that they had a lesson which covered the topic. The student would then need to review their own notes to find this information and apply it as needed.

If a student needs additional help to get started on an initial task that is critical to accessing the rest of the assessment, you can provide this help if you feel it is necessary, but you must not award the student with any assessment criteria directly associated with the part(s) of the task for which they received help. This **must** be recorded on the student's work and recording document (see [Section 7.5.2](#)). You will also need to include this information on the Moderation Service if the student's work is requested as part of the moderation sample. This helps our moderators to understand why such criterion may look to be achieved but have not been awarded. More information about how to record additional help given in these circumstances is in [Section 7.4.1](#).

With the exception of the specific feedback allowed to help students start a critical task, mentioned above, feedback must not provide specific advice and guidance that would be construed as coaching. This would compromise the student's ability to independently perform the task(s) they are doing and constitutes malpractice. Our moderators use a number of measures to assure themselves the work is the student's own.

### Assessing completed work

When students have completed their work on an assignment, you must assess it and give feedback to them on the completed work they submitted to you for assessment. ([Section 7.4.1](#) has more information about how to assess NEA work.) Assessment should not be an iterative process. This means you must not assess work and give feedback on it in stages. You must only assess the work when the assignment is complete.

Feedback **must**:

- be supportive, encouraging and positive.
- tell the student what has been noticed, not what you think (for example, if you have observed the student completing a task, you can describe what happened, what was produced and what was demonstrated).

Feedback **can**:

- identify what task and part of the task could be improved, but not say how to improve it. You could show the student work from a **different** unit that demonstrates higher achievement, but you must not detail to the student how they could achieve that in their work. If you are using another student's work from a different unit as an example, you must anonymise this work and make sure that the potential to plagiarise from this work is minimised. You could remind students that they had a lesson on a specific topic and that they could review their notes, but you must not tell them how they could apply the teaching to improve their work.
- comment on what has been achieved, for example 'the evidence meets the P2 and M2 criteria'.
- identify that the student hasn't met a command word or assessment criteria requirement. For example, 'This is a description, not an evaluation'.
- use text from the specification, assignment or assessment criteria in general guidance to clarify what is needed in the work. For example, 'Research the fundamentals of the genetic disorder and how genes and DNA are affected.'

Feedback **must not**:

- point out specific gaps. For example, you must not prompt the student to include specific detail in their work, such as 'Add the countries the gene therapy is offered in and people's understanding of the gene therapy.'
- be so detailed that it leads students to the answer. For example, you must not give:
  - model answers.
  - step-by-step guidance on what to do to complete or improve work.
  - headings or prompts that include examples which give all or part of what students have to write about or produce.
- talk the student through how to achieve or complete the task.
- give detail on where to find information/evidence.

In other words, feedback must help the student to take the initiative in making changes. It must not direct or tell the student what to do to complete or improve their work in a way that means they do not need to think how to apply their learning. Students need to recall or apply their learning. You must not do the work for them.

Students can reattempt their work on an assignment after you have marked it and provided feedback. This **must** happen before the work is submitted to us for moderation. Neither you nor the student can add, amend or remove any work after the final mark has been submitted for moderation.

[Sections 7.4.4](#) and [7.4.5](#) give more guidance for students who wish to reattempt or resubmit their work following feedback.

### Informing students of their marks

When you have marked students' work, you must inform students of the marks (number of criteria) achieved. **Before** you submit marks to us, you must provide enough time for:

- students to appeal the marks.
- you to complete a review and make changes if needed.
- you to inform students of the outcome of the review.

The marks must be submitted to us **at least 10 working days** before your visit.

### What improper assistance might look like

When we see anything that suggests the teacher has led students to the answer, we become concerned because it suggests students have not worked independently to produce their assignment work. The following are examples of what might indicate improper assistance by the teacher:

- prompts that instruct students to include specific detail in their work, such as, 'You need to include the aims of the activity. Who is it aimed at? What is the purpose of the activity? How will it benefit the specific group/individual?'
- headings or templates that include examples which give all or part of what students have to write about or produce, such as sources of support.

Our moderators will report suspected malpractice when they cannot see differences in content between students' work in the sample they are moderating. An exception is when students have only used and referenced technical facts and definitions. If our moderator is in any doubt, they will report suspected malpractice. The decision to investigate or not is made by us, not the moderators.

#### 7.3.1 Reporting suspected malpractice

It is the responsibility of the head of centre to report all cases of suspected malpractice involving teachers or students.

A JCQ Report of Suspected Malpractice form (JCQ/M1 for student suspected malpractice or JCQ/M2 for staff suspected malpractice) is available to download from the [JCQ website](#). The form must be completed as soon as possible and emailed to us at [compliance@ocr.org.uk](mailto:compliance@ocr.org.uk).

When we ask centres to gather evidence to assist in any malpractice investigation, heads of centres must act promptly and report the outcomes to us.

The JCQ document [Suspected Malpractice Policies and Procedures](#) has more information about reporting and investigating suspected malpractice, and the possible sanctions and penalties which could be imposed. You can also find out more on our [website](#).

### 7.3.2 Student and centre declarations

Both students and teachers must declare that the work is the student's own:

- **each student** must sign a declaration before submitting their work to their teacher. A **candidate authentication statement** can be used and is available to download from our [website](#). You must keep these statements in the centre until all reviews of results, malpractice and appeal issues have been resolved.
- **teachers** must declare the work submitted for centre assessment is the students' own work by completing a [centre authentication form \(CCS160\)](#) for each cohort of students for each unit. You must keep centre authentication forms in the centre until all post-results issues have been resolved.

### 7.3.3 Generating evidence

The set assignments will tell the students what they need to do to meet the assessment criteria for the NEA units. It is your responsibility to make sure that the methods of generating evidence for the assignments are:

- valid
- safe and manageable
- suitable to the needs of the student.

#### Valid

The evidence presented must be valid. For example, it would not be appropriate to present an organisation's equal opportunities policy as evidence towards a student's understanding of how the equal opportunities policy operates in an organisation. It would be more appropriate for the student to incorporate the policy in a report describing the different approaches to equal opportunities.

#### Safe and manageable

You must make sure that methods of generating evidence and approaches taken:

- are safe and manageable
- do not put unnecessary demands on the student
- are appropriate and in line with ethical standards and your centre's safeguarding responsibilities.

#### Suitable to the needs of the student

We are committed to ensuring that achievement of these qualifications is free from unnecessary barriers.

You must follow this commitment through when modifying tasks (where this is allowed) and/or considering assessment and evidence generation. If you are modifying tasks and are not sure what is acceptable, [contact us](#).

#### Observation and questioning

The primary evidence for assessment is the work submitted by the student, however the following assessment methods might be suitable for you to use for some aspects of these qualifications, where identified:

- **observation** of a student doing something
- **questioning** of the student or witness.

## Observation

You and the student should plan observations together, but it is your responsibility to record the observation properly (for example observing a student undertaking a practical task). More information is in the Teacher Observation Records section.

## Questioning

Questioning the student is normally an ongoing part of the formative assessment process and may, in some circumstances, provide evidence to support achievement of the criteria.

Questioning is often used to:

- test a student's understanding of work which has been completed outside of the classroom (where this may be permitted)
- check if a student understands the work they have completed
- collect information on the type and purpose of the processes a student has gone through.

If questioning is used as evidence towards achievement of specific topic areas, it is important that you record enough information about what they asked and how the student replied, to allow the assessment decision to be moderated.

### 7.3.4 Teacher Observation Records

You **must** complete the Teacher Observation Record form in the set assignment for:

**Unit F173 Biomedical techniques** (Task 2, Topic Areas 4 and 5) for each student as evidence of a safely performed planned investigation of unidentified samples. The Teacher Observation Record form must provide evidence of how the student performed the planned investigation safely.

**Unit F175 Human reproduction** (Task 3, Topic Areas 1, 2, 3 and 4) for each student as evidence of delivering a presentation of the plan created either in Task 1 or Task 2. The Teacher Observation Record form must provide evidence of how the student delivered the presentation effectively, with clear explanations of rationale beyond what is included in the presentation documentation.

**Unit F177 Drug development** (Task 3, Topic Areas 1, 2, 3 and 4) for each student as evidence of delivering a pitch of the proposal completed in Task 2. The Teacher Observation Record form must provide evidence of how the student delivered the presentation to the intended audience, with explanations of rationale beyond what is included in the presentation documentation.

Teacher observation **cannot** be used as evidence of achievement for a whole unit. Most evidence **must** be produced directly by the student. Teacher observation **must only** be used where specified as an evidence requirement.

Teacher Observation Records must be individual to each student and suitably detailed to help moderators to determine if the assessment criteria have been met. You must follow the guidance provided in the 'guidance notes' section of the form so that the evidence captured and submitted is appropriate. Both you and the student must sign and date the form to show that you both agree its contents. Electronic signatures are acceptable. The signed form must form part of the students' evidence and be submitted with work requested for moderation.

Where the guidance has not been followed, the reliability of the form as evidence may be called into question. If doubt about the validity of the Teacher Observation Record form exists, it cannot be used as assessment evidence and marks based on it cannot be awarded. Our moderators will be instructed to adjust centre marks accordingly.



### 7.3.5 Presentation of the final piece of work

Students must produce their evidence in the format specified in the tasks where specific formats are given. Written work can be digital (e.g. word processed) or hand-written and tables and graphs (if relevant) can be produced using appropriate ICT. If a student's work is requested as part of the moderation sample, a digital version will need to be uploaded to our Submit for Assessment service. You should consider this before students produce their work for assessment.

More information about how to submit work to Submit for Assessment is on our [Administration](#) pages. This also includes information relating to the operating systems, browsers and devices that are supported by Submit for Assessment, as well as the file formats and sizes that are acceptable.

Any sourced material must be suitably acknowledged. Quotations must be clearly marked and a reference provided.

You will provide information to us relating to the criteria achieved by students on our Cambridge OCR Moderation Service. This is accessed through My Cambridge. More information is available on our [Administration](#) pages.

## 7.4 Assessing NEA units

All NEA units are assessed by teachers and externally moderated by our moderators. Assessment of the set assignments must adhere to JCQ's [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#).

The centre is responsible for appointing someone to act as the internal assessor. This would usually be the teacher who has delivered the programme but could be another person from the centre. The assessment criteria must be used to assess the student's work. These specify the levels of skills, knowledge and understanding that the student needs to demonstrate.

### 7.4.1 Applying the assessment criteria

When students have completed the assignment, they must submit their work to you to be assessed.

You must assess the tasks using the assessment criteria and any additional assessment guidance provided. Each criterion states what the student needs to do to achieve that criterion (e.g. Create an appropriate specialised diet). The command word and assessment guidance provide additional detail about breadth and depth where it is needed.

You must judge whether each assessment criterion has been **successfully achieved** based on the evidence that a student has produced. For the criterion to be achieved, the evidence must show that all aspects have been met in sufficient detail.

When making a judgement about whether a criterion has been **successfully achieved**, you must consider:

- the requirements of the specific NEA task that the student is completing
- the criterion wording, including the command word used and its definition
- any assessment guidance for the criterion
- the unit content that is being assessed.

You must annotate the work to show where evidence meets each criterion (see [Section 7.4.2](#)). You can then award the criterion. Assessment should be positive, rewarding achievement rather than penalising failure or omissions.



The number of criteria needed for each unit grade (pass, merit or distinction) is provided in [Section 6.4](#).

The total number of criteria achieved is the 'raw mark'.

You must be convinced, from the evidence presented, that students have worked independently to the required standard.

If you have given additional, more specific support or guidance to an individual student to get them started on a task, this **must** also be recorded on the student's work and/or your recording sheet. You will also need to include this information on the Moderation Service if the student's work is requested as part of the moderation sample. This ensures the information is available for our moderator to see. You may give this support because the student could not start a task or part of a task that was **critical to them accessing the rest of the task or assignment** (see [Section 7.3](#)). In this situation, the student should **not** be awarded the assessment criteria for the work for which they received help, and the number of criteria achieved must be adjusted appropriately. Recording this on the student's work/recording document (and the Moderation Service where this work is requested in the sample) will help our moderator to understand why the assessment criteria have not been awarded.

There should be clear evidence that work has been attempted and some work produced. If a student does not submit any work for an NEA unit, the student should be identified as being absent from that unit.

If a student completes any work at all for an NEA unit, you must assess the work using the assessment criteria and award the appropriate number of criteria. This might be zero.

Your centre must internally standardise the assessment decisions for the cohort **before** you give feedback to students (see [Section 7.4.3](#)). When you are confident the internal assessment standardisation and appeals process is complete, you can submit work for moderation at the relevant time. You **must not** add, amend or remove any work after it has been submitted to us for final moderation. Work **must** be kept securely until the end of the review of results process.

#### 7.4.2 Annotating students' work

Each piece of NEA work must show how you are satisfied the assessment criteria have been met.

Comments on students' work and your recording document provide a means of communication about assessment decisions made, between teachers during internal standardisation, and with our moderator if the work is part of the moderation sample. (Comments or annotations must not be used as a method of communication with our moderator for any other reason.)

#### 7.4.3 Internal standardisation

It is important that all teachers are assessing work to common standards. For each unit, centres must make sure that internal standardisation of outcomes across teachers and teaching groups takes place using an appropriate procedure.

This can be done in a number of ways. In the first year, reference material and our training meetings will provide a basis for your centre's own standardisation. In following years, this, and/or your own centre's archive material, can be used. We advise you to hold preliminary meetings of staff involved to compare standards through cross-marking a small sample of work. After you have completed most of the assessment, a further meeting at which work is exchanged and discussed will help you make final adjustments.

If you are the only teacher in your centre assessing these qualifications, we still advise you to make sure your assessment decisions are internally standardised by someone else in your centre. Alternatively, this could be a teacher that may be delivering in another local centre or as part of

your Multi Academy Trust (MAT) if relevant. Ideally this person will have experience of these types of qualifications, for example someone who:

- is delivering a similar qualification in another subject.
- has relevant subject knowledge.

You must keep evidence of internal standardisation in the centre for our moderators to see.

We have a **guide** to how internal standardisation can be approached on our [website](#).

#### 7.4.4 Reattempting work to improve the grade before submitting marks to us

As described in [Section 7.2](#), **before** submitting a final outcome to us for external moderation, you can allow students to repeat any element of the assignment and rework their original evidence. We refer to this as a reattempt. A reattempt allows the student to reflect on **internal** feedback, and to improve their work. A reattempt is **not** an iterative process where students make small modifications through ongoing feedback to eventually achieve the desired outcome.

Any feedback **must** be noted by you and a record of this kept in centre. We have provided a feedback form for this purpose, which can be found on our [website](#) and [Teach Cambridge](#). We recommend that you use the feedback form we provide or create your own recording form.

To summarise, a reattempt is a process that is internal to the centre. This allows students to rework their evidence:

- after it has been marked by you as a complete assignment.
- before it is submitted to us as the final work.

A reattempt must be done before submission for external moderation. When a student submits the work to you as final for external moderation, they must not complete any further work on any aspect of it.

#### 7.4.5 Resubmitting moderated work to us to improve the grade

We use the term 'resubmission' when referring to student work that has previously been submitted to us for moderation. Following moderation, if you and the student feel they have not performed at their best during the assessment, the student can, with your agreement, improve their work and resubmit it to you again for assessment and to us for external moderation. You must be sure it is in the student's best interests to resubmit the work for assessment. There is one resubmission opportunity per NEA assignment. If you have submitted the same assignment twice for a student, they will need to use the next live assignment for any further reattempt and resubmission.

Students can only resubmit work using the **same** assignment if the assignment is still live. The live assessment dates and intended cohort will be shown on the front cover of the assignment. We will not accept work based on an assignment that is no longer live. If the assignment is no longer live, students will need to produce work using the new live assignment for the unit for the resubmission.

If students are resubmitting using a new live assignment, they can use the evidence they produced for the previous assignment, but they will need to make any changes that are necessary so that the work meets the requirements of the new scenario and task.

Students can also build on the work to improve it. All work for a resubmission must be completed under the required teacher supervised conditions and marked against the assessment criteria and assessment guidance. You must not over direct students on how to adapt/improve work to meet the requirements of the new assignment. You must adhere to all requirements relating to giving and recording feedback from [Section 7.3](#) and [Section 7.4.4](#).

To summarise, a resubmission is the reworking and submitting of assignment evidence and marks to us, following previous external moderation by us.

## 7.5 Moderating NEA units

The purpose of external moderation is to make sure that the standard of assessment is the same for all centres and that internal standardisation has taken place.

Our [Administration](#) pages give full details about how to submit work for moderation.

This includes the deadline dates for entries and submission of marks. For moderation to happen, you must submit your marks **at least 10 working days** before your visit.

### 7.5.1 Submitting outcomes

When you have assessed the work, given students their marks and the opportunity to review them (see **Informing students of their marks** in [Section 7.3](#)), you must submit the total number of criteria achieved (raw marks) for each student using the Moderation Service. This is accessed via My Cambridge. The marks must be submitted to us **at least 10 working days** before your moderation visit. You must have made NEA unit entries before you can submit marks.

### 7.5.2 Sample requests

Once you have submitted your marks, we will tell you which work will be sampled as part of the moderation process. Samples will include work from across the range of students' attainment. You must submit the sample work **and** the assessment criteria awarded, **at least three working days** before your visit. Failure to submit or late submission of samples will be classed as maladministration and may result in a malpractice investigation.

For students requested in the sample, you **must**:

- submit full details of each individual criterion achieved on the Moderation Service.
  - this must include providing any additional information related to criteria that look to be achieved but you have not awarded for specific reasons (e.g. because plagiarism was detected or because additional support was given to the student to get them started on a task).
- upload work to our Submit for Assessment service.

At the same time as you submit the above information for students in the sample, you must upload to Submit for Assessment the detail of each criterion achieved by every student in the cohort for which you have submitted outcomes. This information may be captured on unit recording sheets (URS) or another recording document.

Any recording document used must include the same information that is captured on the URS for each student:

- The qualification and unit name
- The candidate name(s) and number(s)
- The assignment taken (e.g. Set A) and scenario title
- The date
- The assessment criteria for the unit and whether each criterion has been achieved or not

- Any additional information related to criterion that may look to be achieved but you have not awarded for specific reasons (e.g. because plagiarism was detected or because additional support was given to the student to get them started on a task).

We have provided a unit recording sheet (URS) for each NEA unit which you may use to help capture this information as you are assessing students' work. We have also provided a spreadsheet recording document for each size of unit, which you may prefer to use for your student cohort.

These can be downloaded from the [qualification webpage](#) or [Teach Cambridge](#). The URS are **not** mandatory documents. You may use the spreadsheet we provide or create your own recording document to capture the required information instead.

Students' work must be securely kept until after the unit has been awarded and any review of results and appeals windows are closed.

We need sample work to help us monitor standards. We might ask some centres to release work for this purpose. We will let you know as early as possible if we need this from you. We always appreciate your co-operation.

### **7.5.3 NEA outcomes**

You will know the outcomes of moderation after your moderation visit. On results day, for each student, you will be sent the unit grades and the raw marks (number of criteria achieved). The raw marks will be converted to UMS. (More information about raw marks and UMS is in [Section 6.4](#).)

## 8 Administration

This section gives an overview of the processes involved in administering these qualifications. More information about the processes and deadlines involved at each stage is on our [Administration](#) pages.

### 8.1 Assessment availability

There are two assessment opportunities available each year for the externally assessed units: one in January and one in June. Students can be entered for different units in different assessment series.

All students must take the exams at a set time on the same day in a series.

NEA assignments can be taken by students at any time during the live period shown on the front cover. We strongly recommend you use the set assignment that is released in the same calendar year as the new cohort starts to ensure that students have two years to use the assignment, giving them access to a resubmission opportunity if needed.

There are two series (windows) each year to submit NEA outcomes. All dates relating to NEA moderation are on our [Administration](#) pages.

(More information is in [Section 8.5.2](#).)

Qualification certification is available in each series.

### 8.2 Collecting evidence of student performance to ensure resilience in the qualifications system

Regulators have published guidance on collecting evidence of student performance as part of long-term contingency arrangements to improve the resilience of the qualifications system. You should review and consider this guidance when delivering this qualification to students at your centre.

For more detailed information on collecting evidence of student performance please visit our [website](#).

### 8.3 Equality Act information relating to Cambridge Advanced Nationals

The Cambridge Advanced Nationals require assessment of a broad range of skills and, as such, prepare students for further study and higher-level courses.

The Cambridge Advanced National qualifications have been reviewed to check if any of the competences required present a potential barrier to disabled students. If this was the case, the situation was reviewed again to make sure that such competences were included only where essential to the subject.

### 8.4 Accessibility

There can be adjustments to standard assessment arrangements based on the individual needs of students. It is important that you identify as early as possible if students have disabilities or particular difficulties that will put them at a disadvantage in the assessment situation and that you choose a qualification or adjustment that allows them to demonstrate attainment.

If a student requires access arrangements that need approval from us, you must use [Access arrangements \(online\)](#) to gain approval. You must select the appropriate qualification type(s) when you apply. Approval for GCSE or GCE applications alone does not extend to other qualification types. You can select more than one qualification type when you make an application. For guidance or support please contact our [Special Requirements Team](#).

The responsibility for providing adjustments to assessment is shared between your centre and us. Please read the JCQ document [Access Arrangements and Reasonable Adjustments](#).

If you have students who need a post-exam adjustment to reflect temporary illness, indisposition or injury when they took the assessment, please read the JCQ document [A guide to the special consideration process](#).

If you think any aspect of these qualifications unfairly restricts access and progression, please email [Support@ocr.org.uk](mailto:Support@ocr.org.uk) or call our Customer Support Centre on **01223 553998**.

The following access arrangements are allowed for this specification:

| Access arrangement                    | Type of assessment               |
|---------------------------------------|----------------------------------|
| Reader/Computer reader                | All assessments                  |
| Scribes/Speech recognition technology | All assessments                  |
| Practical assistants                  | All assessments                  |
| Word processors                       | All assessments                  |
| Communication professional            | All assessments                  |
| Language modifier                     | All assessments                  |
| Modified question paper               | Timetabled exams                 |
| Extra time                            | All assessments with time limits |

## 8.5 Requirements for registering students and making unit entries

We provide information on key dates, timetables, moderation and how to submit marks on our [website](#).

Your centre must be registered with us as an approved centre before you register students and can make entries. Centre approval should be in place well in advance of making your first entries. Details on how to register with us are on our [website](#).

### 8.5.1 Registering your students

At the start of the course, you need to register your students for the relevant Cambridge Advanced Nationals subject. The registration code for this subject is A005.

Full information about how to register students is on the [Administration](#) page of our website.

### 8.5.2 Making unit entries

When students are ready for assessment, you must make exam and NEA unit entries for the relevant assessment series. You need to know the unit entry codes. Students submitting work must be entered for the appropriate unit entry code from the table below.

The short title for these Cambridge Advanced Nationals is AAQ CAN. This is the title that will be displayed on Interchange and some of our administrative documents.

**You need to register your students first, before you can make individual unit entries. Individual unit entries should be made for each series in which you intend to submit or resubmit an NEA unit or sit an externally assessed examination.**

| Unit entry code | Component code | Assessment method | Unit titles                   |
|-----------------|----------------|-------------------|-------------------------------|
| F170            | 01             | Written paper     | Fundamentals of human biology |
| F171            | 01             | Written paper     | Health and disease            |
| F172            | 01             | Moderated         | Genetics                      |
| F173            | 01             | Moderated         | Biomedical techniques         |
| F174            | 01             | Moderated         | Nutrition and metabolism      |
| F175            | 01             | Moderated         | Human reproduction            |
| F176            | 01             | Moderated         | The brain                     |
| F177            | 01             | Moderated         | Drug development              |

Full information about how to make entries for units is on our [Administration](#) page.

## 8.6 Making certification entries

You must enter students for qualification certification separately from registration and unit entries. You should make a **certification entry** in the final series, at the end of the student's course. If a certification entry is **not** made, no overall grade can be awarded. These are the qualifications that students should be entered for:

- Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Certificate) - certification code H049.
- Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology (Extended Certificate) - certification code H149.

## 8.7 Unit and qualification resits

Students can resit the assessment for each unit and the best result will be used to calculate the certification result. Students may resit each examined unit (F170 and F171) twice before certification. This means they can have three attempts in total at exams for each of these units. Unit F170 is in both the Certificate and Extended Certificate qualifications. Students may have a **total** of three exam attempts for this unit across the qualifications.

Resit opportunities must be fair to all students and **not** give some students an unfair advantage over other students. For example, the student must not have direct guidance and support from the teacher in producing further evidence for NEA units. When resitting an NEA unit, students must submit new, amended or enhanced work, as detailed in the JCQ [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#).

When you arrange resit opportunities, you must make sure that you do not adversely affect other assessments being taken.

Arranging a resit opportunity is at the centre's discretion. Summative assessment series must not be used as a diagnostic tool and resits should only be planned if the student has taken full advantage of the first assessment opportunity and any formative assessment process.

## 8.8 Post-results services

A number of post-results services are available:

- Reviews of results - if you think there might be something wrong with a student's results, you may submit a review of marking or moderation.
- Missing and incomplete results - if an individual subject result for a student is missing, or the student has been omitted entirely from the results supplied you should use this service.



- Access to scripts - you can ask for access to marked scripts.
- Late certification - following the release of unit results, if you have not previously made a certification entry, you can make a late request, which is known as a **late certification**. This is a free service.

Please refer to the JCQ [Post-Results Services booklet](#) and our [Administration](#) page for more guidance about post-result services.

For each NEA unit, a review of moderation can only be requested for the cohort. It cannot be requested for individual students.



## **Appendix A: Guidance for the production of electronic evidence**

All student work requested as part of the moderation sample will need to be submitted to us digitally. It will need to be uploaded to our Submit for Assessment service.

More information about how to submit work to our Submit for Assessment service is on our [website](#). This includes information relating to the operating systems, browsers and devices that are supported by the service, as well as the file formats and sizes that are acceptable.

You should consider this before students produce their work for assessment.

## Appendix B: Command Words

### External assessment

The table below shows the command words that will be used in exam questions. This shows what we mean by the command word and how students should approach the question and understand its demand. Remember that the rest of the wording in the question is also important.

| Command Word                     | Meaning                                                                                                                                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analyse</b>                   | <ul style="list-style-type: none"> <li>Separate or break down information into parts and identify their characteristics or elements</li> <li>Explain the different elements of a topic or argument and make reasoned comments</li> <li>Explain the impacts of actions using a logical chain of reasoning</li> </ul> |
| <b>Annotate</b>                  | <ul style="list-style-type: none"> <li>Add information, for example, to a table, diagram or graph</li> </ul>                                                                                                                                                                                                        |
| <b>Calculate</b>                 | <ul style="list-style-type: none"> <li>Work out the numerical value. Show your working unless otherwise stated</li> </ul>                                                                                                                                                                                           |
| <b>Choose</b>                    | <ul style="list-style-type: none"> <li>Select an answer from options given</li> </ul>                                                                                                                                                                                                                               |
| <b>Compare</b>                   | <ul style="list-style-type: none"> <li>Give an account of the similarities and differences between two or more items or situations</li> </ul>                                                                                                                                                                       |
| <b>Complete</b>                  | <ul style="list-style-type: none"> <li>Add information, for example, to a table, diagram or graph to finish it</li> </ul>                                                                                                                                                                                           |
| <b>Describe</b>                  | <ul style="list-style-type: none"> <li>Give an account that includes the relevant characteristics, qualities or events</li> </ul>                                                                                                                                                                                   |
| <b>Discuss</b> (how/whether/etc) | <ul style="list-style-type: none"> <li>Present, analyse and evaluate relevant points (for example, for/against an argument) to make a reasoned judgement</li> </ul>                                                                                                                                                 |
| <b>Draw</b>                      | <ul style="list-style-type: none"> <li>Produce a picture or diagram</li> </ul>                                                                                                                                                                                                                                      |
| <b>Explain</b>                   | <ul style="list-style-type: none"> <li>Give reasons for and/or causes of something</li> <li>Make something clear by describing and/or giving information</li> </ul>                                                                                                                                                 |
| <b>Give examples</b>             | <ul style="list-style-type: none"> <li>Give relevant examples in the context of the question</li> </ul>                                                                                                                                                                                                             |
| <b>Identify</b>                  | <ul style="list-style-type: none"> <li>Name or provide factors or features from stimulus</li> </ul>                                                                                                                                                                                                                 |
| <b>Label</b>                     | <ul style="list-style-type: none"> <li>Add information, for example, to a table, diagram or graph until it is final</li> </ul>                                                                                                                                                                                      |
| <b>Outline</b>                   | <ul style="list-style-type: none"> <li>Give a short account or summary</li> </ul>                                                                                                                                                                                                                                   |
| <b>State</b>                     | <ul style="list-style-type: none"> <li>Give factors or features</li> <li>Give short, factual answers</li> </ul>                                                                                                                                                                                                     |

**Non examined assessment (NEA)**

The table shows the command words that will be used in the NEA assignments and/or assessment criteria.

| Command Word                     | Meaning                                                                                                                                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adapt</b>                     | <ul style="list-style-type: none"> <li>Change to make suitable for a new use or purpose</li> </ul>                                                                                                                                                                                                                  |
| <b>Analyse</b>                   | <ul style="list-style-type: none"> <li>Separate or break down information into parts and identify their characteristics or elements</li> <li>Explain the different elements of a topic or argument and make reasoned comments</li> <li>Explain the impacts of actions using a logical chain of reasoning</li> </ul> |
| <b>Assess</b>                    | <ul style="list-style-type: none"> <li>Offer a reasoned judgement of the standard or quality of situations or skills. The reasoned judgement is informed by relevant facts</li> </ul>                                                                                                                               |
| <b>Calculate</b>                 | <ul style="list-style-type: none"> <li>Work out the numerical value. Show your working unless otherwise stated</li> </ul>                                                                                                                                                                                           |
| <b>Classify</b>                  | <ul style="list-style-type: none"> <li>Arrange in categories according to shared qualities or characteristics</li> </ul>                                                                                                                                                                                            |
| <b>Compare</b>                   | <ul style="list-style-type: none"> <li>Give an account of the similarities and differences between two or more items, situations or actions</li> </ul>                                                                                                                                                              |
| <b>Conclude</b>                  | <ul style="list-style-type: none"> <li>Judge or decide something</li> </ul>                                                                                                                                                                                                                                         |
| <b>Describe</b>                  | <ul style="list-style-type: none"> <li>Give an account that includes the relevant characteristics, qualities or events</li> </ul>                                                                                                                                                                                   |
| <b>Discuss</b> (how/whether/etc) | <ul style="list-style-type: none"> <li>Present, analyse and evaluate relevant points (for example, for/against an argument) to make a reasoned judgement</li> </ul>                                                                                                                                                 |
| <b>Evaluate</b>                  | <ul style="list-style-type: none"> <li>Make a reasoned qualitative judgement considering different factors and using available knowledge/experience</li> </ul>                                                                                                                                                      |
| <b>Examine</b>                   | <ul style="list-style-type: none"> <li>To look at, inspect, or scrutinise carefully, or in detail</li> </ul>                                                                                                                                                                                                        |
| <b>Explain</b>                   | <ul style="list-style-type: none"> <li>Give reasons for and/or causes of something</li> <li>Make something clear by describing and/or giving information</li> </ul>                                                                                                                                                 |
| <b>Interpret</b>                 | <ul style="list-style-type: none"> <li>Translate information into recognisable form</li> <li>Convey one's understanding to others, e.g. in a performance</li> </ul>                                                                                                                                                 |
| <b>Investigate</b>               | <ul style="list-style-type: none"> <li>Inquire into (a situation or problem)</li> </ul>                                                                                                                                                                                                                             |
| <b>Justify</b>                   | <ul style="list-style-type: none"> <li>Give valid reasons for offering an opinion or reaching a conclusion</li> </ul>                                                                                                                                                                                               |
| <b>Research</b>                  | <ul style="list-style-type: none"> <li>Do detailed study in order to discover (new) information or reach a (new) understanding</li> </ul>                                                                                                                                                                           |
| <b>Summarise</b>                 | <ul style="list-style-type: none"> <li>Express the most important facts or ideas about something in a short and clear form</li> </ul>                                                                                                                                                                               |

We might also use other command words but these will be:

- commonly used words whose meaning will be made clear from the context in which they are used (e.g. create, improve, plan)
- subject specific words drawn from the unit content.

## Appendix C: How Science Works Concepts and Skills

The concepts and skills set out in this section are intended to develop learners as critical and creative thinkers, and to enable learners to solve problems in a variety of contexts. The concepts and skills are set out as references and associated statements.

The concepts and skills in this section will be assessed in the examined assessment (EA) and non examined assessment (NEA) units where appropriate.

Terms associated with measurement and data analysis are used in accordance with their definitions in the Association of Science Education publication *The Language of Measurement* (2010).

| How Science Works Reference | How Science Works Statement                                                                                                                        | To include understanding of:                                                                                                                                                                                                                                                                                                                                                                                                                                    | Areas of the specification covering the HSW concepts and skills                                                                                                                                                                       |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HSW1</b>                 | Use theories, models, and ideas to develop scientific explanations                                                                                 | <ul style="list-style-type: none"> <li>• Peer review</li> <li>• Use of a variety of models (representational, spatial, descriptive, computational, and mathematical) to solve problems</li> <li>• Hypotheses and predictions</li> </ul>                                                                                                                                                                                                                         | <b>F170:</b> 1.1.1, 2.1.2, 3.2.2<br><b>F171:</b> 2.1, 3.1<br><b>F172:</b> 1.1, 2.2.1, 2.3.2, 3.2.2, 4.2.2<br><b>F173:</b> 5.2<br><b>F174:</b> 1.3.2<br><b>F176:</b> 1.1, 1.2, 1.3, 2.1.2, 2.2.1, 5.1<br><b>F177:</b> 2.2              |
| <b>HSW2</b>                 | Use knowledge and understanding to pose scientific questions, define scientific problems, present scientific arguments and ideas                   | <ul style="list-style-type: none"> <li>• Use of online and offline research skills</li> <li>• Correctly citing sources of information</li> <li>• How to present reasoned explanations, including relating data to hypotheses</li> </ul>                                                                                                                                                                                                                         | <b>F170:</b> 1.1.2, 3.3.1<br><b>F171:</b> 4.1, 5.1.1<br><b>F172:</b> 1.3.2<br><b>F173:</b> 1.1.1, 2.4, 5.1.2<br><b>F176:</b> 1.1                                                                                                      |
| <b>HSW3</b>                 | Use appropriate methodology, including information and communication technology (ICT) to answer scientific questions and solve scientific problems | <ul style="list-style-type: none"> <li>• Experimental design, including to solve problems in a practical context</li> <li>• Control variables, dependent variables, and independent variables</li> <li>• Appropriateness of an experimental method to meet expected outcomes</li> <li>• Importance of scientific quantities and how they are determined</li> <li>• How to determine an appropriate sample size and/or range of values to be measured</li> </ul> | <b>F170:</b> 2.1.3<br><b>F171:</b> 4.2.2, 5.2<br><b>F173:</b> 1.1.2, 1.2, 2.1, 2.2.2, 2.4, 3.2.1, 3.3.1, 3.5.1, 4.2.2, 4.3<br><b>F174:</b> 3.3.3, 4.2.1<br><b>F175:</b> 2.1<br><b>F176:</b> 2.2.1, 3.1.1<br><b>F177:</b> 2.4.1, 2.4.2 |

| <b>How Science Works Reference</b> | <b>How Science Works Statement</b>                                                                                 | <b>To include understanding of:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Areas of the specification covering the HSW concepts and skills</b>                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HSW4</b>                        | Carry out experimental and investigative activities, including appropriate risk management, in a range of contexts | <ul style="list-style-type: none"> <li>How to use the apparatus, techniques and procedures correctly, skilfully and safely</li> <li>Apply investigative approaches and methods to practical work</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>F170:</b> 1.1.2, 1.2.3, 2.2.3, 4.1.3, 4.1.4<br><b>F171:</b> 2.1<br><b>F172:</b> 3.2.2, 3.2.3<br><b>F173:</b> 1.2, 2.1, 2.3, 3.3.2, 3.5.3, 4.2.4<br><b>F174:</b> 4.2.3<br><b>F177:</b> 1.4, 2.1.1 |
| <b>HSW5</b>                        | Use data to provide evidence, and recognise correlations and causal relationships                                  | <ul style="list-style-type: none"> <li>Appropriate units for measurements (this already exists as part of Maths skills)</li> <li>How to present observations and data in an appropriate format</li> <li>How to process data using appropriate prefixes (e.g. tera, giga, mega, kilo, centi, milli, micro and nano) and powers of ten for orders of magnitude</li> <li>How to distinguish between a correlation and a cause-effect link</li> <li>How to translate data from one form to another</li> <li>How to identify the presence/absence of a mechanism as reasonable grounds for accepting/rejecting a claim that a factor is a cause of an outcome</li> </ul>                                                                    | <b>F170:</b> 1.1.2, 1.1.3, 2.1.3, 4.1.1<br><b>F171:</b> 1.2.2, 2.1<br><b>F173:</b> 2.1, 3.3.1, 4.2.3<br><b>F174:</b> 2.1.2, 2.2.3<br><b>F175:</b> 1.4, 3.3<br><b>F176:</b> 3.1.2                    |
| <b>HSW6</b>                        | How to evaluate methodology, evidence and data, and resolve conflicting evidence                                   | <ul style="list-style-type: none"> <li>How to interpret and make judgments and draw conclusions from qualitative and quantitative experimental results (including observations and graphs)</li> <li>Anomalies in experimental measurements</li> <li>How to use appropriate maths skills for analysis of quantitative data</li> <li>Limitations in experimental procedures</li> <li>Precision, accuracy, repeatability, reproducibility, and validity of measurements and data, including margins of error and percentage uncertainties in apparatus</li> <li>How to refine experimental design by suggestion of improvements to the apparatus, procedures, and techniques</li> <li>Confidence in a prediction or hypothesis</li> </ul> | <b>F170:</b> 1.1.2<br><b>F171:</b> 5.2<br><b>F173:</b> 1.1.2, 2.2.1, 2.2.2, 3.1, 3.2.1, 3.5.2, 4.2.1, 5.1.1, 5.2, 5.3<br><b>F174:</b> 2.2.1, 3.3.2<br><b>F175:</b> 3.2, 4.3<br><b>F177:</b> 1.2     |

| <b>How Science Works Reference</b> | <b>How Science Works Statement</b>                                                                      | <b>To include understanding of:</b>                                                                                                                                                                                                                                                                                             | <b>Areas of the specification covering the HSW concepts and skills</b>                                                                                                                                                                                                                                                                      |
|------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HSW7</b>                        | How scientific knowledge and understanding develops over time                                           | <ul style="list-style-type: none"> <li>How theories have developed over time and been modified when new evidence has become available</li> <li>Problems that science cannot currently answer</li> </ul>                                                                                                                         | <b>F170:</b> 3.2.1<br><b>F172:</b> 2.3.1<br><b>F175:</b> 2.5<br><b>F177:</b> 2.5.2                                                                                                                                                                                                                                                          |
| <b>HSW8</b>                        | How to communicate information and ideas in appropriate ways using appropriate scientific terminology   | <ul style="list-style-type: none"> <li>Use of diagrammatical, graphical, numerical and symbolic forms in communication</li> <li>Paper based and electronic forms of presentation</li> <li>Accurate representation and labelling of objects observed</li> </ul>                                                                  | <b>F170:</b> 1.1.1<br><b>F171:</b> 1.3.2<br><b>F172:</b> 3.1.2, 3.3.2<br><b>F173:</b> 2.5, 3.2.2<br><b>F174:</b> 3.3.1, 4.1.1, 4.3.1<br><b>F177:</b> 1.1, 2.5.1                                                                                                                                                                             |
| <b>HSW9</b>                        | Consider applications and implications of science and evaluate their associated benefits and risks      | <ul style="list-style-type: none"> <li>Examples of technological applications of science that have made significant positive differences to people's lives</li> <li>Risks that have arisen from new scientific or technological advances</li> <li>Perceived and calculated risk in relation to data and consequences</li> </ul> | <b>F170:</b> 1.1.4, 1.1.5, 3.3.3<br><b>F171:</b> 1.3.1, 3.2.1, 3.2.2<br><b>F172:</b> 1.2, 2.3.1, 3.1.1, 3.3.1, 3.3.2, 4.1, 4.2.1, 4.2.2<br><b>F173:</b> 1.2, 2.5, 3.4<br><b>F174:</b> 1.3.1, 4.2.2<br><b>F175:</b> 1.2, 2.3, 2.4, 3.2, 4.1<br><b>F176:</b> 1.2, 1.3, 5.2<br><b>F177:</b> 1.1, 3.2, 3.3                                      |
| <b>HSW10</b>                       | Consider impact of science and technology on humans, other organisms, and the environment               | <ul style="list-style-type: none"> <li>Reasons why different decisions on the same issue might be appropriate in view of differences in personal, social, economic or environmental context, and be able to make decisions based on the evaluation of evidence and arguments</li> </ul>                                         | <b>F170:</b> 1.1.5, 1.2.3, 2.1.1, 2.2.3, 3.1.2<br><b>F171:</b> 1.2.1, 2.2, 3.2.1, 3.2.2, 5.1.1<br><b>F172:</b> 1.3.1, 3.3.1<br><b>F173:</b> 1.3<br><b>F174:</b> 1.2, 3.2.1, 3.2.2, 4.3.3<br><b>F177:</b> 2.4.3                                                                                                                              |
| <b>HSW11</b>                       | How to evaluate the role of the scientific community in validating new knowledge and ensuring integrity | <ul style="list-style-type: none"> <li>Reasons why scientists should communicate their work to a range of audiences</li> </ul>                                                                                                                                                                                                  | <b>F170:</b> 1.1.1, 1.1.5, 2.1.4, 3.1.1, 4.2<br><b>F171:</b> 4.2.1, 5.3<br><b>F172:</b> 2.2.2, 3.2.1, 4.2.2<br><b>F173:</b> 2.2.1, 3.4, 4.1<br><b>F174:</b> 2.1.1, 2.2.2, 3.2.3, 4.1.2, 4.3.2<br><b>F175:</b> 1.1, 1.3, 2.2, 3.1, 3.4<br><b>F176:</b> 2.2.1, 2.2.2, 3.2.1, 3.3, 4.1, 4.2.1, 4.2.2<br><b>F177:</b> 1.3, 2.1.2, 2.6, 3.1, 4.1 |
| <b>HSW12</b>                       | How to evaluate the ways in which society uses science to inform decision making                        | <ul style="list-style-type: none"> <li>How to distinguish between questions that could be answered using a scientific approach, from those that could not</li> </ul>                                                                                                                                                            | <b>F171:</b> 1.1, 2.3<br><b>F172:</b> 2.3.3, 3.2.3<br><b>F173:</b> 2.3<br><b>F174:</b> 1.1<br><b>F176:</b> 3.2.2, 4.1<br><b>F177:</b> 2.3                                                                                                                                                                                                   |

## Appendix D: Mathematical skills for Human Biology

In order to be able to develop their skills, knowledge and understanding in Cambridge OCR Level 3 Alternative Academic Qualification Cambridge Advanced National in Human Biology, students need to have acquired competence in the mathematical skills listed in the table of coverage.

Students will be required to apply their knowledge and understanding of these mathematical skills to the examined assessment (EA) and non examined assessment (NEA) units where appropriate.

A minimum of 10% of the marks available in the externally assessed units will be for the assessment of mathematical skills. These skills will be applied in the context of Human Biology.

| Mathematical skill to be assessed                |                                                                      | Exemplification of the mathematical skill in context                                                                                        | Areas of the specification which exemplify the mathematical skill (assessment is not limited to the examples below)                                                |
|--------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M0 - Arithmetic and numerical computation</b> |                                                                      |                                                                                                                                             |                                                                                                                                                                    |
| M0.1                                             | Recognise and make use of appropriate units in calculations          | e.g. converting $\mu\text{m}$ to mm as part of cell size calculations                                                                       | <b>F170:</b> 1.1.5, 1.2.2, 4.1.2, 4.1.5<br><b>F171:</b> 3.1<br><b>F173:</b> 2.1, 2.2.1, 4.3<br><b>F174:</b> 1.3.1, 2.2.3<br><b>F176:</b> 1.2, 1.3, 5.2             |
| M0.2                                             | Recognise and use expressions in decimal, ordinary and standard form | e.g. carrying out calculations using numbers expressed in standard form, such as use of magnification                                       | <b>F170:</b> 1.2.2, 4.1.5<br><b>F172:</b> 4.1<br><b>F173:</b> 1.3, 2.1, 3.3.2, 4.3<br><b>F174:</b> 3.1.1, 4.3.1<br><b>F175:</b> 3.2<br><b>F176:</b> 5.2            |
| M0.3                                             | Use ratios, fractions and percentages                                | e.g. calculating surface area to volume ratios                                                                                              | <b>F170:</b> 2.1.2<br><b>F171:</b> 5.1.2<br><b>F174:</b> 1.2, 2.1.2<br><b>F176:</b> 4.1, 5.2<br><b>F177:</b> 2.1.1                                                 |
| M0.4                                             | Estimate results                                                     | e.g. estimating effect of changing experimental parameters on measurable values                                                             | <b>F170:</b> 2.2.3, 3.1.1<br><b>F171:</b> 2.1, 4.1, 5.1.2<br><b>F172:</b> 3.3.2<br><b>F173:</b> 2.3, 2.5, 3.3.1<br><b>F174:</b> 2.1.1, 2.1.2                       |
| M0.5                                             | Use calculators to find and use power functions                      | e.g. estimating the number of bacteria grown over a certain length of time                                                                  | <b>F170:</b> 4.1.4<br><b>F172:</b> 4.1<br><b>F173:</b> 3.3.2, 3.5.1<br><b>F174:</b> 3.1.1                                                                          |
| <b>M1 - Handling data</b>                        |                                                                      |                                                                                                                                             |                                                                                                                                                                    |
| M1.1                                             | Use an appropriate number of significant figures                     | e.g. reporting calculations to an appropriate number of significant figures given raw data quoted to varying numbers of significant figures | <b>F170:</b> 4.1.2<br><b>F171:</b> 4.1<br><b>F172:</b> 2.1.2, 4.1<br><b>F173:</b> 2.1, 3.2.2, 4.3<br><b>F175:</b> 2.1<br><b>F176:</b> 1.2, 1.3<br><b>F177:</b> 2.2 |
| M1.2                                             | Understand the terms mean, median and mode                           | e.g. calculating or comparing the mean, median and mode of a set of data such as height or mass of a group of organisms                     | <b>F171:</b> 1.2.1, 2.3, 4.2.1<br><b>F173:</b> 1.3, 5.1.1<br><b>F174:</b> 4.1.1<br><b>F175:</b> 2.3<br><b>F177:</b> 2.2, 2.4.1                                     |

|                     |                                                                                                                                                                                     |                                                                                                                                                           |                                                                                                                                                                                                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1.3                | Understand simple probability                                                                                                                                                       | e.g. understanding probability in context of monohybrid crosses                                                                                           | <b>F170:</b> 1.1.5<br><b>F171:</b> 1.2.2, 5.1.1<br><b>F172:</b> 1.3.1, 2.1.1, 2.1.2, 2.3.2, 3.1.2, 3.2.3<br><b>F173:</b> 4.1<br><b>F175:</b> 2.3, 2.4, 4.1                                                                                                           |
| M1.4                | Make order of magnitude calculations                                                                                                                                                | e.g. making order of magnitude calculations in relation to magnification                                                                                  | <b>F170:</b> 1.1.2, 3.2.2<br><b>F172:</b> 2.3.2<br><b>F173:</b> 2.1                                                                                                                                                                                                  |
| M1.5                | Uncertainties in measurements and use of simple techniques to determine uncertainty when data are combined by addition, subtraction, multiplication, division and raising to powers | e.g. determining uncertainty when two burette readings are used to calculate a titre value                                                                | <b>F170:</b> 1.1.2, 3.1.2<br><b>F172:</b> 3.3.1<br><b>F173:</b> 3.2.2, 3.5.3, 4.2.2<br><b>F174:</b> 1.1, 3.3.2, 4.2.3<br><b>F175:</b> 2.1                                                                                                                            |
| M1.6                | Frequency tables and diagrams, bar charts, line graphs, scatter plots, pie charts, and histograms                                                                                   | e.g. interpret data for a variety of graphs such as electrocardiogram traces                                                                              | <b>F170:</b> 2.2.3, 3.3.2<br><b>F171:</b> 1.2.2, 1.3.1, 2.2, 4.2.2, 5.2<br><b>F172:</b> 1.3.1, 3.2.2<br><b>F173:</b> 2.3, 2.5<br><b>F174:</b> 2.2.2, 3.3.2, 4.1.1<br><b>F175:</b> 3.2, 3.3, 4.2, 4.3<br><b>F176:</b> 3.3, 4.2.1<br><b>F177:</b> 1.2, 2.3, 2.4.1, 3.3 |
| M1.7                | Understand the principles of sampling as applied to scientific data, including representative sampling                                                                              | e.g. how to ensure sampling is representative in a population                                                                                             | <b>F170:</b> 4.1.3<br><b>F171:</b> 2.3, 4.2.2, 5.1.1<br><b>F173:</b> 2.2.1, 2.2.2, 5.3<br><b>F176:</b> 4.2.1<br><b>F177:</b> 3.1                                                                                                                                     |
| M1.8                | Understand measures of dispersion, including standard deviation and range                                                                                                           | e.g. understanding why standard deviation might be a more useful measure of dispersion for a given set of data, such as where there is an outlying result | <b>F173:</b> 5.1.2<br><b>F174:</b> 2.1.1, 4.3.1                                                                                                                                                                                                                      |
| <b>M2 - Algebra</b> |                                                                                                                                                                                     |                                                                                                                                                           |                                                                                                                                                                                                                                                                      |
| M2.1                | Understand and use the symbols:<br>=, <, >, <<, >>, α, ~                                                                                                                            | e.g. calculating surface area to volume ratios                                                                                                            | <b>F171:</b> 2.1<br><b>F173:</b> 3.4<br><b>F174:</b> 1.3.1, 4.2.1<br><b>F176:</b> 3.1.1, 3.1.2<br><b>F177:</b> 1.2                                                                                                                                                   |
| M2.2                | Change the subject of an equation, including non-linear equations                                                                                                                   | e.g. carrying out magnification and cell size calculations                                                                                                | <b>F170:</b> 1.1.2<br><b>F171:</b> 1.3.2<br><b>F173:</b> 2.1                                                                                                                                                                                                         |
| M2.3                | Substitute numerical values into algebraic equations using appropriate units for physical quantities                                                                                | e.g. carrying out pulmonary ventilation rate calculations                                                                                                 | <b>F170:</b> 2.2.3<br><b>F173:</b> 3.4<br><b>F174:</b> 3.1.2                                                                                                                                                                                                         |
| M2.4                | Solve algebraic equations                                                                                                                                                           | e.g. solving equations in a biological context, such as pulmonary ventilation rate                                                                        | <b>F170:</b> 2.2.3<br><b>F174:</b> 3.1.2, 4.2.1                                                                                                                                                                                                                      |
| M2.5                | Understand power, exponential and logarithmic functions                                                                                                                             | e.g. interpreting a graph of bacterial growth                                                                                                             | <b>F170:</b> 4.1.4                                                                                                                                                                                                                                                   |



| <b>M3 - Graphs</b>                    |                                                                                                              |                                                                                |                                                                                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M3.1                                  | Translate information between graphical, numerical, and algebraic forms                                      | e.g. interpreting and analysing spectra                                        | <b>F170:</b> 2.1.2, 3.1.2<br><b>F171:</b> 2.2, 5.1.2<br><b>F172:</b> 1.3.2, 2.3.1, 2.3.3, 3.1.1, 4.2.2<br><b>F173:</b> 3.1, 3.3.1, 4.2.4, 5.1.1<br><b>F174:</b> 1.2, 2.1.1, 3.3.1<br><b>F175:</b> 3.1, 3.2, 4.1<br><b>F176:</b> 3.3, 5.1 |
| M3.2                                  | How to plot two variables from experimental or other data                                                    | e.g. plotting calibration curves                                               | <b>F170:</b> 2.2.3, 3.1.2<br><b>F173:</b> 3.1, 3.5.2, 5.1.1<br><b>F174:</b> 3.3.1                                                                                                                                                        |
| M3.3                                  | Understand that $y = mx + c$ represents a linear relationship                                                | e.g. interpreting the effect of stroke volume and heart rate on cardiac output | <b>F171:</b> 2.2<br><b>F173:</b> 3.5.2<br><b>F174:</b> 3.3.1<br><b>F176:</b> 2.1.1, 2.2.2                                                                                                                                                |
| M3.4                                  | The slope and intercept of a linear graph                                                                    | e.g. reading off and interpreting rate of diffusion                            | <b>F170:</b> 1.1.4<br><b>F174:</b> 3.3.3<br><b>F175:</b> 1.1<br><b>F176:</b> 2.1.1, 2.2.2                                                                                                                                                |
| M3.5                                  | Rate of change from a graph showing a linear relationship                                                    | e.g. calculating diffusion rate                                                | <b>F170:</b> 2.1.2<br><b>F171:</b> 3.2.2<br><b>F175:</b> 1.1                                                                                                                                                                             |
| M3.6                                  | Sketch relationships for graphs                                                                              | e.g. sketching the relationship between exercise and breathing rate            | <b>F170:</b> 2.2.2, 3.2.1, 4.1.4<br><b>F171:</b> 1.2.2, 2.1, 3.2.1<br><b>F172:</b> 2.2.2<br><b>F174:</b> 3.3.3<br><b>F176:</b> 5.1                                                                                                       |
| <b>M4 - Geometry and trigonometry</b> |                                                                                                              |                                                                                |                                                                                                                                                                                                                                          |
| M4.1                                  | Visualise and represent two-dimensional representations of 3D objects                                        | e.g. drawing biological molecules                                              | <b>F170:</b> 1.1.4, 1.1.5<br><b>F171:</b> 3.1<br><b>F174:</b> 1.1, 1.3.1                                                                                                                                                                 |
| M4.2                                  | Circumferences and areas of circles, surface areas and volumes of rectangular blocks, cylinders, and spheres | e.g. calculating the surface area or volume of a cell                          | <b>F170:</b> 1.1.2, 3.3.3<br><b>F173:</b> 2.4                                                                                                                                                                                            |

The questions and tasks across all units that are used to target mathematical skills will be at a level of demand that is appropriate to our Cambridge Advanced Nationals in Human Biology. The questions that assess mathematical skills will not be of a lower demand than that of questions and tasks in the assessment for Level 1/Level 2 GCSE (9-1) in Mathematics.

The list of examples provided in the table is not exhaustive and is not limited to Level 2 examples. These skills could be developed in other areas of the specification content from those indicated.

Students will not be expected to memorise mathematical formulas. Any necessary mathematical formulas will be provided in the examination paper.

Mathematical skills should be taught using both theoretical and practical contexts.

## Appendix E: Units in science

It is expected that learners will show and be able to apply understanding of the physical quantities and corresponding units, and SI base units and derived units listed below. The tables also include symbols commonly used for these quantities; use of symbols by learners is optional. Learners will be able to use them in qualitative work and calculations.

| Physical quantity     | Common symbol(s)<br>(use of these symbols is optional)                                                                                                                                                                                                                                                                                                    | SI base unit | Unit abbreviation |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------|
| Length                | $d$ – diameter<br>$h$ – height (e.g. height raised above ground level to calculate gravitational potential energy)<br>$l$ – length (e.g. of a wire)<br>$s$ – displacement (e.g. displacement of a force)<br>$x$ – extension (e.g. of a spring) or distance travelled (e.g. for attenuation of X-rays through a medium)<br>$\lambda$ (lambda) – wavelength | metre        | m                 |
| Mass                  | $m$                                                                                                                                                                                                                                                                                                                                                       | kilogram     | kg                |
| Time                  | $t$<br>$t_E$ – effective half-life<br>$t_{1/2}$ – physical half-life<br>$t_B$ – biological half-life<br>$T$ – time period                                                                                                                                                                                                                                 | second       | s                 |
| Temperature           | $T$ – for Kelvin temperature<br>$\Delta T$ – for change in Kelvin temperature                                                                                                                                                                                                                                                                             | kelvin       | K                 |
| Amount of a substance | $n$                                                                                                                                                                                                                                                                                                                                                       | mole         | mol               |

The following table includes SI derived or SI accepted units for quantities which will be commonly used across the qualification:

| Physical quantity | Common symbol(s)<br>(use of these symbols is optional)                                                   | SI derived / accepted unit                                  | Unit abbreviation                                     |
|-------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|
| Area              | $A$                                                                                                      | squared metre                                               | m <sup>2</sup>                                        |
| Concentration     | $c$                                                                                                      | mole per cubic decimetre<br>gram per cubic decimetre        | mol dm <sup>-3</sup><br>g dm <sup>-3</sup>            |
| Temperature       | $\theta$ (theta) – for Celsius temperature<br>$\Delta\theta$ (theta) – for change in Celsius temperature | degree Celsius                                              | °C                                                    |
| Volume            | $V$                                                                                                      | cubic metre; litre;<br>cubic decimetre;<br>cubic centimetre | m <sup>3</sup> ; L; dm <sup>3</sup> ; cm <sup>3</sup> |



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